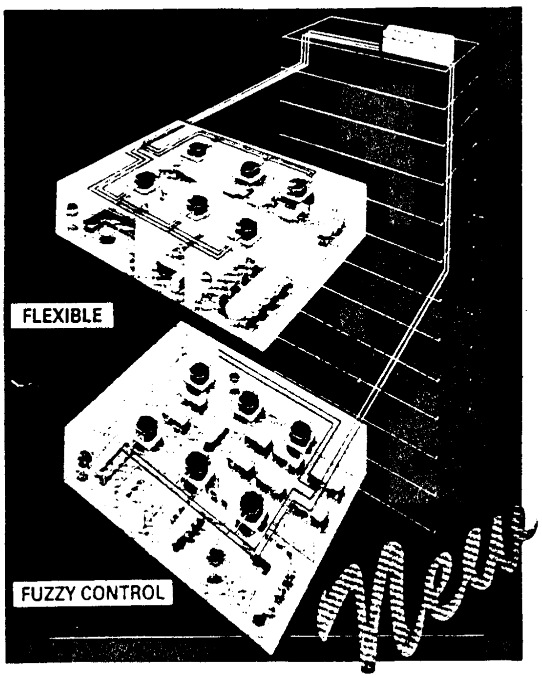
# HITACHI INVERTER-DRIVEN MULTI-SPLIT SYSTEM HEAT PUMP AIR CONDITIONERS

## HI-MULTI SET-FREE FS2 SERIES AND HI-MULTI SET-FREE FX2 SERIES

## SERVICE MANUAL




Models:

**Indoor Units**

- **In-the-Ceiling Type**
  RPI-0.8FS
  RPI-1FS
  RPI-1.5FS
  RPI-2FS
  RPI-2.5FS
  RPI-4FS
  RPI-5FS
- **4-Way Cassette Type**
  RCI-1FSE
  RCI-1.5FSE
  RCI-2FSE
  RCI-2.5FSE
- **2-Way Cassette Type**
  RCD-1FS
  RCD-1.5FS
  RCD-2FS
  RCD-2.5FS
- **Wall Type**
  RPK-1.5FS
- **Floor Type**
  RPF-1.5FS
- **Floor Concealed Type**
  RPFI-1.5FS
- **Ceiling Type**
  RPC-2.5FS(E)

**Outdoor Units**

- **FS2 SERIES**
  RAS-5FS2
  RAS-8FS2
  RAS-10FS2
- **FX2 SERIES**
  RAS-5FX2
  RAS-8FX2
  RAS-10FX2

This service manual provides the technical information for the HITACHI Inverter-driven Multi-Split System Heat Pump Air Conditioners,"*HI-MULTI SET-FREE, FS2 and FX2 Series*".
Read this manual carefully before service activities are started.

**HITACHI**

P5412838

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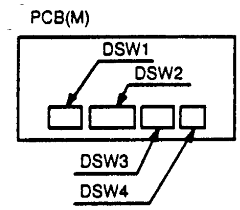
# 1. TROUBLESHOOTING

## 1.1 Initial Troubleshooting

### 1.1.1 Dip Switch Setting

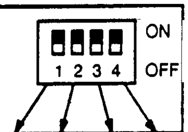
(1) Indoor Unit




The PCB in the indoor unit is equipped with 4 dip switches. Before testing unit, set these dip switches according to the following instructions. Unless these dip switches are set in the field, the unit can not be operated.

a. Unit No. Setting (DSW1)

Setting is required. Set the unit No. of all indoor units respectively and serially, by following setting position shown in the table below. Numbering must start from "0" for every outdoor unit except the case using the central station.

| Unit No. | Remote Control Switch PC-2H Display |  1 | 2 | 3 | 4 ON/OFF |
|---|---|---|---|---|---|
| 0 | A0 | ON | ON | ON | ON |
| 1 | A1 | OFF | ON | ON | ON |
| 2 | A2 | ON | OFF | ON | ON |
| 3 | A3 | OFF | OFF | ON | ON |
| 4 | A4 | ON | ON | OFF | ON |
| 5 | A5 | OFF | ON | OFF | ON |
| 6 | A6 | ON | OFF | OFF | ON |
| 7 | A7 | OFF | OFF | OFF | ON |
| 8 | A8 | ON | ON | ON | OFF |
| 9 | A9 | OFF | ON | ON | OFF |
| 10 | AA | ON | OFF | ON | OFF |
| 11 | Ab | OFF | OFF | ON | OFF |
| 12 | AC | ON | ON | OFF | OFF |
| 13 | Ad | OFF | ON | OFF | OFF |
| 14 | AE | ON | OFF | OFF | OFF |
| 15 | AF | OFF | OFF | OFF | OFF |

Setting before Shipment

NOTES:

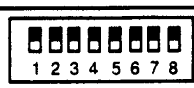
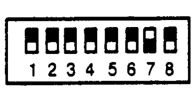
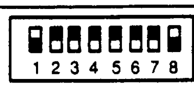
1. Automatic Address
   (Dip switch setting is not required.)
   The unit number for all indoor units in the same refrigerant cycle will be addressed automatically, when the remote control switch or the central station is used to the indoor units in same refrigerant cycle and the dip switches of all indoor units are set at "No.0" position.

2. For Other Cases
   See the chapter 4 for other cases of setting using the remote control switch or the central station.

b. Optional Function Setting (DSW2)

When setting the switch for optional function, refer to the chapter 4 for each optional functions

Setting before Shipment

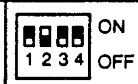
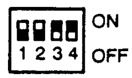
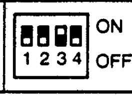
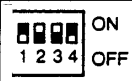
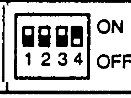
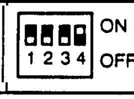
| DSW2 | Indoor Unit Model |
|---|---|
|  ON / OFF 1 2 3 4 5 6 7 8 | RPC-2.5FS(E)<br>RPF-1.5FS<br>RPFI-1.5FS |
|  ON / OFF 1 2 3 4 5 6 7 8 | RPI-0.8FS, RPI-1FS, RPI-1.5FS, RPI-2FS, RPI-2.5FS,<br>RPI-4FS, RPI-5FS<br>RCI-1FSE, RCI-1.5FSE, RCI-2FSE, RCI-2.5FSE<br>RCD-1FS, RCD-1.5FS, RCD-2FS, RCD-2.5FS |
|  ON / OFF 1 2 3 4 5 6 7 8 | RPK-1.5FS |

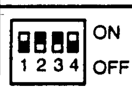
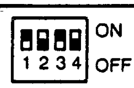
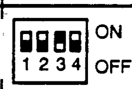
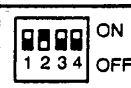
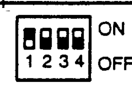
NOTE:

The "■" mark indicates position of dip switches. Figures show the setting before shipment.

c. Capacity Code Setting (DSW3)

No setting is required, due to setting before shipment. This switch is used for setting the capacity code which corresponds to the Horse-Power of the indoor unit.

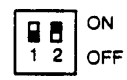
| Horse-Power | 0.8 | 1 | 1.3 | 1.5 | 1.8 | 2 |
|---|---|---|---|---|---|---|
| Setting Pattern |  |  |  |  |  |  |

| Horse-Power | 2.2 | 2.5 | 2.8 | 4 | 5 |
|---|---|---|---|---|---|
| Setting Pattern |  |  |  |  |  |

d. Capacity Adjustment Setting (DSW4)

No setting is required, due to setting before shipment.

This switch is used for setting the capacity adjustment.

| Horse-Power | 0.8~5 |
|---|---|
| Setting Pattern |  |

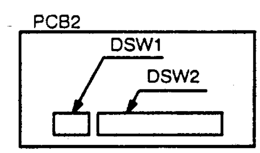
CAUTION:

1. Before setting dip switches, firstly turn OFF power source and set the position of the dip switches. If the switches are set without turning OFF the power source, the switches can not function.
2. Check to ensure that the connector from transformer is correctly connected to the PCN7 on PCB, corresponding with the power supply voltage.
   - CN29 (brown line) for 220V
   - CN30 (red line) for 240V

(2) CH Unit

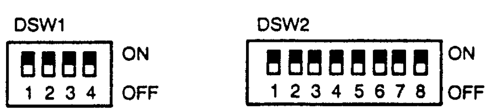
a. Position of Dip Switches

The PCB2 in the electrical box of CH unit is equipped with 2 types of dip switches (DSW1 DSW2), as figure below.




b. Setting of Dip Switches

No setting is required, due to setting before shipment as figure below. These are utilized for the self-diagnosis of PCB. (Refer to chapter 1.3.4.)




NOTE:

The "■" mark indicates position of dip switches. Figures show the setting before shipment.

CAUTION:

1. Before setting dip switches, firstly turn OFF power source and set the position of the dip switches. If the switches are set without turning OFF the power source, the switches can not function.
2. Check to ensure that the connector from transformer is correctly connected to the PCN7 on PCB, corresponding with the power supply voltage.
   - CN21 (brown line) for 220V
   - CN22 (red line) for 240V

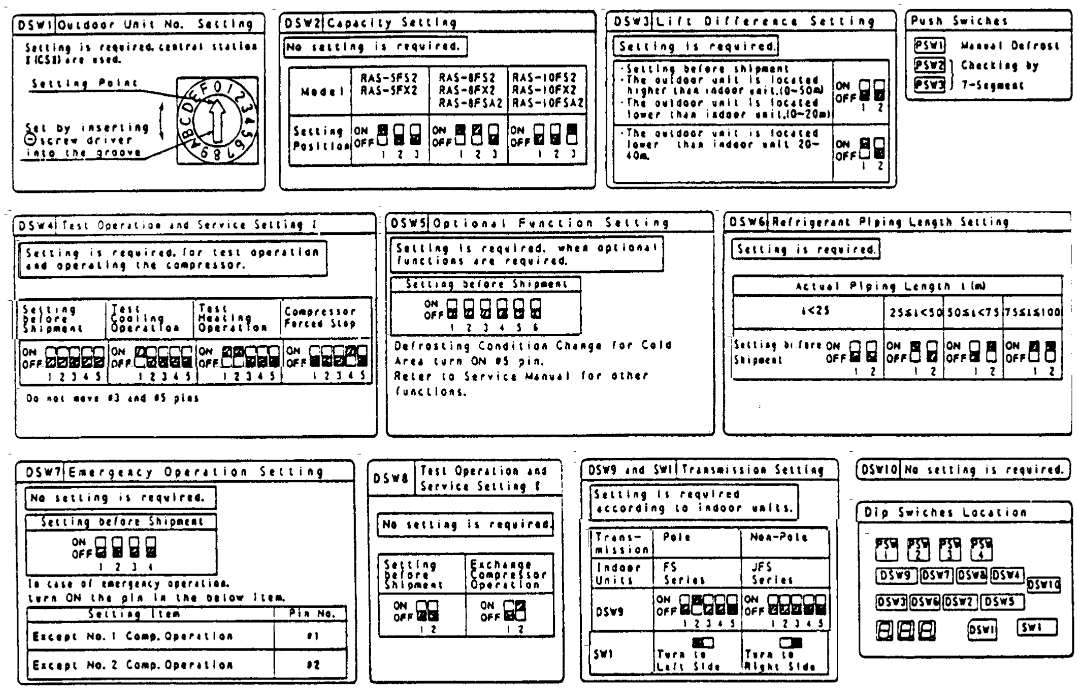
(3) Outdoor Unit

TURN OFF all power sources before setting. Without turning OFF, the switches do not work and the contents of the setting are invalid. Mark of "■" indicates the position of dip switches. Set the dip switches according to the figure below.

NOTES:

By using switch DSW4, the unit is started or stopped 10 to 20 seconds after the switch is operated.

2 Make the outdoor unit No. clear to distinguish from other outdoor units for service and maintenance.

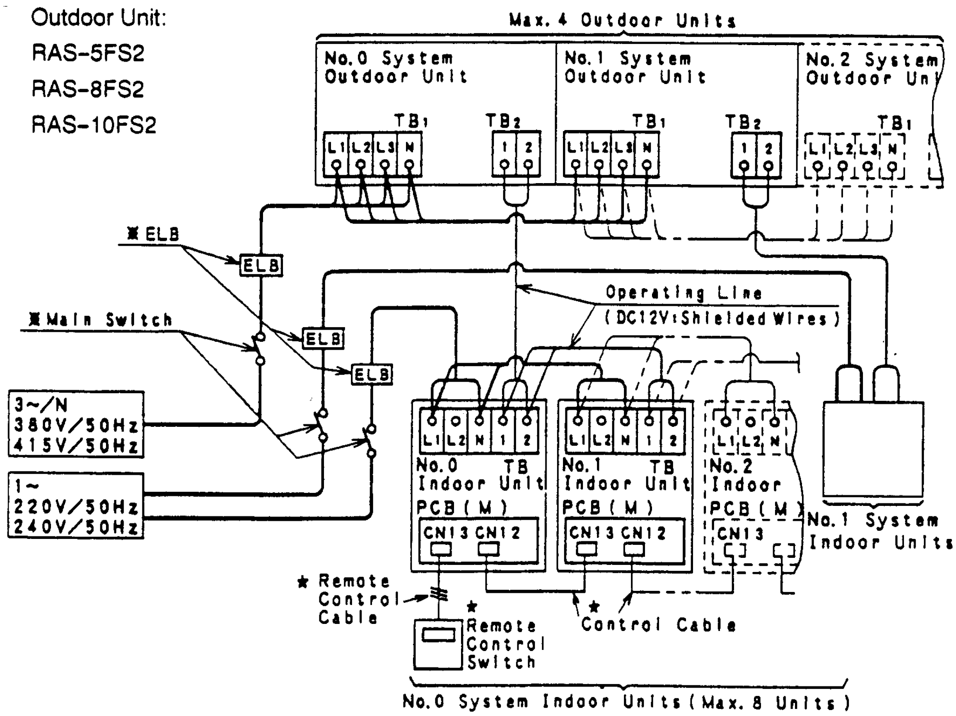


1 2 Checking of Electrical Wiring

- Check to ensure that the terminal for power source wiring (terminals "L1" to "L1" and "N" to "N" of each terminal board: AC380-415V) and intermediate wiring (Operating Line: terminals "1" to "1" and "2" to "2" of each terminal board: DC12V) between the indoor unit and the outdoor unit coincide correctly, as figure below.
- Check to ensure that the shielded wires are used for intermediate wiring to protect noise obstacle at length of less than 300m and size complied with local code.
- Check to ensure that the wirings and the breakers are chosen correctly, as shown in Table 1.1
- All the field wiring and equipment must comply with local code.

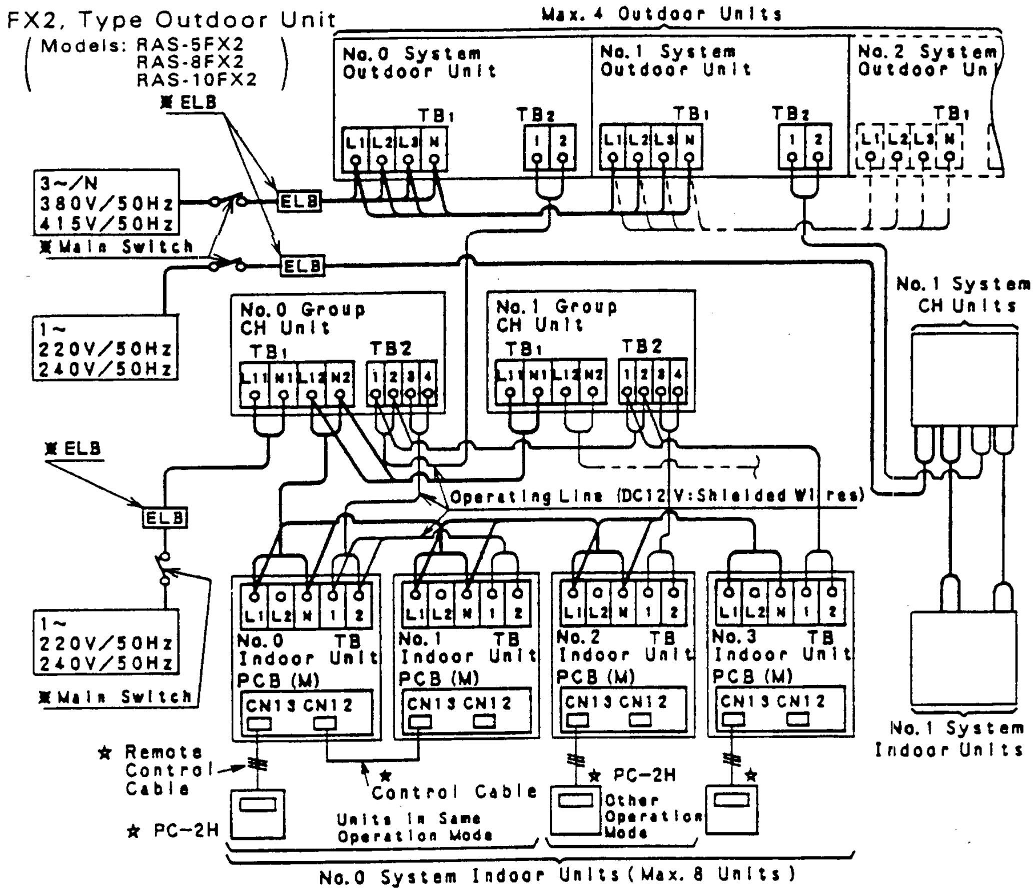
a. Example for Electrical Wiring Connection of FS2 System

Outdoor Unit:
RAS-5FS2
RAS-8FS2
RAS-10FS2




TB: Terminal Board
PCB: Printed Circuit Board
---: Field Wiring
---: Field Wiring
(Further Extension)
※ : Field-Supplied
☆ : Optional Accessory

b. Example for Electrical Wiring Connection of FX2 System




c. Table for Terminal Connection

O.U.: Outdoor Unit, U.: Indoor Unit, CH: CH Unit

| Wiring | System | [Connection (Connection of Terminals)] |
|---|---|---|
| Power Supply | FS2, FX2 | [O.U.-O.U. (L1-L1, L2-L2, L3-L3, N-N)] |
| | | [I.U.-I.U. (L1-L1, N-N)] |
| | FX2 | [CH-CH (L12-L11, N2-N1)], [CH-I.U. (L12-L1, N2-N1)] |
| Operating | FS2 | [O.U.-I.U., I.U.-I.U. (1-1, 2-2)] |
| | FX2 | [O.U.-CH, CH-CH (1-1, 2-2)] |
| | | [CH-I.U. (3-1, 4-2)], [CH-I.U. (1-1, 2-2: Cooling Only)] |
| | | [I.U.-I.U. (1-1, 2-2: Between Same Operation Mode Units)] |

Table 1.1 Electrical Data and Recommended Wiring, Breaker Size/1 Outdoor Unit

| Model | Max. Running Current (A) | Power Supply Line ($\phi$ mm) | ELB Nominal Current (A) | ELB Nominal Sensitive Current (mA) | Fuse (A) |
|---|---|---|---|---|---|
| RAS-5FS2, RAS-5FX2 | 17 | MLFC2.0SQ | 20 | 30 | 20 |
| RAS-8FS2, RAS-8FX2 | 32 | MLFC3.5SQ | 40 | 30 | 40 |
| RAS-10FS2, RAS-10FX2 | 36 | | | | |

ELB: Earth Leakage Breaker, MLFC: Flame Retardant Polyflex Wire

NOTE: Regarding the wiring or breakers, follow to the local code.

### 1.1.3 Checking by 7-Segment Display

#### (1) Simple Checking by 7-Segment Display

1 ※Turn on All Indoor Units

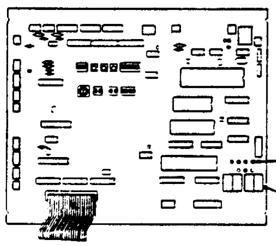
※ All the Indoor Units Connected to the Outdoor Unit

2 Turn on the Outdoor Unit

3 Auto-addressing Starts

During auto-addressing, the following items can be checked using the outdoor unit's on-board 7-segment LED display.
(1) Disconnection of power supply to the indoor unit.
(2) Reverse connection of the operating line between the outdoor and indoor units.
(3) Duplication of indoor unit number.

Outdoor Unit Printed Circuit Board PCB1




| | |
|---|---|
| Normal Case | (1) The outdoor unit's on-board 7-segment LED display indicates as follows normally.<br>①<br>101 ~ 17F, 201 ~ 27F: will be indicated a few times repeatedly.<br>401, 801: may or may not be indicated momentarily.<br>During this time, 2~5 minutes will pass (it depends on the number of indoor units.).<br>② LED's off. |
| Abnormal Case | (2) The outdoor unit's on-board 7-segment LED display indicates as follows if there is something wrong.<br>(A) The indoor units are not supplied with power. ➡ ① 101 ~ 17F will be indicated continuously.<br>(B) Disconnection of the operating line between the outdoor and indoor units. ➡ ① 101 ~ 17F will be indicated continuously.<br>(C) Reverse connection of the operating line between the outdoor and indoor units. ➡ ① 101 continues to appear for 60 seconds. ② 03 continues to flash after then, for 60 seconds.<br>(D) Duplicated settings of the indoor unit number DIP switch DSW1 (Refer to the description of alarm code "35", in the item 1.2.2, "Troubleshooting by Alarm Code".) |

1.4 Emergency Operation when Compressor is Damaged

(1) Failure of Inverter Compressor

a. Contents of Operation

① This operation is an emergency operation by a constant compressor, when the inverter compressor is failed.

② By turning ON "#1" of DSW7 on PCB1 and "#1" of DSW1 on PCB3, emergency operation is started.

③ Control of the emergency operation is same with normal control except the inverter compressor stoppage.

b. Operating Condition

The constant speed compressor is forced to be stopped for compressor protection under the condition below.

Total Capacity of Thermo-ON indoor units<50% of Outdoor Unit Capacity

In case of the above condition, the compressor is operated at a low frequency and stopped repeatedly. Therefore, the compressor is forced to be stopped to protect it.

NOTE:

If the printed circuit board for inverter (PCB3, 4) is damaged, this is not available.

c. Method of Emergency Operation

● Checking Before Emergency Operation.

① Measure insulation resistance of the inverter compressor.
If the insulation resistance is 0Ω, this emergency operation is not available.
There is a possibility that refrigerant oil may be oxidized, if the emergency operation is performed, the other compressor is damaged.

② In case of total capacity of Thermo-ON indoor units are more than 50% of outdoor unit capacity, emergency operation is available.

③ In this emergency operation, frequency of the compressor is not controlled at each 1Hz. Therefore, alarm code "07", "43", "44", "45" or "47" may be indicated on LCD. Details of alarm codes are shown in the alarm code table (page 1-16).

④ This emergency operation does not provide sufficient cooling capacity.

⑤ This method is an emergency operation temporarily when the inverter compressor is damaged. Therefore, change the new one as soon as possible.

⑥ Turn OFF "#1" of DSW1 on PCB7 and "#1" of DSW1 on PCB3 after changing the new compressor. If this setting is performed the inverter compressor will be damaged.

d. Emergency Operation

① Turn OFF all the power source switches.

② Disconnect the wiring from the inverter compressor. Insulate the faston terminals for inverter compressor wires by insulation tape.

③ Set the No.1 of DSW7 on the PCB1 at the "ON" side.

④ Turn ON all the power source switches.

⑤ Operate the system by remote control switches.

⑥ The system is stopped by turning OFF all the remote control switches or turning OFF all the power source switches.

(2) Failure of Constant Speed Compressor

a. Contents of Operation

① This operation is an emergency operation by the inverter compressor, when the constant speed compressor is failed.

② This operation is controlled by a normal control.

b. Operating Condition

① Set the No.2 of DSW7 on the PCB1 at the "ON" side.

② Temperature of THM2 & THM3 on the top of compressors are not ignored by setting DSW7. If the thermistor is short-circuited or cut, this operation is available.

### 1.1.5 Failure of Power Supply to Indoor Unit and Remote Control Switch

- Lights and LCD is not Indicated
- Not Operated

If fuses are melted or a breaker is activated, investigate the cause of over current and take necessary action.

| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Power Failure or Power is Not ON | | Measure voltage by volt-meter. | Supply power. |
| Melted Fuse or Activation of Breaker at Power Source | Short Circuit Supplied between Wiring | Check for any uncovered part of wiring. | Remove the cause of short-circuit and replace the fuse. |
| | Short Circuit of Wiring to Earth | Measure insulation resistance. | Remove the cause of short-circuit and replace the fuse. |
| | Failure of Indoor Unit Fan Motor | Measure resistance between wiring and insulation resistance. | Replace the fan motor and fuse. |
| Melted Fuse at Control Circuit | Short Circuit Supplied between Wiring | Check for any uncovered part of wiring. | Remove the cause of short-circuit and replace the fuse. |
| | Short Circuit of Control Circuit to Earth | Measure insulation resistance. | Remove the cause of short-circuit and replace the fuse. |
| | Failure of Indoor Unit Fan Motor | Measure resistance between wiring and insulation resistance. | Replace the fan motor and fuse. |
| Failure of Transformer at Indoor Unit Side | | Measure voltage at secondary side. | Replace the transformer. |
| Disconnected Cable of Remote Control Switch | | Connect the cable. | Replace cable or repair It. |

(1.1.5 Failure of Power Supply to Indoor Unit and Remote Control Switch)

| | | | |
|---|---|---|---|
| Insufficient Contacting at Connectors of Remote Control Switch | Insufficient Connection or Incorrect Connection of Indoor Unit PCB | Check connectors. | Correctly connect It. |
| | Insufficient Connection or Incorrect Connection of PCB in Remote Control Switch | | |
| Failure of Remote Control Switch | | Check it by self-checking mode *1). | Replace remote control switch if failed. |
| Failure of PCB | Unconnected Wiring to PCB | Check connectors. | Correctly connect wires. |
| | Failure of PCB | Check it by self-checking mode *2). | Replace PCB if failed. |
| Incorrect Wiring Connection | | Take action according to the procedure indicated in "TEST RUN". | |

*1): Refer to Item 1.3.2.
*2): Refer to Item 1.3.1.

1.1.6 Abnormal Transmission of Remote Control Switch

- "RUN" Lamp on Remote Control Switch:
  Flickering every 2 seconds

| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Disconnection or Insufficient Contacting of Remote Control Cable | | Check cable and connections. | Repair cable or connect it. |
| Incorrect Wiring Connection (Incorrect Polarity) | | Check wiring and connections. | Repairing |
| Failure of Remote Control Switch | | Check it by self-check mode *1). | Replace it if remote control switch is faulty. |
| Failure of PCB (in Indoor Unit and Remote Control Switch) | Disconnected Wire to PCB | Check connections. | Correctly connect wires. |
| | Failure of PCB | Check it by self-check mode *2). | Replace PCB if failed. |
| Immediate Starting within 5 minutes after Power Supply *3) | | | Restart the system after completion of automatic setting of unit No. |

*1): Refer to Item 1.3.2.
*2): Refer to Item 1.3.1.
*3): This phenomenon occurs when the system is newly installed and is started within 5 minutes after power supply, since automatic setting of unit No. is performed during this period.

### 1.1.7 Abnormalities of Devices

In the case that no abnormality (Alarm Code) is indicated on the remote control switch, and normal operation is not available, take necessary action according to the procedures mentioned below.

| Phenomenon | Cause | | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|---|
| "RUN" light is ON and LCD is indicated. However, the system does not operated. (ex. The indoor, outdoor fans or compressor does not operate.) | Failure of Indoor Unit Fan Motor | Disconnected Coil | Measure coil resistance by tester. | Replace Indoor unit fan motor. |
| | | Burnt-Out Coil | Measure insulation resistance. | Replace Indoor unit fan motor. |
| | Failure of Outdoor Unit Fan Motor | Disconnected Coil | Measure coil resistance by tester. | Replace outdoor unit fan motor. |
| | | Burnt-Out Coil | Measure insulation resistance. | Replace outdoor unit fan motor. |
| | Failure of Magnetic Switch for Outdoor Unit Fan Motor | Insufficient Contacting | Measure voltage between contacting parts. | Replace PCB for outdoor unit. |
| | Failure of Comp. Motor | | Measure resistance between two wires. | Replace compressor. |
| | Failure of Comp. | | Check for abnormal sound from comp. | Replace compressor. |
| | Failure of Magnetic Switch for Comp. | Insufficient Contacting | Check magnetic switch activates correctly or not. | Replace magnetic switch |
| | Failure of One of PCBs | Disconnected Wiring to PCB | Check connections. | Correctly connect wiring. |
| | | Failure of PCB | Check PCB by self-checking *1). | Replace PCB if failed. |
| The Comp. does not Stop or Start even if the Setting temperature on LCD is Changed *3) | Failure of Air Inlet Thermistor | Failure of Thermistor | Check it by self-checking *2). | Replace or correctly connect wires if abnormality exists. |
| | | Disconnection of Thermistor | Check it by self-checking *2). | Replace or correctly connect wires if abnormality exists. |
| | Abnormality of Remote Control Switch Cord | | Check it by self-checking *2). | Replace or correctly connect wires if abnormality exists. |
| | Failure of Indoor Unit PCB | | Check it by self-checking *1). | Replace PCB if failed. |

*1): Refer to Item 1.3.1 ~ 1.3.4.

*2): Refer to Item 1.3.1.

*3): Even if controllers are normal, the compressor does not operate under the following conditions.

① Indoor Air Temp. is lower than 21℃ or Outdoor Air Temp. is lower than -5℃ for FS2 and FX2 during cooling operation.

② Indoor Air Temp. is higher than 30℃ or Outdoor Air Temp. is higher than 23℃ during heating operation.

③ When a cooling (or heating) operation signal is given to the outdoor unit and a different mode as heating (or cooling) operation signal is given to indoor units for FS2.

④ When a emergency stop signal is given to outdoor unit.

⑤ When the dip switch, DSW4 on the outdoor unit PCB is set at the "ON" side and the compressor is stopped

(1.1.7 Abnormalities of Devices)

| Phenomenon | Cause | | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|---|
| Indoor Fan Speed is Not Changed | Failure of Discharge Air Temp. Thermistor | Failure of Thermistor | Check thermistor by self-checking *2). | Replace or correctly connect wiring when it is abnormal. |
| | | Disconnected Wire of Thermistor | | |
| | Failure of Remote Control Switch | | Check it by self-checking *1). | Replace if failed. |
| | Failure of PCB for Indoor Unit | | | Replace if PCB is failed. |
| No Defrosting Operation Mode is Available during Heating Operation or Defrosting Operation is Continued | Failure of Thermistor for Outdoor Evaporating Temp. during Heating | Failure of Thermistor | Check it by self-checking *1). | Replace or correctly connect when it is abnormal. |
| | | Disconnected Wire of Thermistor | | |
| | Failure of 4-Way Valve | Disconnected 4-Way Valve Coil | Measure resistance of coil. | Replace 4-way valve. |
| | | Incorrect Activation of 4-Way Valve | Enforced power supply. | |
| | Disconnected Control Wire between Indoor Unit, CH Unit and Outdoor Unit | | Check connectors. | Correctly connect wiring. |
| | Failure of Outdoor Unit PCB | Disconnected Wire to PCB | Check connectors. | Correctly connect wiring. |
| | | Failure of PCB | Check it by self-checking *1). | Replace it when check mode is not available. |
| | Failure of Indoor Unit PCB | Disconnected Wire to PCB | Check connectors. | Correctly connect wiring. |
| | | Failure of PCB | Check PCB by self-checking *1). | Replace if PCB is failed. |
| Light and LCD Indication on Remote Control Switch are kept ON | Failure of PCB in Indoor Unit or Remote Control Switch | | | |

*1): Refer to Item 1.3.1 ~ .3.4.
*2): Refer to Item 1.3.1.

## TROUBLESHOOTING

(1.1.7 Abnormalities of Devices)

| Phenomenon | Cause | | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|---|
| Insufficient Cooling Operation | Indoor Heat Load is Greater than Cooling Capacity | | Calculate heat load. | Use a bigger unit. |
| | Excessively Low Suction Pressure | Gas Leakage or Shortage of Refrigerant | Check refrigerant quantity *1). | Correctly charge refrigerant after repairing gas leakage. |
| | | Excessively Small Diameter Tube or Long Piping | Measure and check field-supplied pipes. | Use correct pipes. |
| | | Incorrect Activation of Check Valve of Outdoor Unit | Check whether or not temp. difference exists before/after check valve. | Replace check valve for outdoor unit. |
| | | Failure or Malfunction of Expansion Valve | Check for clogging. | Remove clogging. |
| | | | Check for connection cord and connector. | Replace connector. |
| | | | Is there operation sound from coil? | Replace coil. |
| | | | Is thermistor on compressor normal? | Replace thermistor. |
| | | | Is thermistor installed correctly on compressor? | Correctly install it. |
| | | Clogged Strainer in Indoor Unit Clogging at Low Pressure Piping | Check temp. difference at inlet and outlet of strainer. | Replace strainer in indoor unit. |
| | | Clogging at Low Pressure Piping | Check temp. difference. | Remove clogging. |
| | | Insufficient Air to Indoor Unit Heat Exchanger | Check for clogged air filter. | Clean air filter. |
| | | | Check for obstacle at inlet or outlet. | Remove obstacles. |
| | | Excessively Low Air Temp. to Indoor Unit Heat Exchanger | Insufficient revolution of indoor unit Fan motor? | Replace fan motor. |
| | | | Short-circuited indoor unit air? | Remove cause of short-circuited air. |

*1): Refer to item 9.3 of TC II.

(1.1.7 Abnormalities of Devices)

| Phenomenon | Cause | | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|---|
| Insufficient Cooling Operation | Excessively High Discharge Pressure | Insufficient Air Flow to Outdoor Unit Heat Exchanger | Clogging of Outdoor Unit Heat Exchanger? | Remove Clogging. |
| | | | Obstacles at Inlet or outlet of outdoor unit heat exchanger. | Remove obstacles. |
| | | | Is service space for outdoor unit sufficient? | Secure service space. |
| | | | Correct Fan Speed? | Replace fan motor. |
| | | Excessively Low Air Temp. to Outdoor Unit Heat Exchanger | Short-Circuited Air to Outdoor Unit? | Remove cause of short-circuited air. |
| | | | Any Other Heat Load near Outdoor Unit? | Remove heat source. |
| | | Excessively Charged Refrigerant | Check each Temp. | Correctly Charge Refrigerant. |
| | | Non-Condensed Gas in Cycle | Check each Temp. and Pressure. | Charge Refrigerant after Vacuum Pumping. |
| | | Clogging of Discharge Piping | Check for Clogging. | Remove Clogging. |
| | | Failure or Malfunction of Expansion Valve | Check for clogging. | Remove clogging. |
| | | | Check for connecting cord and connector. | Replace connector. |
| | | | Is there operation sound from coil? | Replace coil. |
| | | | Is thermistor on compressor normal? | Replace thermistor. |
| | | | Is thermistor installed correctly on compressor? | Correctly install it. |
| | Excessively Low Suction Pressure | Failure of Bypass Solenoid Valve | Check for Leakage of Solenoid Valve. | Replace Solenoid Valve. |
| | | Malfunction or Internal Leakage of 4-Way Valve | Check Temp. Difference between Inlet and Outlet of 4-Way Valve. | Replace 4-Way Valve. |
| | Malfunction or Internal Leakage of 4-Way Valve | | Check temp. difference at inlet and outlet of 4-way valve. | Replace 4-way valve. |
| | Discharge Temp. of Indoor Unit Is Unstable. | | Check for expansion valve of indoor unit in the same system. | Replace failed expansion valve of indoor unit. |

(1.1.7 Abnormalities of Devices)

| Phenomenon | Cause | | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|---|
| Insufficient Heating Operation | Indoor Heat Load is Greater than Heating Capacity | | Calculate Heat Load. | Replace the Unit with a Bigger Unit. |
| | Excessively Low Suction Pressure | Gas Leakage or Insufficient Refrigerant Charge | Measure Superheat. | Correctly Charge Refrigerant after Gas Leakage Check and Repairing. |
| | | Excessively Small Diameter or Long Piping | Measure Field-Supplied Piping. | Use Specified Pipes. |
| | | Failure or Malfunction of Expansion Valve | Check for clogging. | Remove clogging. |
| | | | Check for connecting cord and connector. | Replace connector. |
| | | | Is there operation sound from coil? | Replace coil. |
| | | | Is thermistor on compressor normal? | Replace thermistor. |
| | | | Is thermistor installed correctly on compressor? | Correctly install it. |
| | | Clogging of Indoor Unit/ Outdoor Unit Strainer | Check Temp. Difference between Inlet and Outlet of Strainer. | Replace Strainer for Outdoor Unit or Indoor Unit. |
| | | Clogging of Suction Piping | Check Temp. Difference of each Part. | Remove Clogging. |
| | | Insufficient Air Flow through Outdoor Unit Heat Exchanger | Is Outdoor Unit Heat Exchanger Clogged? | Remove Clogging. |
| | | | Is there any Obstacles at Inlet or Outlet of Outdoor Unit? | Remove Obstacles. |
| | | | Is Service Space for Outdoor Unit Sufficient? | Secure Sufficient Service Space. |
| | | | Check for Speed of Outdoor Unit Fan. | Replace Fan Motor. |
| | | Excessively Low Air Temp. through Outdoor Unit Heat Exchanger | Check for any Short-Circuited Air to Outdoor Unit. | Remove Cause of Short-Circuited Air. |
| | | Defrosting is Insufficiently Completed | Check Thermistor for Defrosting. | Replace Thermistor for Defrosting. |

(1.1.7 Abnormalities of Devices)

| Phenomenon | Cause | | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|---|
| Insufficient Heating Operation | Excessively High Discharge Pressure | Insufficient Air Flow to Outdoor Unit Heat Exchanger | Check for filter clogging. | Remove Clogging. |
| | | | Check for any obstacles at inlet or outlet of indoor unit. | Remove Obstacles. |
| | | | Check indoor fan speed. | Replace fan motor. |
| | | Excessively High Air Temp. to Indoor Unit Heat Exchanger | Check whether or not short-circuited air exists. | Remove cause of short circuited air. |
| | | Excessively Charged Refrigerant | Check refrigerant quantity *1). | Correctly charge refrigerant. |
| | | Non-Condensate Gas in Ref. Cycle | Check refrigerant quantity *1). | Recharge refrigerant after vacuum pumping. |
| | | Clogging of Discharge Piping | Check for clogging. | Remove clogging. |
| | Excessively High Suction Pressure | Failure of Bypass Solenoid Valve | Check for leakage of solenoid valve. | Replace solenoid valve. |
| | | Malfunction or Internal Leakage of 4-Way Valve | Check temp. difference at inlet and outlet of 4-way valve. | Replace 4-way valve. |
| | Malfunction or Internal Leakage of 4-Way Valve | | Check temp. difference at inlet and outlet of 4-way valve. | Replace 4-way valve. |
| | Malfunction of Reverse Protection Valve of Outdoor Unit | | Check temp. difference at inlet and outlet of reverse protection valve | Replace reverse protection valve. |
| | Discharge Temp. of Indoor Unit Is Unstable. | | Check for expansion valve of indoor unit in the same system. | Replace failed expansion valve of indoor unit. |

*1): Refer to item 9.3 of TC II

(1.1.7 Abnormalities of Devices)

| Phenomenon | Cause | | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|---|
| Cooling or Heating Operation with Abnormal Sound | Foreign Particles Inside of Fan Casing | | Check it by viewing. | Remove foreign Particles. |
| | Indoor Unit Fan Runner is Hitting Casing | | Check it by viewing. | Adjust position of fan runner. |
| | Outdoor Unit Propeller Fan is Hitting Shroud | | Check it by viewing | Adjust Position of Propeller Fan |
| | Abnormal Sound from Compressor | Faulty Installation | Check each Part is Tightly Fixed | Tightly Fix it |
| | | Liquid Ref. Compression | Check refrigerant quantity *1). | Correctly Charge Refrigerant |
| | | Wear or Breakage of Internal Comp. Parts | | Replace Compressor |
| | | No Heating by Oil Heater | Check Resistance | Replace Oil Heater or Fuse |
| | Hamming Sound from Magnetic Conductor | | Check Surface of Contacts | Replace Magnetic Switch |
| | Abnormal Vibration by Cabinets | | Check each Fixing Screws | Tightly Fix it |

*1): Refer to item 9.3 of TC II

## 1.2 Troubleshooting Procedure

### 1.2.1 Alarm Code Table

| Code No. | Category | Content of Abnormality | Leading Cause |
|---|---|---|---|
| 01 | Indoor Unit | Tripping of Protection Device | Failure of Fan Motor, Drain Discharge, PCB, Relay. |
| 02 | Outdoor Unit | Tripping of Protection Device | Failure of Compressor, Refrigerant Quantity, Inverse Phase. |
| 03 | Transmission | Abnormality between Indoor and Outdoor (or Indoor) | Incorrect Wiring, Failure of PCB, Tripping of Fuse. |
| 04 | Inverter | Inverter Trip of Outdoor Unit | Failure in Transmission of PCB for Inverter. |
| 05 | Transmission | Abnormality of Power Source Wiring | Reverse Phase Incorrect Wiring. |
| 06 | Voltage Drop | Voltage Drop in Outdoor Unit Excessively Low or High Voltage to Outdoor Unit | Voltage Drop, Incorrect Wiring, Tripping of Fuse. |
| 07 | Cycle | Decrease in Discharge Gas Superheat | Excessive Refrigerant Charge, Failure of Thermistor and Wiring |
| 08 | | Increase in Discharge Gas Temperature | Insufficient Refrigerant , Failure of Thermistor, Wiring. |
| 09 | Outdoor Unit | Tripping of Protection Device | Failure of Fan Motor, Incorrect Wiring. |
| 11 | Sensor on Indoor Unit | Inlet Air Thermistor | Failure of Thermistor, Sensor, Connection. |
| 12 | | Outlet Air Thermistor | |
| 13 | | Freeze Protection Thermistor | |
| 14 | | Gas Piping Thermistor | |
| 19 | | Tripping of Protection Device | Failure of Fan Motor, Incorrect Wiring. |
| 21 | Sensor on Outdoor Unit | High Pressure Sensor | Failure of Thermistor, Sensor, Connection. |
| 22 | | Outdoor Air Thermistor | |
| 23 | | Discharge Gas Thermistor | |
| 24 | | Evaporating Thermistor | |
| 29 | | Low Pressure Sensor | |
| 30 (FX2 Only) | System | Incorrect Setting of Outdoor and Indoor Unit Dip Switches | Incorrect Setting of Capacity Code. |
| 31 | | Incorrect Setting of Outdoor and Indoor Unit | Incorrect Setting of Capacity Code. |
| 32 | | Abnormal Transmission of Other Indoor Unit | Failure of Power Supply, PCB in Other Indoor Unit. |
| 35 | | Incorrect Setting in Indoor Unit No. | Existence of the same Indoor Unit No. |
| 38 | | Abnormality of Protective Circuit in Outdoor Unit | Incorrect Connection to PCB in Outdoor Unit. |
| 39 | | Abnormality of Running Current at Constant Compressor | Incorrect Wiring. |
| 43 | Pressure | Pressure Ratio Decrease Protection Activating | Failure of Compressor, Inverter, Power Supply. |
| 44 | | Low Pressure Increase Protection Activating | Overload to Indoor in Cooling, High Temperature or Outdoor Air. |
| 45 | | High Pressure Increase Protection Activating | Overload Operation, Excessive Refrigerant. |
| 46 | | High Pressure Decrease Protection Activating | Insufficient Refrigerant. |
| 47 | | Low Pressure Decrease Protection Activating | Vacuum Condition in Cycle Failure of Expansion Valve. |
| 51 | Inverter | Abnormality of Current Sensor for Inverter Thermal Protection Activating | Failure of Compressor, PCB, IPM. |
| 52 | | Overcurrent Protection Activating | Overload, Overcurrent, Locking of Compressor. |
| 53 | | IPM Protection Activating | Overheating of Inverter, PCB. |
| EE | | Compressor Protection | Failure of Compressor. |

### 1.2.2 Troubleshooting by Alarm Code

| Alarm Code 01 | Activation of Safety Device in Indoor Unit |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★ This alarm code is indicated when AC220V or 240V is not given between PCN1(FS) on PCB and L2(N) phase during the cooling, fan or heating operation.

Is 220VAC or 240VAC applied between PCN1(FS) on PCB(P) and L2 phase ? — No → Activation of Safety Device (such as Float Switch FS and Internal Thermostat ITI) or Incorrect Wiring

↓ Yes

Check for looseness between CN16 of PCB(M) and CN20 of PCB(P). — No → Check connections of connector.

↓ Yes

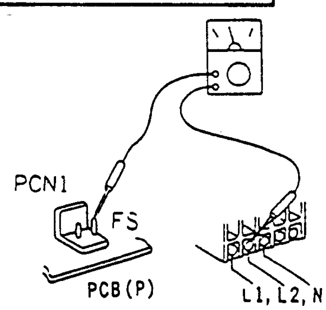
Check for wire continuity between CN16 of PCB(M) and CN20 of PCB(P). — No → Disconnected Cable

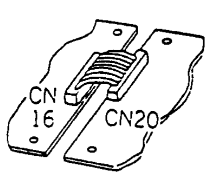
↓ Yes

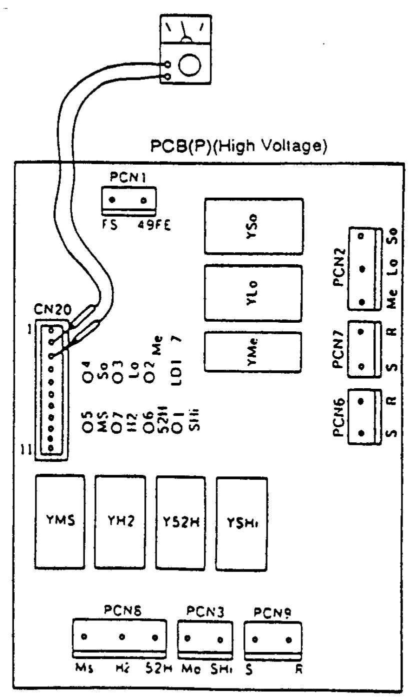
Is the voltage between #2 and #3 of CN20 on PCB(P) 0V? — No → Faulty PCB(P)

↓ Yes

Faulty PCB(M)




NOTE:
Above figure shows a separated-board type PCB, some unit has a one-board type PCB which has the same functions and devices with the separated-board type.



HITACHI

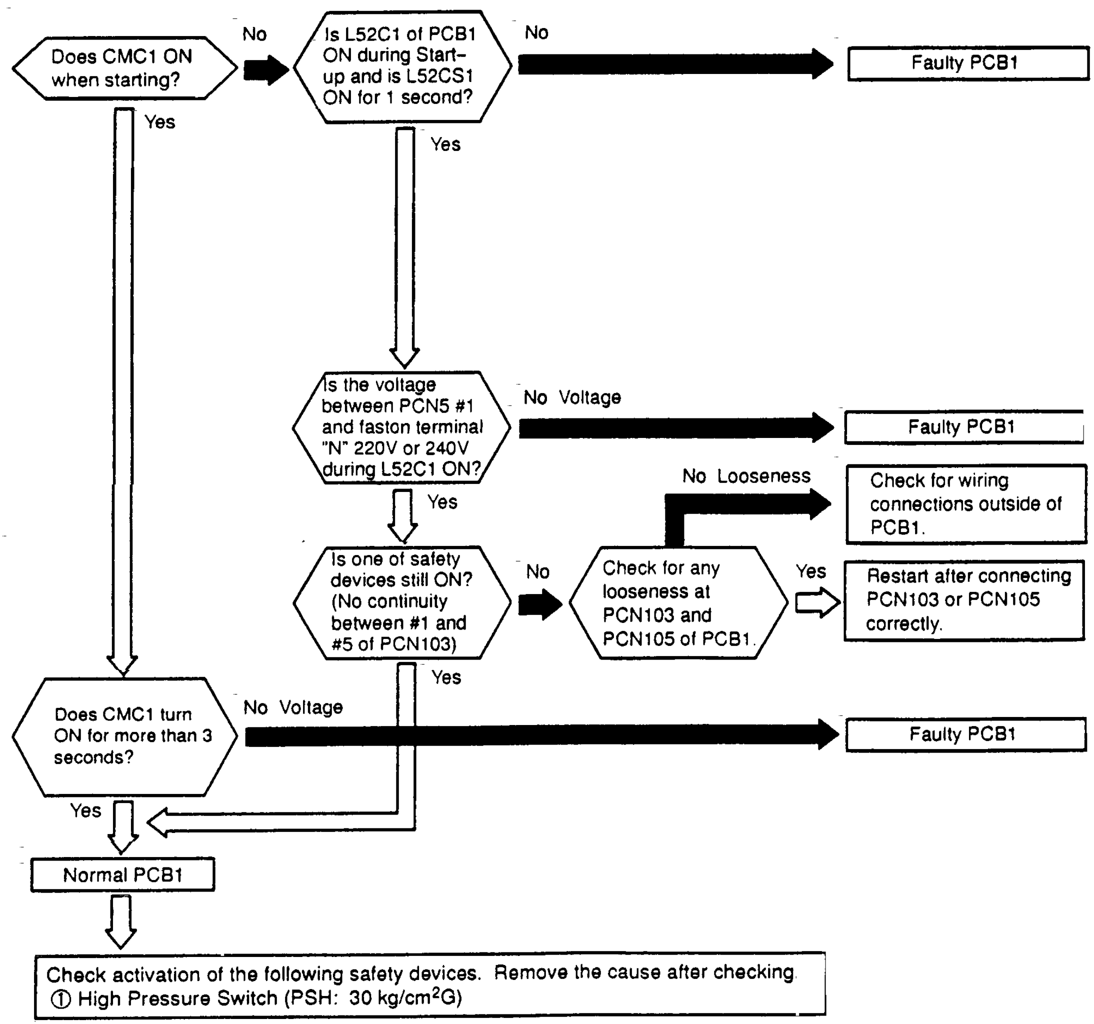
| Phenomenon | Cause | | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|---|
| Tripping of Internal Thermostat for Indoor Unit Fan Motor | Faulty Indoor Unit Fan Motor | | Measure coil resistance and insulation resistance. | Replace indoor unit fan motor if faulty. |
| | Faulty Internal Thermostat | Fault | Check continuity after indoor unit fan motor decreases to normal temp. | Indoor Unit Fan Motor if Faulty |
| | | Faulty Contacting | Measure resistance by tester. | Repair looseness and replace connector. |
| | | Faulty Connection | Check connections. | Repair connection. |
| Activation of Float Switch | High Drain Level | Clogging of Drainage | Check drain pan. | Remove clogged foreign particles. |
| | Faulty Float Switch | Fault | Check continuity when drain level is low. | Replace float switch if faulty. |
| | | Faulty Contacting | Measure resistance by tester. | Repair looseness and replace connector. |
| | | Faulty Connection | Check connections. | Repair connection. |
| | Opening of Indoor Unit Ex. Valve is Locked Open | | Check whether or not cool air discharged by setting at "FAN" when other indoor units are operated under cooling operation. | If cool air discharged, ① Repair wiring of ex. valve, connector or connector. ② Replace ex. valve, if faulty. |
| Faulty Indoor Unit PCB | | | Check PCB by self-checking *1). | Replace it if faulty. |

*1): Refer to Item 1.3.1 and 1.3.3

| Alarm Code 02 | Activation of Safety Device for Outdoor Unit |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

This alarm is indicated when one of safety devices is activated during compressor running (CMC1: ON)



| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Activation of High Pressure Switch due to Excessively High Discharge Pressure | Insufficient Air Flow to Heat Exchanger (Outdoor Heat Exchanger during Cooling Operation or Indoor Heat Exchanger during Heating Operation) | Check for dust on heat exchanger. | Remove dust. |
| | | Check for dust on air filter | Remove dust. |
| | | Check for any obstacles at inlet or outlet of heat exchanger. | Remove obstacles. |
| | | Check the service space. | Secure service space. |
| | | Check for speed. (Outdoor Fan: Cooling; Indoor Fan: Heating) | Replace fan motor if faulty. |
| | Malfunction of Electronic Ex. Valve | Check for looseness of connectors. | Repair looseness. |
| | | Check opening of ex. Valve. *1) | Replace Ex. Valve. |

*1): Refer to item 3.3

## TROUBLESHOOTING

| Phenomenon | Cause | | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|---|
| Activation of High Pressure Switch due to Excessively High Discharge Pressure | Excessively High Temp. Air to Indoor Unit Heat Exchanger | | Calculate heat load. | Reduce heat load or use a bigger unit. |
| | | | Check for hot air near the ceiling (Heating). | Make good circulation. |
| | | | Check for short circuited air (Heating). | Remove short-circuited air. |
| | | | Check for other heat source. | Remove heat source. |
| | Faulty High Pressure Switch | Faulty Pressure Switch | Measure discharge pressure. Check continuity after decreasing of pressure. | Replace it if faulty. |
| | | Insufficient Contacting | Measure resistance by tester. | Repair looseness. Replace connector. |
| | | Incorrect Connection | Measure connections. | Repair connections. |
| | Faulty Bypass Solenoid Valve | | Check for clogging | Replace gas bypass solenoid valve. |
| | Overcharged Refrigerant | | Check refrigerant quantity *1). | Charge refrigerant correctly. |
| | Mixture of Non-Condensable Gas in Refrigeration Cycle | | Check ambient temp. and pressure | Recharge refrigerant after vacuum pumping. |
| | Clogging of Discharge Piping | | Check for clogging. | Remove clogging. |
| | Clogging of Check Valve | | Check for clogging. | Replace check valve |
| | [FX2 System] Faulty CH Unit | Faulty Solenoid Valve | Check Activation of SVD1. | Replace solenoid coil or CH unit. |
| | | Insufficient Contacting | Check for connection at PCN8 of PCB1 and PCN8 of PCB3. Check for connection at connector 20D1. | Repair looseness. |
| | | Incorrect Connection | | Replace connector. |
| | | | | Repair connections. |

*1): Refer to Item 9.3 of TC II

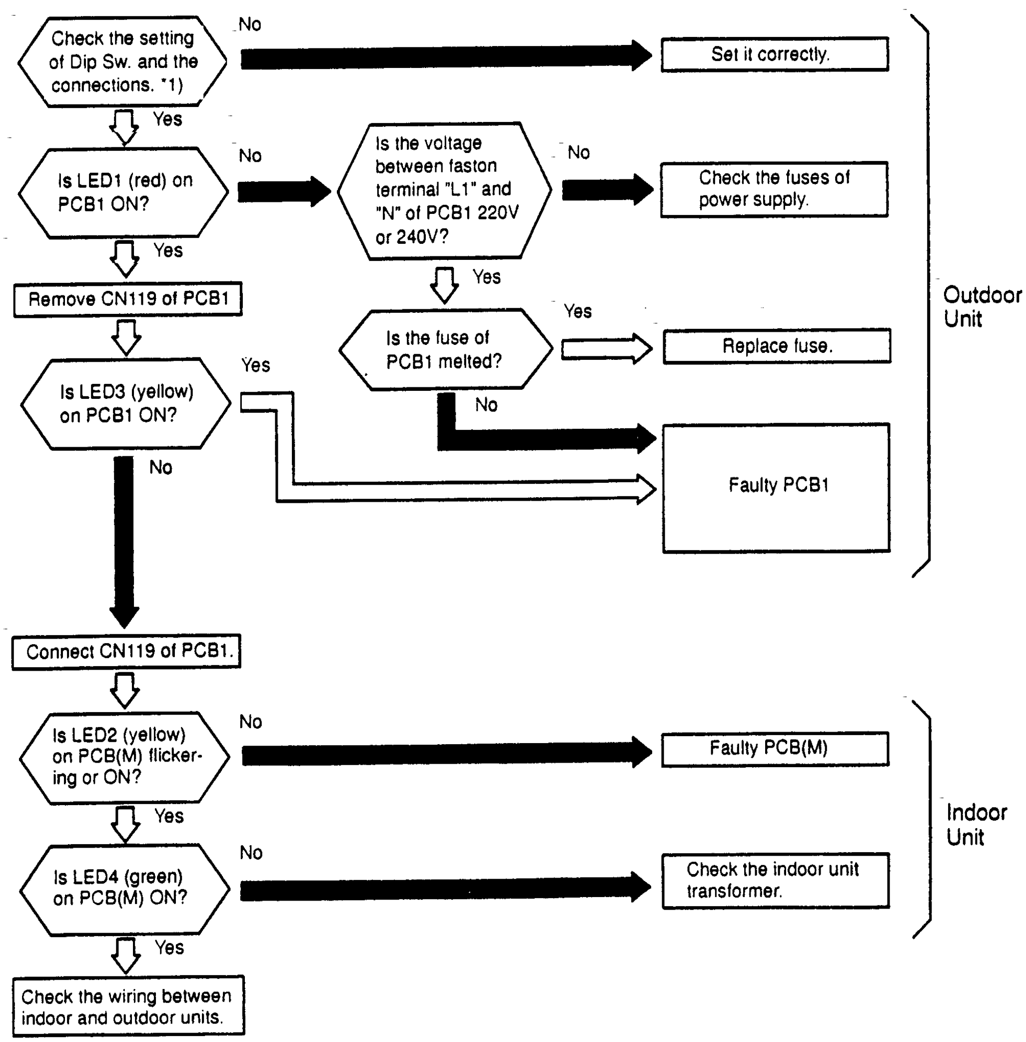
## FS2 SYSTEM

| Alarm Code 03 | Abnormal Transmitting between Indoor Units and Outdoor Unit |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★ This alarm is indicated when abnormality is maintained for 3 minutes after normal transmitting between indoor units and outdoor unit, and also abnormality is maintained for 30 seconds after the micro-computer is automatically reset, or the alarm is indicated when the abnormal transmitting is maintained for 30 seconds from starting of the outdoor unit.

★ Investigate the cause of overcurrent and take necessary action when fuses are melted or the breaker for the outdoor unit are activated.




PCB1: Outdoor Unit PCB (for Control)

PCB(M): Indoor Unit PCB (Low Voltage)

*1): Refer to item 1.1.1.

## TROUBLESHOOTING

### FS2 SYSTEM

| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Power Failure or No Power Supply | | Measure voltage by tester. | Supply power. |
| Melted Fuse for Power Source or Activation of Outdoor Unit Breaker | Short Circuit between Wires | Check for breakage of insulation. | Remove short circuit and replace fuse. |
| | Short Circuited Wire to Ground | Measure insulation resistance. | Remove short circuit to ground and replace fuse. |
| | Faulty Comp. Motor | Measure resistance between wires and insulation resistance | Replace comp. and fuse. |
| | Faulty Outdoor Unit Fan Motor | Measure resistance between wires and insulation resistance. | Replace outdoor unit fan motor and fuse. |
| Melted Fuse for Control Circuit or Activation of Outdoor Unit Breaker | Short Circuit between Wires | Check for breakage of insulation. | Remove short circuit and replace fuse. |
| | Short Circuit of Control Circuit | Measure insulation resistance. | Remove short circuit and replace fuse. |
| | Faulty Solenoid Coil for Magnetic Switch for Comp. Motor | Measure resistance of coil. | Replace magnetic switch and fuse. |
| Faulty PCB4 (Outdoor Unit) | | Measure output voltage of CN401 on PCB4 *2). | Replace PCB. |
| Disconnected Wires Insufficient Contacting or Incorrect Connection | Between Outdoor Unit and Indoor Unit | Check continuity of wires. Check for looseness of Connection screws. Check terminal Nos. | Replacing wires, repairing tightening screws, correct wiring. |
| | Power Source Wiring for Outdoor Unit | | |
| Faulty PCB (Outdoor Unit, Indoor Unit) | Disconnected Wires to PCB | Check connections. | Correctly connect wires. |
| | Faulty PCB | Check it by self-checking *1) | Replace it if faulty. |
| Incorrect Wiring *1) | Disconnected Wire Insufficient Contacting | Check continuity and looseness of connection screws | Replacing Wires, Repairing Tightening Screws |
| | Incorrect Wiring | Check terminal Nos. | Correctly Connect Wires |
| Incorrect Indoor Unit Address Setting | | Check dip switch, DSW1 on indoor unit PCB. | Correctly set unit No. on indoor unit PCB(M). Correctly set dip switch, DSW1. |

*1): Refer to Item 1.3.1 ~ 1.3.4.

*2): Output Voltage at CN401 on PCB4.
(5VDC between #1 and #4, 15VDC between #1 and #6, −15VDC between #1 and #5)

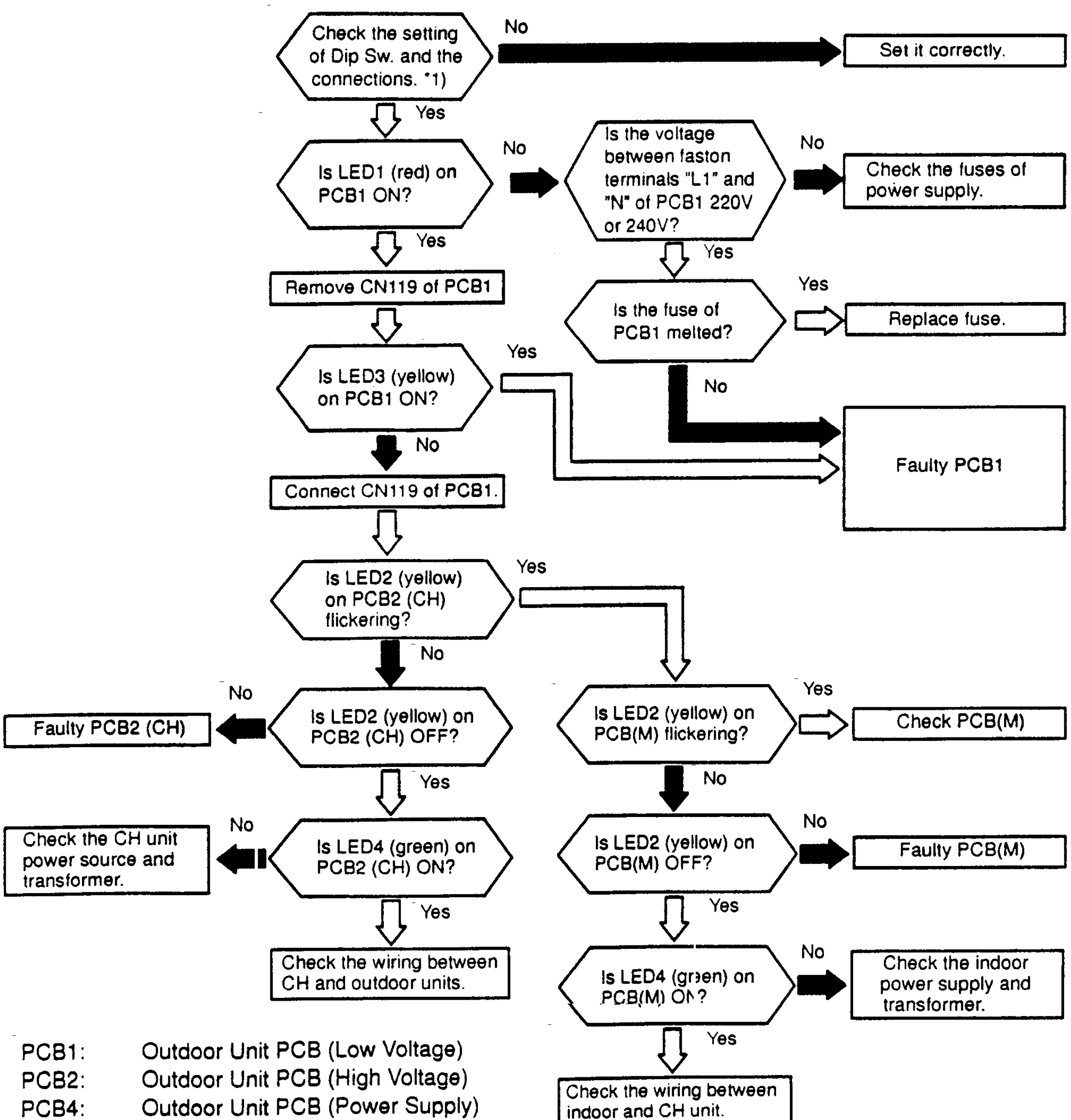
FX2 SYSTEM

| Alarm Code 03 | Abnormal Transmitting between Indoor Units, CH Units and Outdoor Unit |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★ This alarm is indicated when abnormality is maintained for 3 minutes after normal transmitting between indoor units, CH units and outdoor unit, and also abnormality is maintained for 30 seconds after the micro-computer is automatically reset, or the alarm is indicated when the abnormal transmitting is maintained for 30 seconds from starting of the outdoor unit.

★ Investigate the cause of overcurrent and take necessary action when fuses are melted or the breaker for the outdoor unit are activated.




PCB1: Outdoor Unit PCB (Low Voltage)
PCB2: Outdoor Unit PCB (High Voltage)
PCB4: Outdoor Unit PCB (Power Supply)
PCB2(CH): CH Unit PCB (Low Voltage)

## TROUBLESHOOTING

### FX2 SYSTEM

| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Power Failure or No Power Supply | | Measure voltage by tester. | Supply power. |
| Melted Fuse for Power Source or Activation of Outdoor Unit Breaker | Short Circuit between Wires | Check for breakage of insulation. | Remove short circuit and replace fuse. |
| | Short Circuited Wire to Ground | Measure insulation resistance. | Remove short circuit to ground and replace fuse. |
| | Faulty Comp. Motor | Measure resistance between wires and insulation resistance | Replace comp. and fuse. |
| | Faulty Outdoor Unit Fan Motor | Measure resistance between wires and insulation resistance. | Replace outdoor unit fan motor and fuse. |
| Melted Fuse for Control Circuit or Activation of Outdoor Unit Breaker | Short Circuit between Wires | Check for breakage of insulation. | Remove short circuit and replace fuse. |
| | Short Circuit of Control Circuit | Measure insulation resistance. | Remove short circuit and replace fuse. |
| | Faulty Solenoid Coil for Magnetic Switch for Comp. Motor | Measure resistance of coil. | Replace magnetic switch and fuse. |
| Faulty PCB4 (Outdoor Unit) | | Measure output voltage of CN401 on PCB4 *2). | Replace PCB. |
| Faulty Transformer of CH Unit | | Measure input voltage of CN1 on PCB2 (CH) *3). | Replace transformer. |
| Disconnected Wires Insufficient Contacting or Incorrect Connection | Between Outdoor Unit, CH Unit and Indoor Unit | Check continuity of wires. Check for looseness of Connection screws. Check terminal Nos. | Replacing wires, repairing tightening screws, correct wiring. |
| | Power Source Wiring for Outdoor Unit | | |
| Faulty PCB (Outdoor Unit, Indoor Unit and CH Unit) | Disconnected Wires to PCB | Check connections. | Correctly connect wires. |
| | Faulty PCB | Check it by self-checking *1) | Replace it if faulty. |
| Incorrect Wiring *1) | Disconnected Wire Insufficient Contacting | Check continuity and looseness of connection screws | Replacing Wires, Repairing Tightening Screws |
| | Incorrect Wiring | Check terminal Nos. | Correctly Connect Wires |
| Incorrect Indoor Unit Address Setting | | Check dip switch, DSW1 on indoor unit PCB. | Correctly set unit No. on indoor unit PCB(M). Correctly set dip switch, DSW1. |

*1): Refer to Item 1.3.1 ~ 1.3.4.

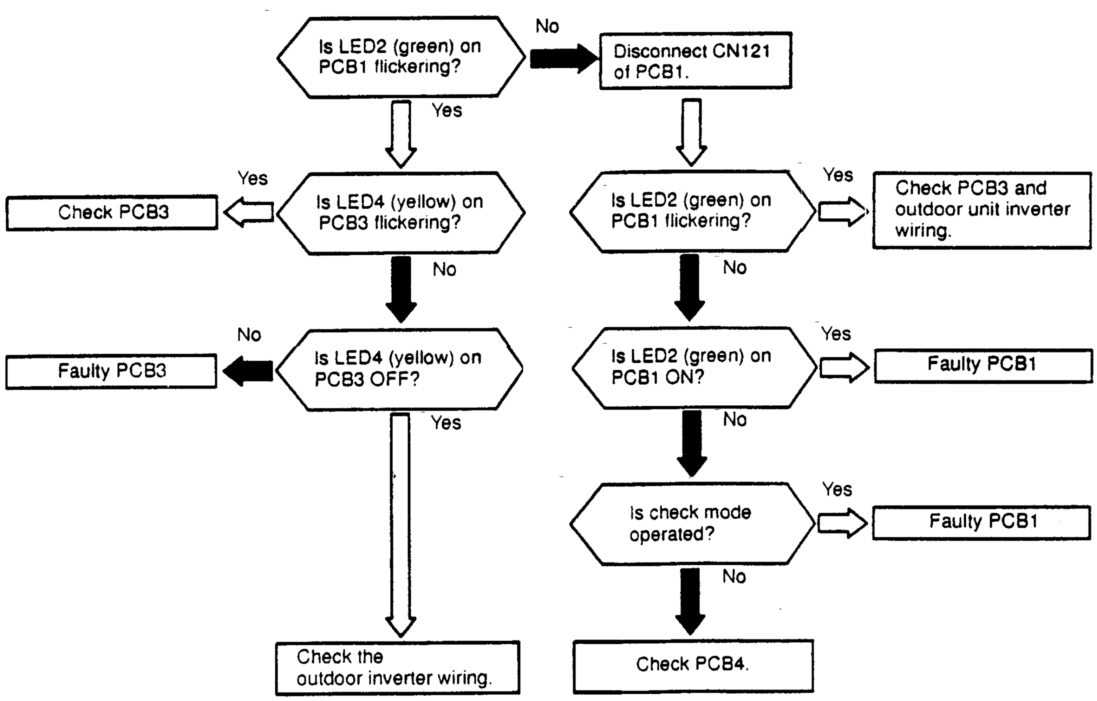
*2): Output Voltage at CN401 on PCB4.
(5VDC between #1 and #4, 15VDC between #1 and #6, −15VDC between #1 and #5)

*3): Input Voltage at CN1 on PCB2 (CH).
(22.2VAC between #1 and #2, 14.5VAC between #3 and #4)

| Alarm Code 04 | Abnormal Transmitting between Inverter and Outdoor PCB1, 3, 4 |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB1.

★ This alarm is indicated when abnormality is maintained for 30 seconds after normal transmitting between the outdoor unit PCB1 and inverter PCB3, and also abnormality is maintained for 30 seconds after the micro-computer is automatically reset, or the alarm is indicated when the abnormal transmitting is maintained for 30 seconds from starting of the outdoor unit.




PCB1: Control PCB
PCB3, 4 Inverter PCB (High Voltage)

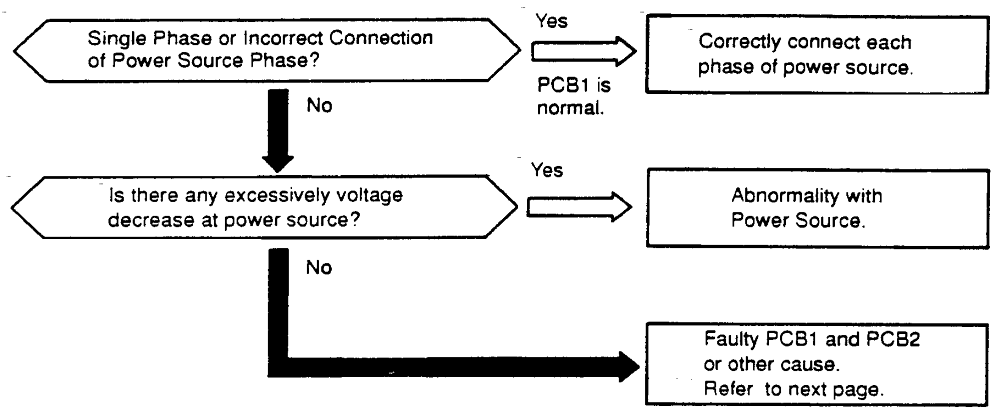
## TROUBLESHOOTING

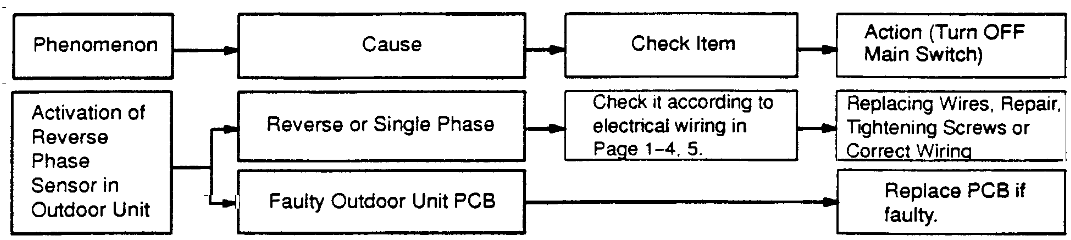
| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Disconnected Wires, Insufficient Contacting Incorrect Connection | Between Control PCB and Inverter PCB | Check continuity of wires. Check for looseness of connection screws. Check connection No. | Replacing Wires, Repairing, Tightening Screws, Incorrect Wiring. |
| | Power Source Wiring for Outdoor Unit | | |
| Faulty PCB (PCB1 for Control, PCB3 For Inverter) | Disconnected Wires to PCB | Check connections. | · Repair wiring connections. |
| | Faulty PCB | | Replace PCB if faulty. |
| Incorrect Wiring | Disconnected Wires Insufficient Contacting | Check continuity. Check for looseness of connection screws. | Replacing Wires, Repairing, Tightening Screws |
| | Incorrect Wiring | Check connection Nos. | Correct connect Wires. |

| Alarm Code 05 | Abnormality of Picking up Phase Signal |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★ This alarm is indicated when the main power source phase is reversely connected or one phase is not connected during running of the constant speed compressor.

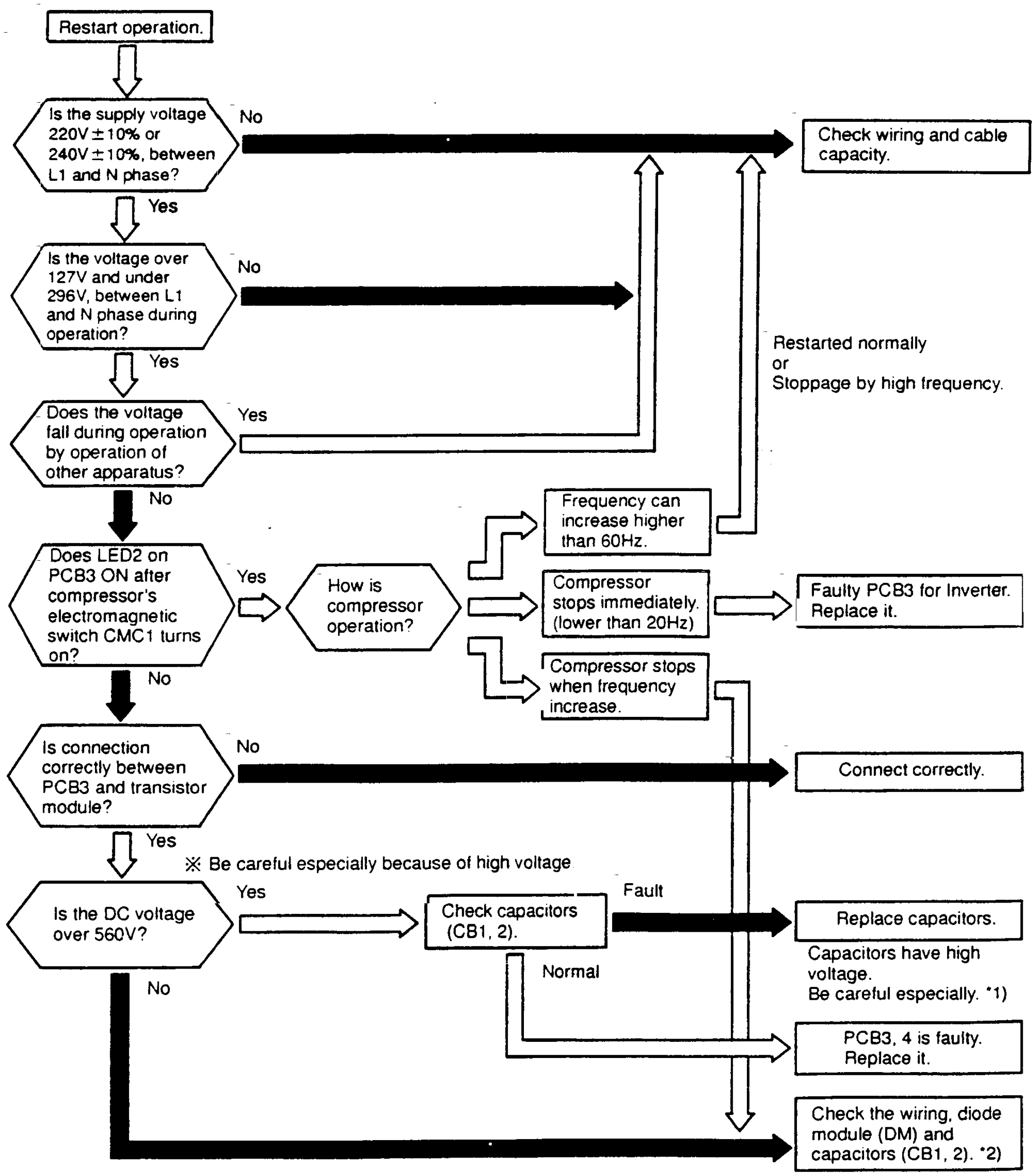


| Alarm Code 06 | Excessively Low or High Voltage for Inverter |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

This alarm code is indicated when voltage between terminal "P" and "N" of transistor module (IPM) is insufficient and its occurrence is three times in 30 minutes. In the case that the occurrence is smaller than 2 times, retry is performed.



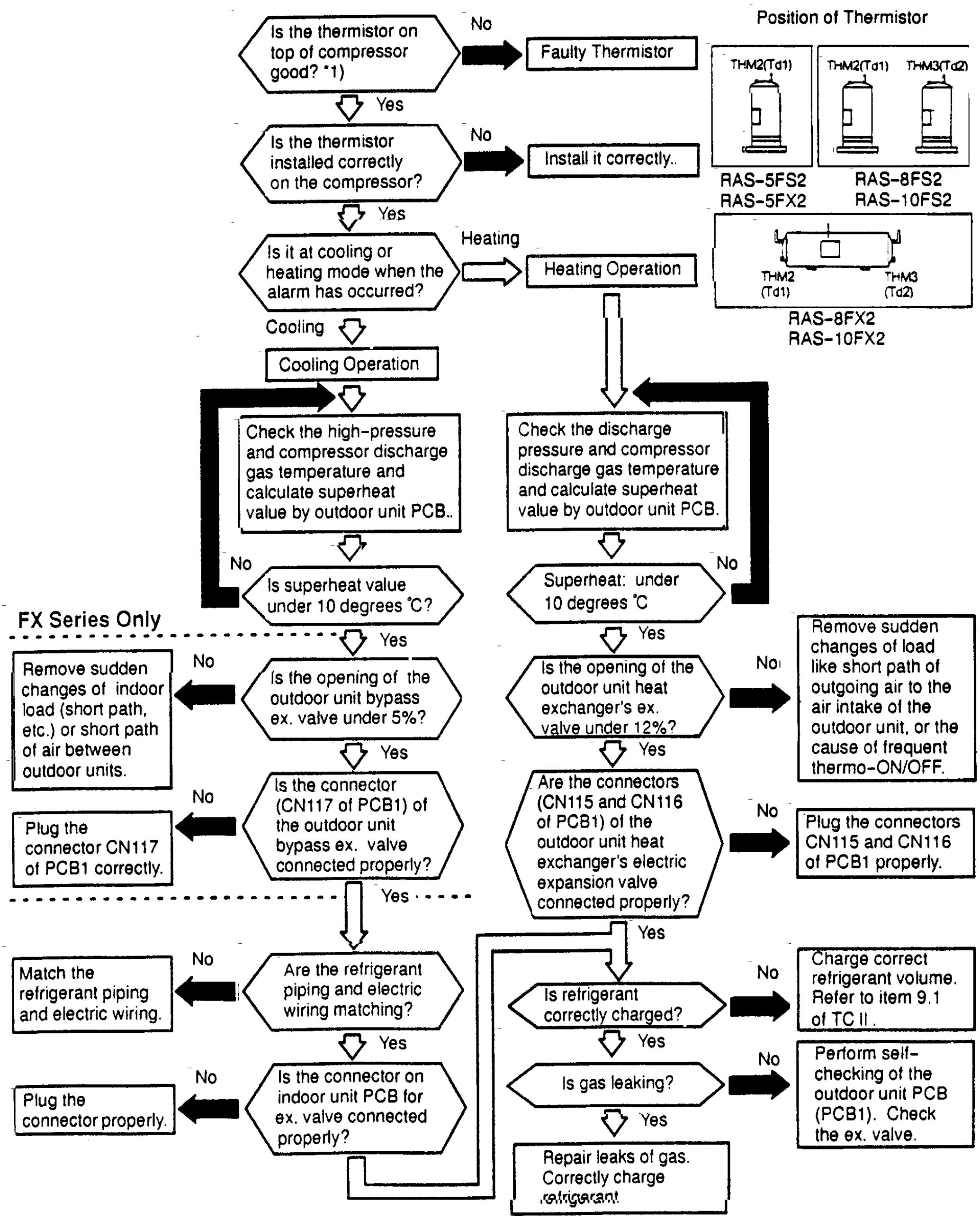

*1): If capacitor has high voltage, perform the high voltage discharge work refer to item 1.3.7.

*2): Checking procedures of diode module is indicated in item 1.3.7.

| Alarm Code 07 | Decrease of Discharge Gas Superheat |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★In the case that the discharge gas superheat less than 10 deg. at the top of the compressor is maintained for 1 hours, retry operation is performed. However, the alarm occurs again within two hours, this alarm code is indicated.




*1): Refer to "Characteristics of Thermistor" of page 1-42

HITACHI

| Phenomenon | Cause | | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|---|
| Decrease of Discharge Gas Superheat | Ref. Cycle is Different from the Electrical System | | Check ref. cycle and electrical system. | Repair wiring. |
| | Overcharged Refrigerant | | Measure pressure. | Correctly charge refrigerant. |
| | Faulty Ex. Valve | | Refer to 1.3.7. | Replace ex. valve if faulty. |
| | Faulty PCB | Fault | Replace PCB and check operation. | Replace PCB if faulty. |
| | | Disconnected Wires for Ex. Valve Control | Check connections. | Repair wiring connection. |
| | Faulty Discharge Gas Thermistor | Fault | Measure resistance. | Replace thermistor if faulty. |
| | | Incorrect Mounting | Check mounting state. | Correctly mount thermistor. |
| | | Incorrect Connection | Check connections. | Remove looseness, replace connector or repair connections. |

| Alarm Code 08 | Excessively High Discharge Gas Temperature at Top of Compressor Chamber |
|---|---|

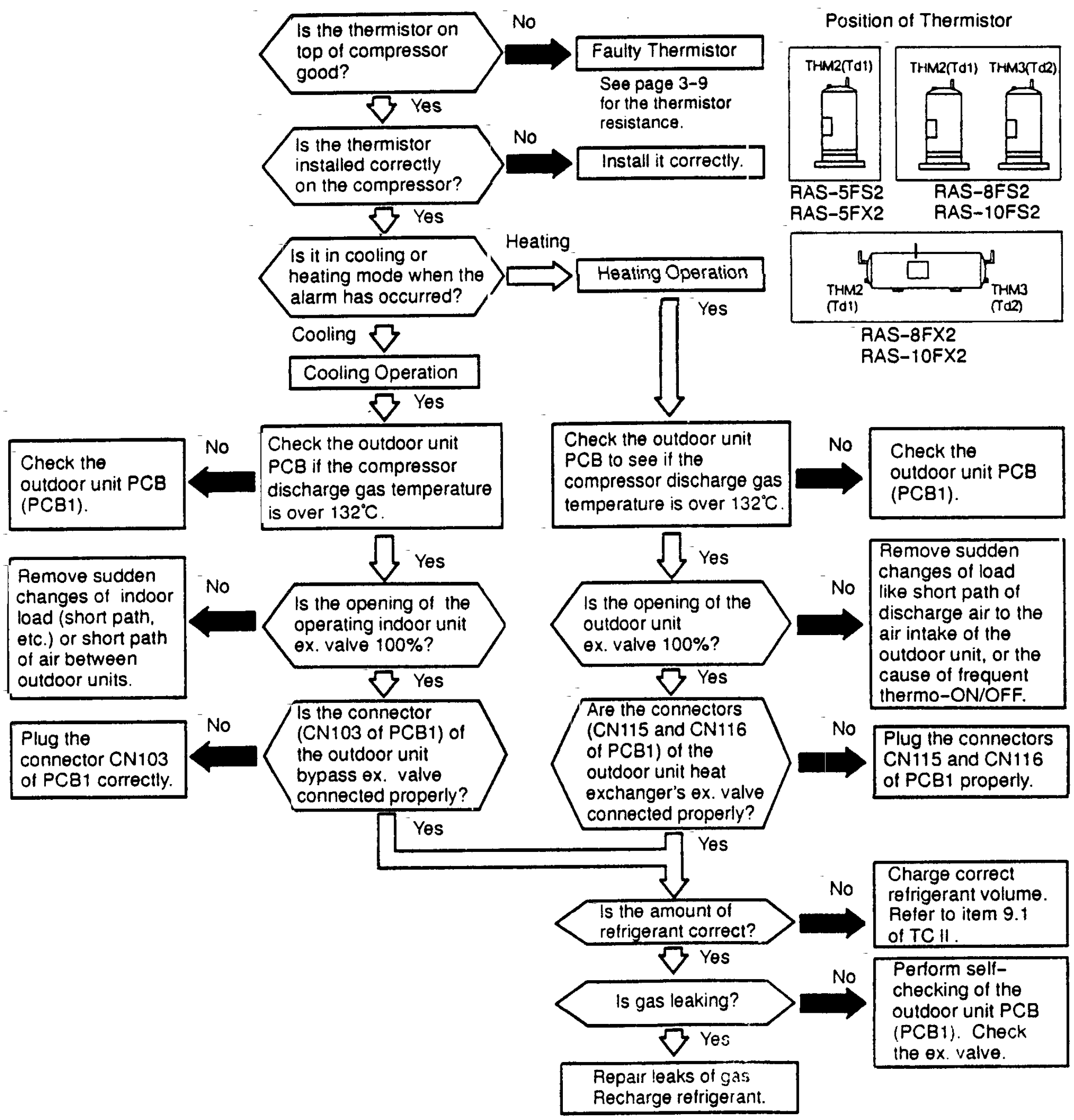
- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm is indicated when the following conditions occurs three times within one hour;

□The temperature of the thermistor on the top of the compressor is maintained higher than 132℃ for 10 minutes.

or

□The temperature of the thermistor on the top of the compressor is maintained higher than 140℃ for 5 seconds.

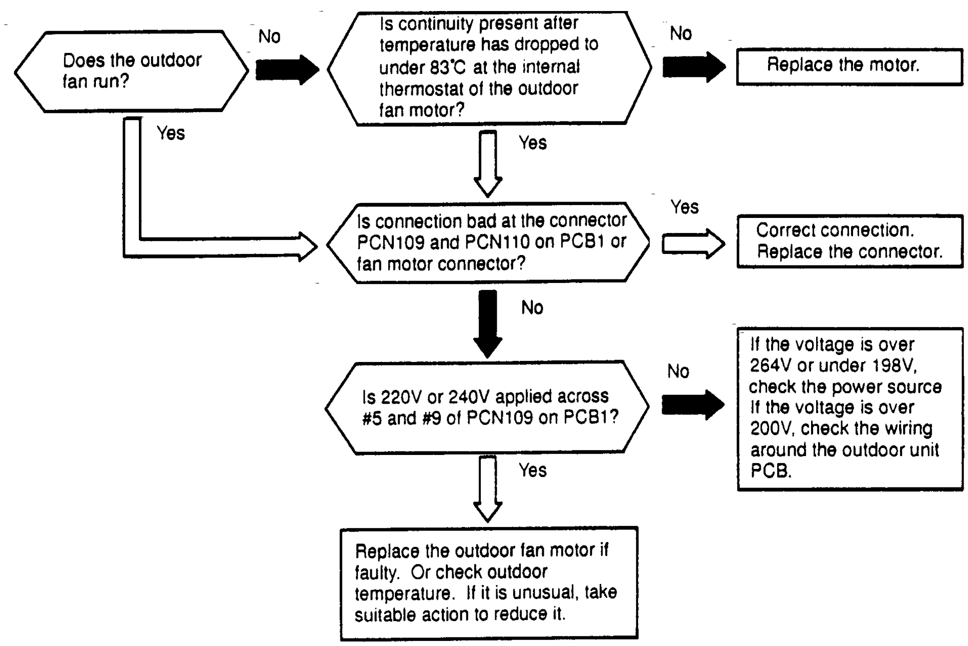


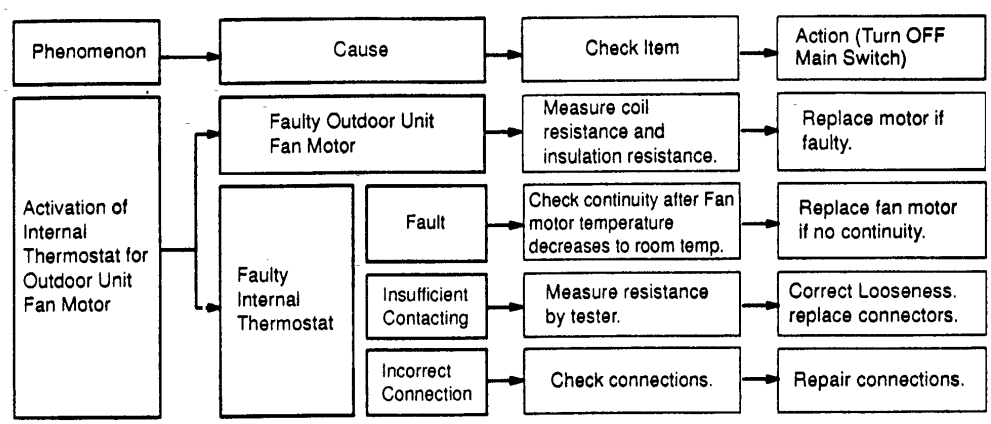
| Phenomenon | Cause | | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|---|
| Excessively High Discharge Gas Temperature (at Top of Compressor) | Ref. Cycle is Different from the Electrical System | | Check ref. cycle and the electrical system. | Repair wiring. |
| | Shortage of Refrigerant | | Check pressures. | Correctly charge refrigerant. |
| | Faulty Ex. Valve | | Refer to 1.3.7. | Replace ex. valve if faulty. |
| | Faulty PCB | Fault | Replace PCB and check operation. | Replace PCB if faulty. |
| | | Disconnected Wires for Ex. Valve | Check connections. | Repair wiring connections. |
| | Faulty Gas bypass Solenoid Valve | | Ref. Leakage | Replace solenoid valve if faulty. |
| | Faulty Discharge Gas Thermistor | Fault | Check resistance of thermistor. | Replace thermistor if faulty. |
| | | Incorrect Mounting of Thermistor | Check mounting state. | Correctly mount thermistor. |
| | | Incorrect Wiring | Check connections. | Remove looseness, replace connectors or repair connections. |

| Alarm Code 07 | Activation of Protection Device for Outdoor Fan Motor |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when the temperature of the internal thermostat for the outdoor fan motor is higher than 130℃



| Alarm Code 11 | Abnormality of Thermistor for Indoor Unit Inlet Air Temperature (Air Inlet Thermistor) |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when the thermistor is short-circuited (less than 0.24 kΩ) or cut (greater than 840 kΩ) during the cooling or heating operation.

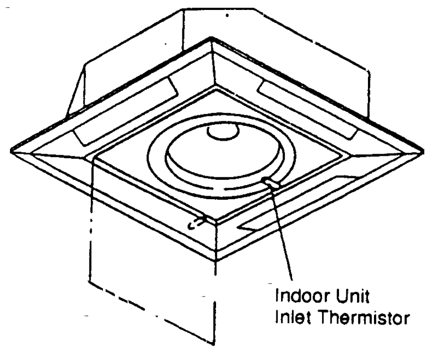
Is it connected to THM3 (intake) of PCB(M). — No → Connect it.

Yes ↓

Is resistance of THM3 between 0.24kΩ and 840kΩ? — No → Faulty Thermistor

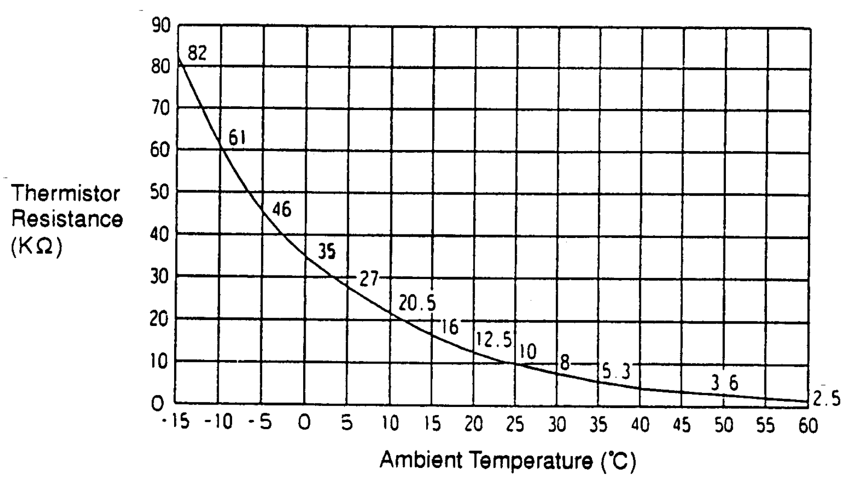
Yes ↓

Faulty PCB(M)




In Case of 4-Way Casette

| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Faulty Intake Air Thermistor | Fault | Check resistance. | Replace thermistor if faulty. |
| | Incorrect Connection | Check connection.. | Repair wiring and connections. |
| Faulty PCB | | | Replace PCB. |



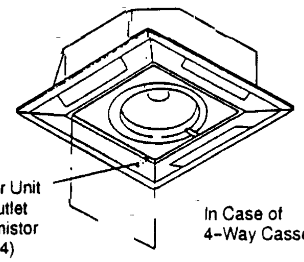

NOTE: This data is applicable to the following thermistors;
1. Indoor Unit Discharge Air Temperature, 2. Indoor Unit Liquid Refrigerant Temperature, 3. Indoor Unit Intake Air Temperature 4. Outdoor Air Temperature 5. Outdoor Unit Evaporating Temperature, 6. Indoor Unit Gas Piping

| Alarm Code 12 | Abnormality of Thermistor for Indoor Unit Discharge Air Temperature (Air Outlet Thermistor) |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when the thermistor is short-circuited (less than 0.24 kΩ) or cut (greater than 840 kΩ) during the heating operation.

Is the thermistor connected to THM4 of PCB(M)? — No → Connect it.
↓ Yes
Is resistance between 0.24 and 840KΩ? — No → Faulty Thermistor
Refer to page 1-36.
↓ Yes
Faulty PCB(M)


Indoor Unit Air Outlet Thermistor (THM4)
In Case of 4-Way Cassette

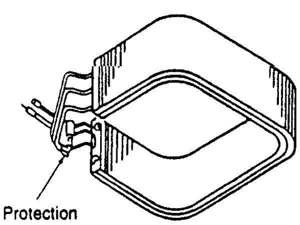
| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Faulty Air Outlet Thermistor | Fault | Check resistance. | Replace thermistor if faulty. |
| | Incorrect Connection | Check wiring to PCB. | Repair wiring and connections. |
| Faulty PCB | | | Replace PCB. |

| Alarm Code 13 | Abnormality of Thermistor for Indoor Unit Heat Exchanger Liquid Refrigerant Pipe Temperature (Freeze Protection Thermistor) |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when the thermistor is short-circuited (less than 0.24 kΩ) or cut (greater than 840 kΩ) during the cooling operation(or heating operation at start-up only) and this state is maintained for 10 minutes.

Is the thermistor connected to THM5 of PCB(M)? — No → Connect it.
↓ Yes
Is resistance between 0.24 and 840KΩ? — No → Faulty Thermistor
Refer to page 1-36.
↓ Yes
Faulty PCB(M)

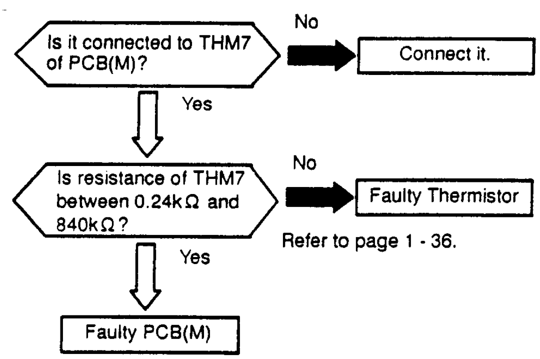

Freeze Protection Thermistor (THM5)
In Case of 4-Way Cassette

| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Faulty Freeze Protection Thermistor | Fault | Check resistance. | Replace thermistor if faulty. |
| | Incorrect Connection | Check wiring to PCB. | Repair wiring and connections. |
| Faulty PCB | | | Replace PCB. |

| Alarm Code 14 | Abnormality of Thermistor for Indoor Unit Heat Exchanger Gas Refrigerant Pipe Temperature (Gas Piping Thermistor) |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when the thermistor is short-circuited (less than 0.24 kΩ) or cut (greater than 840 kΩ) during the cooling or heating operation. The system is automatically restarted when the fault is removed.

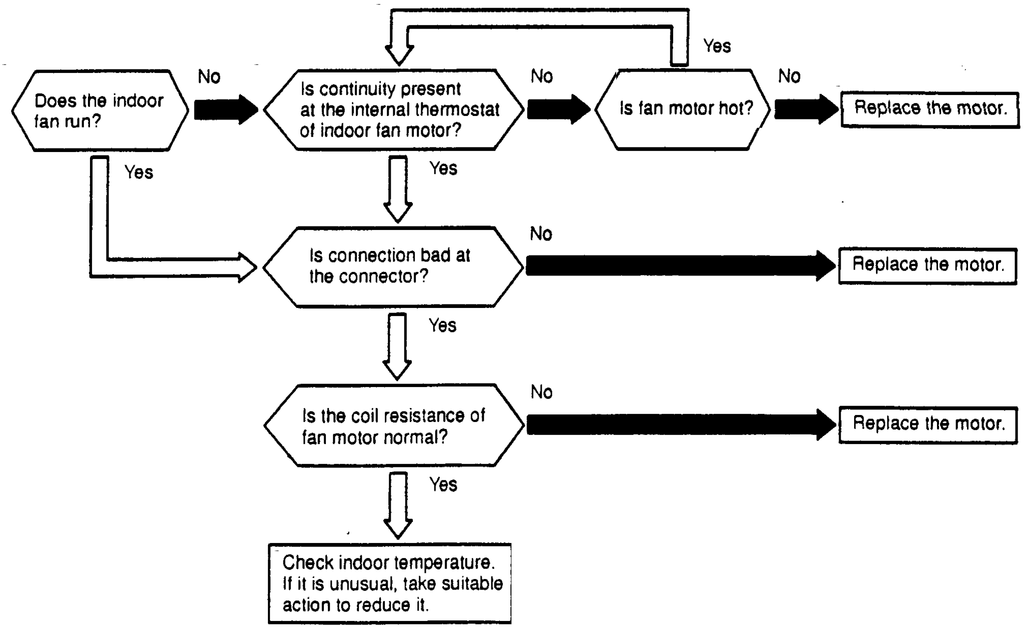



| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Faulty Thermistor for Indoor Unit Heat Exchanger Gas Pipe Temp. | Fault | Check resistance. | Replace thermistor if faulty. |
| | Incorrect Connection | Check wiring to PCB. | Repair wiring and connections. |
| Faulty PCB | | | Replace PCB. |

| Alarm Code 19 | Activation of Protection Device for Indoor Fan Motor |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No and alarm code are indicated on the display of the outdoor unit PCB.

★ This alarm code is indicated when the temperature of the internal thermostat for the indoor fan motor is higher than 130°C

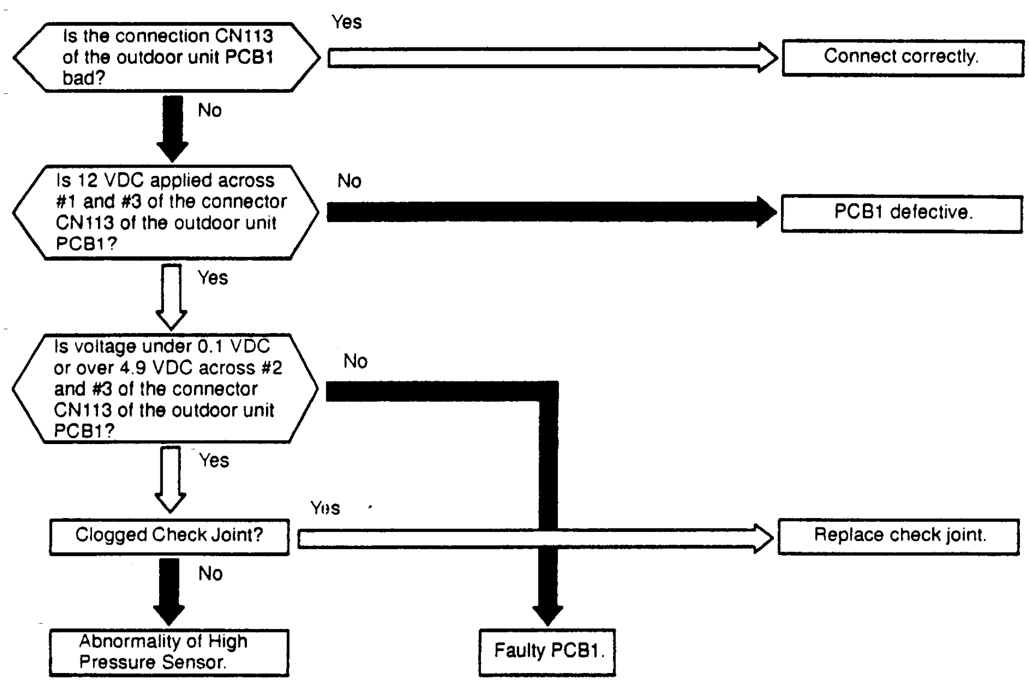



| Phenomenon | Cause | | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|---|
| Activation of Internal Thermostat for Indoor Unit Fan Motor | Faulty Indoor Unit Fan Motor | | Measure coil resistance and insulation resistance. | Replace motor if faulty. |
| | Faulty Internal Thermostat | Fault | Check continuity after fan motor temperature decreases to room temp. | Replace fan motor if no continuity. |
| | | Insufficient Contacting | Measure resistance by tester. | Correct looseness. Replace connectors. |
| | | Incorrect Connection | Check connections. | Repair connections. |

| Alarm Code 21 | Abnormality of High Pressure Sensor for Outdoor Unit |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★ This alarm code is indicated when the pressure sensor voltage decreases lower than 0. V or ncreases higher than 4.9V during running.




| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Faulty High Pressure Sensor | Fault | Check output voltage is correct | Replace pressure sensor if faulty |
| | Incorrect Connection | Check connections. | Repair wiring and connections. |
| Faulty PCB | | | Replace PCB. |
| Indicated Pressure Value is Excessively High or Low | Malfunction of Pressure Sensor due to Faulty Check Joint | Check for clogging of check joint. | Replace check joint. |

| Alarm Code 22 | Abnormality of Thermistor for Outdoor Air Temperature (Outdoor Unit Ambient Thermistor) |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when the thermistor is short-circuited (less than 0.2 kΩ) or cut (greater than 500 kΩ) during running. However, this alarm occurs during test running mode only. In the case that the thermistor is abnormal during running, operation continues based on the assumption that the outdoor temperature, is 35 ℃(Cooling) / 6 ℃(Heating).

Is it connected to CN102 (Ta) of PCB1? — No → Connect it.

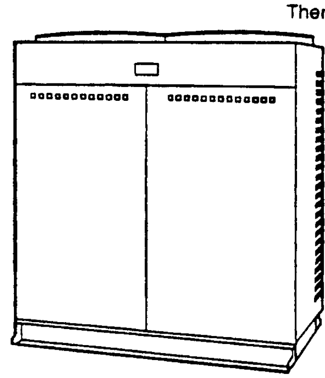
↓ Yes

Is resistance of THM2 between 0.2kΩ and 500kΩ? — No → Faulty Thermistor

Refer to page 1-36.

↓ Yes

Faulty PCB1



Outdoor Unit Ambient Thermistor (THM1)

Regarding the thermistor characteristics, refer to page 1-42.

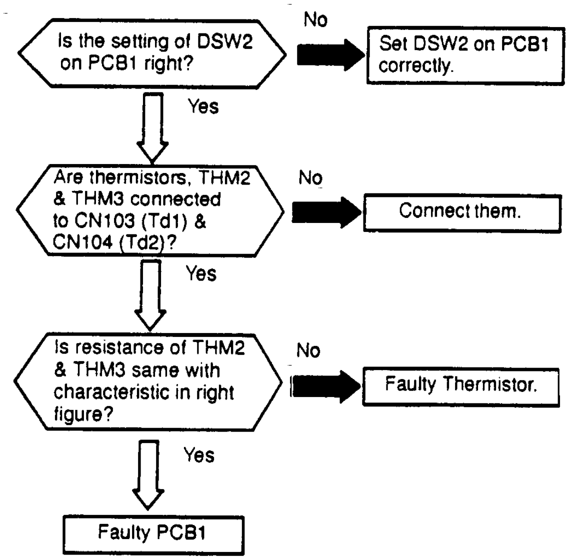
| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Faulty Thermistor for Outdoor Unit Ambient | Fault | Check resistance. | Replace thermistor if faulty. |
| | Incorrect Connection | Check wiring to PCB. | Repair wiring and connections. |
| Faulty PCB | | | Replace PCB. |

| Alarm Code 23 | Abnormality of Thermistor for Discharge Gas Temperature on Top of Compressor Chamber |
|---|---|

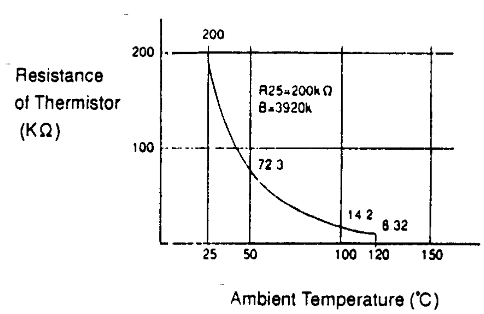
- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when the thermistor is short–circuited (less than 0.9 kΩ) or cut (greater than 2,350 kΩ) during running.

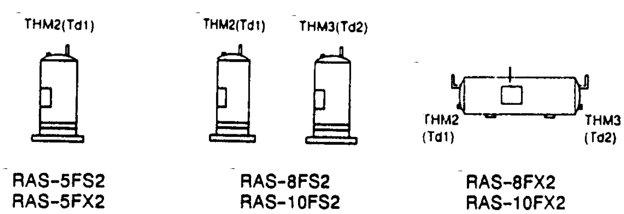
If abnormality with the thermistor is found, check all the thermistors as shown below.




Characteristics of Thermistor




Position of Thermistor

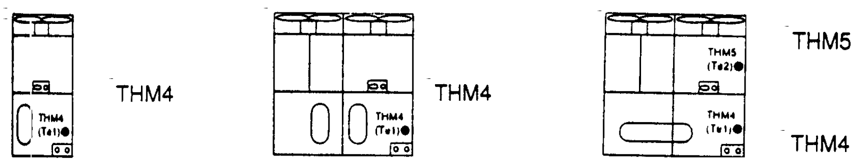



| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Faulty Discharge Gas Thermistor | Fault | Check resistance. | Replace thermistor if faulty. |
| | Incorrect Connection | Check wiring to PCB. | Repair wiring and connections. |
| Faulty PCB | | | Replace PCB. |
| Incorrect Setting of DSW2 on PCB1 | | Check setting of DSW2 on PCB1. | Correctly set DSW2 on PCB1. |

| Alarm Code 24 | Abnormality of Thermistor for Evaporating Temperature during Heating Operation (Outdoor Unit) |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.
- One evaporating thermistor (FS2 Series) during the heating operation is attached to the heat exchanger. Two evaporating thermistors (FX2 Series) during the heating operation are attached to the upper heat exchanger (for liquid) and the lower heat exchanger (for liquid). If one of the two is faulty, this alarm is indicated. Therefore, if this alarm occurs check two thermistors at the same time. The position is indicated below.

★This alarm code is indicated when the thermistor is short-circuited (less than 0.2 kΩ) or cut (greater than 500 kΩ) during running.




RAS-5FS2 and RAS-5FX2 RAS-8FS2, RAS-10FS2 RAS-8FX2 and RAS-10FX2

Are THM1 and THM3 connected to CN105 (Te1) and CN106 (Te2) of PCB1? — No → Connect them.

↓ Yes

Is resistance of the thermistor between 0.2kΩ and 500kΩ? — No → Faulty Thermistor

Regarding the thermistor characteristics, refer to page 1-42.

↓ Yes

Faulty PCB1

| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Faulty Thermistor for Evaporating Temperature during Heating | Fault | Check resistance. | Replace thermistor if faulty. |
| | Incorrect Connection | Check wiring to PCB. | Repair wiring and connections. |
| Faulty PCB | | | Replace PCB. |

| Alarm Code 29 | Abnormality of Low Pressure Sensor (Outdoor Unit) |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when the pressure sensor voltage decreases lower than 0.1 V or increases higher than 4.9V during running.

Is the connection CN114 of the outdoor unit PCB1 mated loose?
- Yes → Connect it correctly.
- No ↓

Is 5 VDC applied across #1 and #3 of the connector CN114 of the outdoor circuit board PCB1?
- No → Faulty PCB2
- Yes ↓

Is voltage under 0.1 VDC or over 4.9 VDC across #2 and #3 of the connector CN114 of the outdoor circuit board PCB1?
- No → Faulty PCB1
- Yes ↓

Is the Check Joint clogged.
- Yes → Replace the check joint.
- No ↓

Abnormality with High Pressure Sensor.

| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Faulty High Pressure Sensor | Fault | Check output voltage is correct. | Replace pressure sensor if faulty. |
| | Incorrect Connection | Check connections. | Repair wiring and connections. |
| Faulty PCB | | Replace PCB and check operation. | Replace PCB if faulty. |
| Indicated Pressure Value is Excessively High or Low | Malfunction of Pressure Sensor due to Faulty Check Joint | Check for clogging of check joint. | Replace Check joint. |

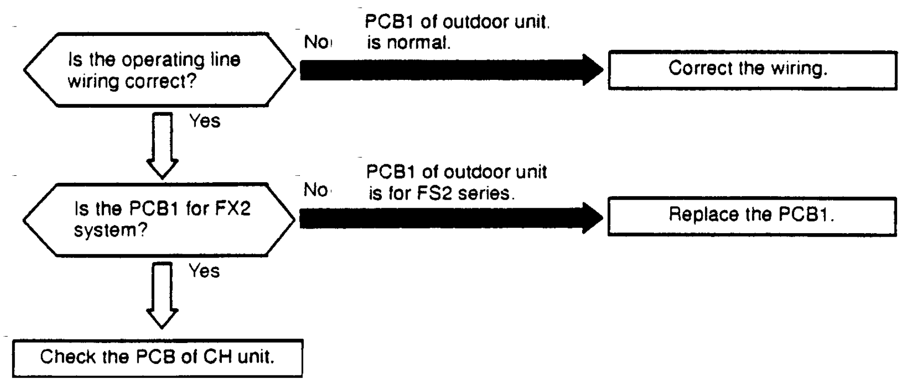
## FX2 SYSTEM

| Alarm Code 30 | Incorrect Wiring Connection |
|---|---|

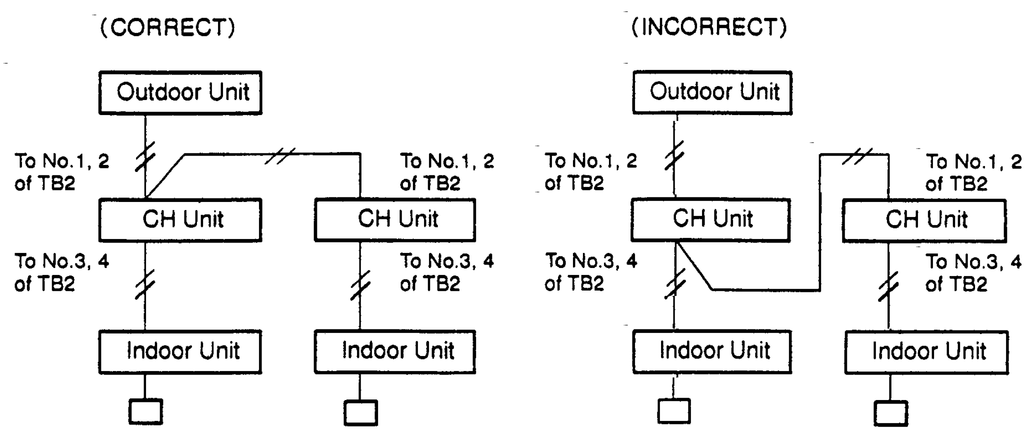
- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when two or more CH units exist between outdoor unit and indoor unit in the operating line, caused by incorrect wiring.

★This alarm code is indicated when PCB1 for FS2 series outdoor unit is mounted by mistake.




<Example of Operating Line Wiring>



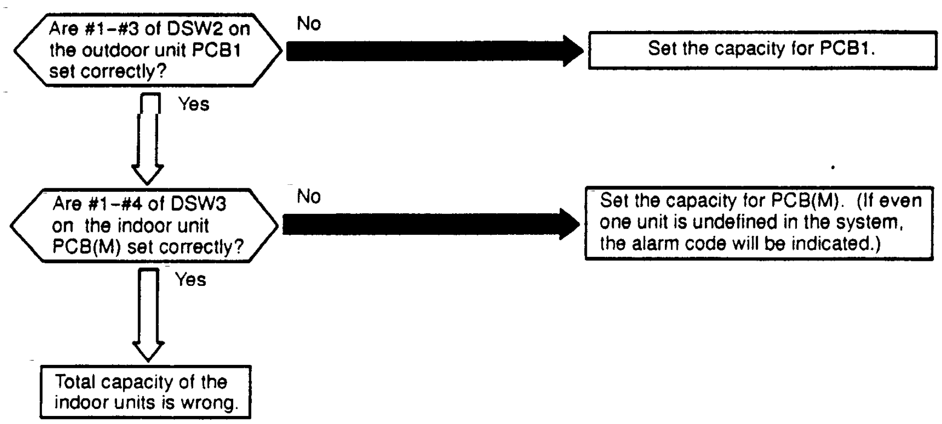
## TROUBLESHOOTING

| Alarm Code 31 | Incorrect Capacity Setting or Combined Capacity between Indoor Units and Outdoor Unit |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when the capacity setting dip switch, DSW2 on the outdoor unit PCB, PCB1 is not set(all the settings from #1 to #3 are OFF).

★This alarm code is indicated when the total indoor unit capacity is smaller than 50% or greater than 130% of the combined outdoor unit capacity.




| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Incorrect Capacity Setting of Indoor Unit | | Check combination of indoor units and capacity setting on PCB. | Correctly set dip switch, DSW3. |
| Incorrect Capacity Setting of Outdoor Unit | | Check capacity setting on outdoor unit PCB. | Correctly set dip switch, DSW2. |
| Total Indoor Unit Capacity Connected to the Outdoor Unit is Beyond Working Range | | Check outdoor unit model by calculating total indoor units capacity. | Ensure that total indoor unit capacity is from 50% to 130%. |

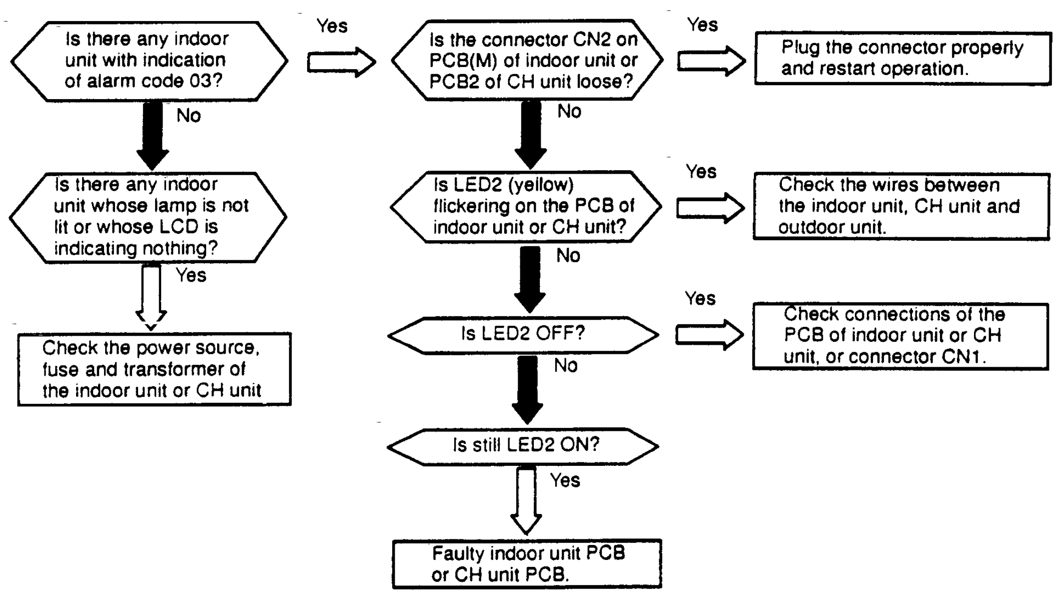
Combination

| Outdoor Unit Model | Combined Indoor Unit | | Notes |
|---|---|---|---|
| | Total Q'ty | Total Capacity (HP) | |
| RAS-5FS2, RAS-5FX2 | 2-8 sets *1) | 2.5-6.5 | *1): If the capacity of one indoor unit is the same as of the outdoor unit, the system can be operated. |
| RAS-8FS2 | 2-12 sets | 4-10 | |
| RAS-10FS2 | | 5-13 | |
| RAS-8FX2 | 2-8 sets | 4-10 | |
| RAS-10FX2 | | 5-13 | |

| Alarm Code 32 | Abnormality in Transmitting of Other Indoor Units |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated on the remote control switch of other indoor unit when no transmitting data is issued from a malfunctioning indoor unit or CH unit for more than 60 minutes after receiving transmitting data from the indoor unit.




| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Abnormal Transmitting from Other Units | Abnormal Transmitting Indoor Unit→Outdoor Unit Outdoor Unit→Indoor Unit | Check indoor unit showing alarm code "03". | Take action according to Pages 1-23 to 1-26. |
| | Incorrect Power Supply (to Indoor Unit or CH Unit) *1) | Check indoor unit showing no light ON on remote control switch. | Take action according to Page 1-8, 9. |
| | Faulty Indoor Unit PCB or CH Unit PCB | If all indoor unit show alarm code 32, check PCB by self-checking mode for all indoor unit or CH unit. | Replace PCB if faulty. |

*1): This abnormality is indicated on an indoor unit which is normal, since this is due to abnormality of transmitting of the indoor units or power source abnormality in the same refrigeration cycle and electrical system. If the power source is abnormal, abnormality can not be indicated by its indoor unit, so, this alarm is indicated on the remote control switch of other indoor unit.

| Alarm Code 35 | Incorrect Indoor Unit No. Setting |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated 10 minutes after power is supplied to the outdoor unit when the indoor unit No. connected to the outdoor unit is duplicated by setting of DSW1.

## Indication of 7-Segment in Outdoor Unit when Unit Nos. are Duplicated

(1)The followings are indicated for 10 minutes after power is supplied to the outdoor unit.

1) The following is repeatedly indicated when NO.1 indoor unit setting is duplicated

[ ][1][0][1] ~ [ ][1][0][2]

2) The following is repeatedly indicated when NO.2 indoor unit setting is duplicated

[ ][1][0][1] [ ][1][0][3]

3) The following is repeatedly indicated when NO.3 indoor unit setting is duplicated

: :

[ ][1][0][1] [ ][1][0][4]

13) The following is repeatedly indicated when NO.D indoor unit setting is duplicated

[ ][1][0][1] [ ][1][0][E]

14) The following is repeatedly indicated when NO.E indoor unit setting is duplicated

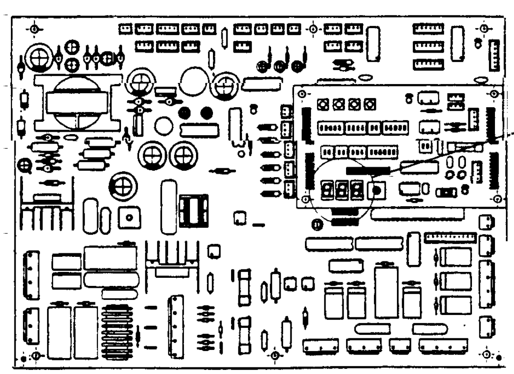
[ ][1][0][1] [ ][1][0][F]

15) The following is repeatedly indicated when NO.F indoor unit setting is duplicated

[ ][1][0][1] [ ][1][1][0]

(2)After 10 minutes, 「35」 flickers. This 「35」 is also flickered on the remote control switch

Outdoor Unit PCB
(PCB1: Control)

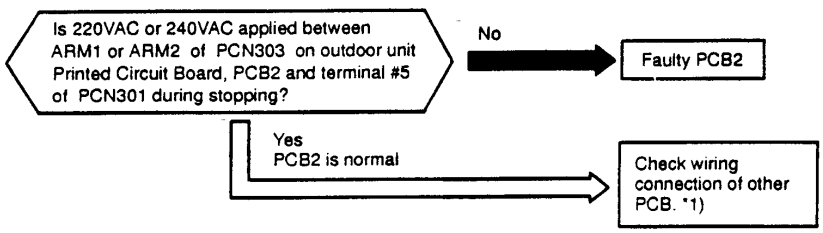


7-Segment

| Alarm Code 38 | Abnormality of Picking up Circuit for Protection (Outdoor Unit) |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when AC 220V or 240V is supplied to voltage between terminal #3 of PCN105 on PCB1 and faston terminal N on PCB1 in the outdoor unit during stoppage.




| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Abnormality of Picking up Circuit for Protection | Incorrect Connection | Check voltage supplied to connectors. | Repair wiring connections. |
| | Faulty PCB | | Replace PCB. |

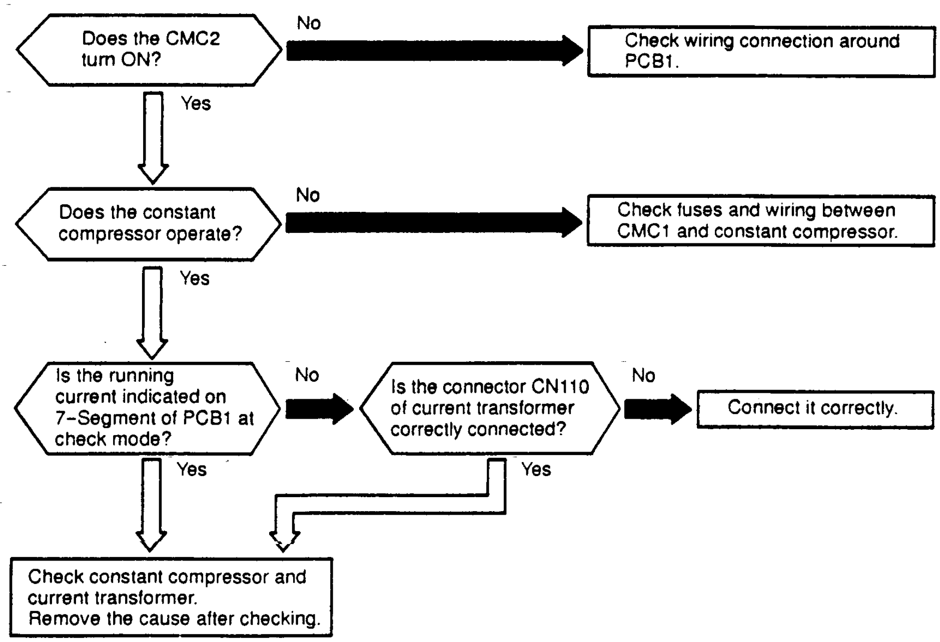
*1): Check wiring system connecting to PCN105 on PCB1

| Alarm Code 39 | Abnormality of Running Current at Constant Compressor |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★ This alarm code is indicated when the following conditions occurs three times within one hour.

- The running current of the constant compressor exceeds the value of overcurrent limitation during operating.
- The running current of the constant compressor is less than 1A during operating.




Indication of 7-Segment

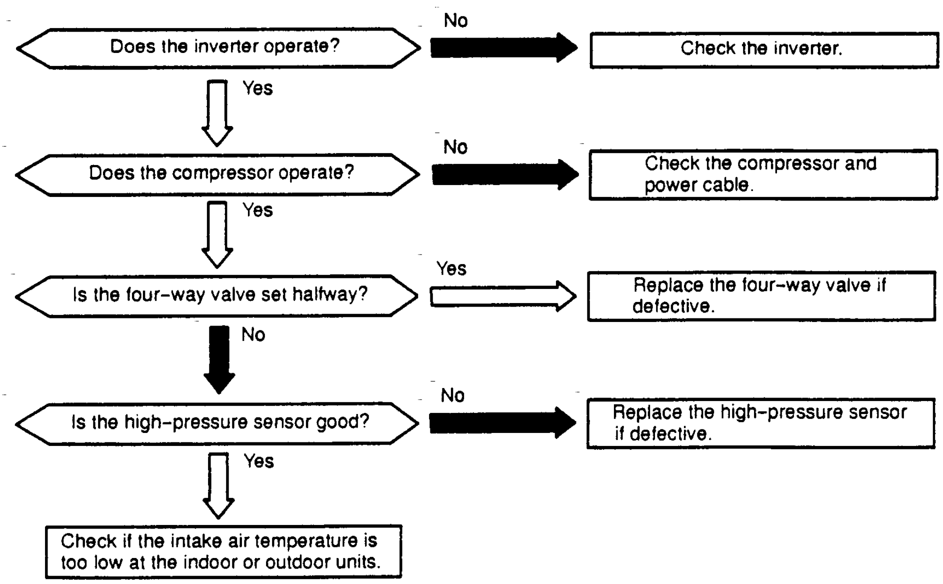
| 239 | Abnormality of Constant Compressor |
|---|---|

| Phenomenon | Cause | | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|---|
| Overcurrent of Compressor Motor | Excessively Large Running Current of Compressor Motor | | Excessively High or Low Voltage of Power Source | Operate within 380~415V±10%. |
| | | | Imbalance Among Phase at Power Source | Adjust voltage between each phase. |
| | | | Excessively High Discharge Pressure | Check for causes. |
| | Single Phase Operation | | Melted Fuse for Power Source | Replace the fuses. |
| | | | Looseness of Screws for Power Source | Tightly fix screws. |
| | | | Deterioration of Contact of Magnetic Switch for Compressor | Replace magnetic contactor. |
| | Seizure of Compressor Bearing | | Locked-up? | Replace compressor. |
| | Insulation Failure of Compressor Motor | | Measure insulation resistance. | Replace compressor.. |
| | Faulty CT | Fault | Check continuity during stoppage. | Replace CT when faulty. |
| | | Incorrect Connection | Measure resistance by tester. | Remove looseness and replace connectors. |
| | | Incorrect Wiring | Check connections. | Correctly connect. |

| Alarm Code 43 | Activation to Protect System from Low Compression Ratio |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when a compression ratio , $\varepsilon$ = {Pd /(Ps + 1.03)} is calculated from a discharge pressure(kg/cm$^2$G) and suction pressure(kg/cm$^2$G) and the condition lower than $\varepsilon$ = 1.8 occurs more than 3 times including 3 in one hour.

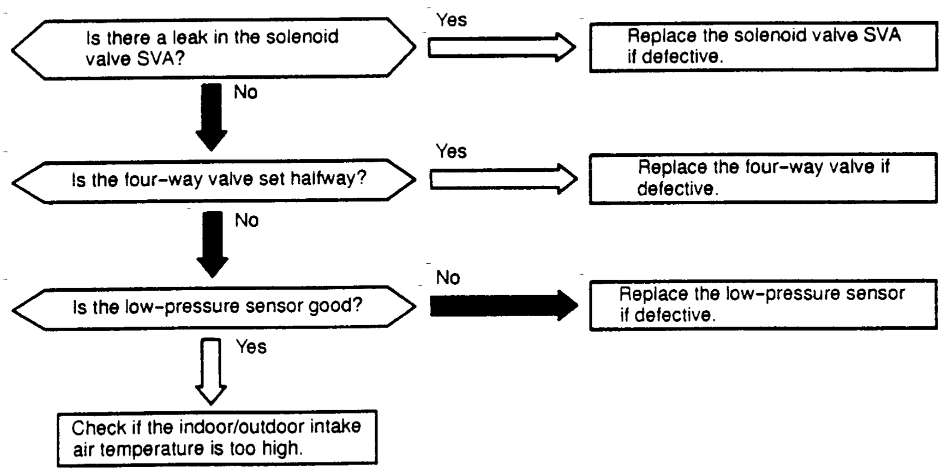



| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Excessively Low Compression Ratio | Inverter is not Functioning | Check inverter. | Repair faulty part. |
| | Compressor is not Operating | Check compressor. | Replace comp. if faulty. |
| | Valve Stoppage at Medium Position of 4-Way Valve | Measure suction pipe temp. of 4-way valve. | Replace 4-way valve if faulty. |
| | Abnormality of High or Low Pressure Sensor | Check connector for PCB, power source and pressure indication. | Replace sensor if faulty. |
| | Excessively Low Indoor Intake Air Temperature | Check indoor unit and outdoor unit air temp. thermistor. | Replace thermistor if faulty. |
| | Leakage from Solenoid Valve: SVA Outdoor Unit or SVD1, SVD2 of CH Unit | Check refrigerant leakage. | Replace SVA or CH unit if leaking. |

| Alarm Code 44 | Activation to Protect System from Excessively High Suction Pressure |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when the compressor is operated under the conditions higher than 0.88 MPa (9 kg/cm2G) of suction pressure and its occurrence is more than 3 times including 3 in one hour.




| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Excessively High Suction Pressure | Leakage of Solenoid Valve SVA | Check outlet pipe temp. of solenoid valve, SVA. | Check connecting wires. Replace solenoid valve, SVA if faulty. |
| | Valve Stoppage at Medium Position of 4-Way Valve | Measure suction gas pipe temp. of 4-way valve. | Replace 4-way valve if faulty. |
| | Abnormal Suction Pressure Sensor | Check connectors of PCB and power source. | Replace sensor if faulty. |
| | Excessively High Indoor Unit and Outdoor Unit Air Temperature | Check indoor unit and outdoor unit air temp. thermistor. | Replace thermistor if faulty. |
| | Leakage from Solenoid Valve: SVD1, SVD2 of CH Unit | Check refrigerant leakage. | Replace CH unit if leaking. |

| Alarm Code 45 | Activation to Protect System from Excessively High Discharge Pressure |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when the compressor is operated under the conditions higher than 2.65 MPa (27 kg/cm2G) of discharge pressure ( in case of frequency 20 to 30Hz, Pd is higher than 2.45 MPa (25 kg/cm2G) and its occurrence is more than 3 times including 3 in one hour).

Is there a leakage of the solenoid valves SVA or SVB?
- Yes → Replace the solenoid valve SVA or SVB if faulty.
- No ↓

Is the outdoor unit stop valve closed?
- Yes → Open the stop valve.
- No ↓

Is the high-pressure sensor good?
- No → Check connection of the circuit board connector. Replace the high-pressure sensor if faulty.
- Yes ↓

Are ex. valves fully closed and locked?
- Yes → Check the circuit board and connection of the circuit board connector. Replace the electronic expansion valves if faulty.
- No ↓ Check the outdoor electronic expansion valve during cooling and the indoor electronic expansion valve during heating.

Are electric wiring and refrigerant piping between the indoor and outdoor unit incorrectly connected?
- Yes → Correctly connect it.
- No ↓

Check if indoor or outdoor intake air temperature is too high.

| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Excessively High Discharge Pressure | Leakage of Solenoid Valve SVA, SVB (Outdoor Unit) | Check outlet temp. of solenoid valve, SVA & SVB. | Check connection. Replace solenoid valve SVA or SVB if faulty. |
| | Closed Stop Valve | Check stop valve. | Open stop valve. |
| | Abnormal High Pressure Sensor | Check connectors for PCB. | Replace pressure sensor if faulty. |
| | Excessively High Indoor Unit and Outdoor Unit Inlet Air Temp. | Check thermistor for indoor unit and outdoor unit inlet air temp. | Replace thermistor if faulty. |
| | Incorrect Connection between Indoor Unit and Outdoor Unit | Check electrical system and ref. cycle. | Correctly Connect. |
| | Locked Ex. Valve | Check connector for PCB. | Repair connector for PCB or ex. valve. Replace it if faulty. |
| | Fully Open of Solenoid Valve SVD1 (CH Unit for FX) | Check solenoid valve SVD1 open in heating. | Replace coil or CH unit if faulty. |

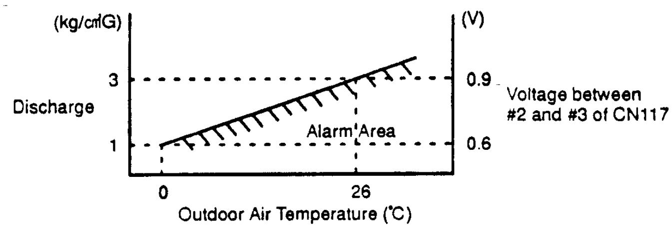
| Alarm Code 46 | Activation to Protect System from Excessively Low Discharge Pressure (for Protection of Refrigerant Shortage) |
|---|---|

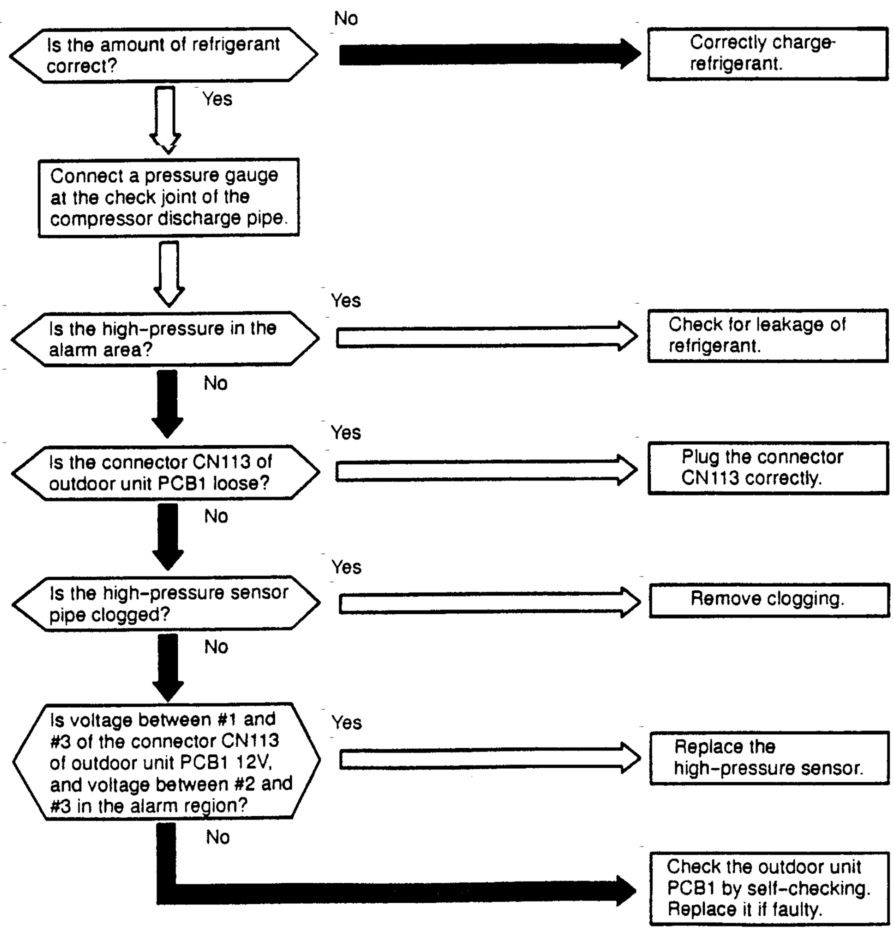
- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★ This alarm code is indicated when the discharge pressure is decreased due to insufficient refrigerant charge. Refer to the figure below.

★ This alarm code is indicated when the discharge gas superheat at the top of the compressor is higher than 80 deg. during running and bypass expansion valve: MVB is fully opened and its state is maintained for more than 25 minutes.

★ This alarm code is indicated when the compressor is operated under the condition less than 0.69 MPa (6 kg/cm2G) of discharge pressure and its occurrence is continued one hour.



| Alarm Code 47 | Activation to Protect System from Excessively Low Suction Pressure (Protection from Vacuum Operation) |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when a suction pressure is lower than 0.02 MPa (0.2 kg/cm2G) for over 10 minutes and its state occurs more than 3 times including 3 in one hour.

- Is the amount of refrigerant correctly?
  - No → Correctly charge refrigerant.
  - Yes ↓
- Is the outdoor stop valve open?
  - No → Open the stop valve.
  - Yes ↓
- Is refrigerant leaking?
  - Yes → Repair leaks.
  - No ↓
- Are the electronic ex. valves fully closed and locked?
  - Yes → Check connection of the connectors and the circuit board. Replace the electronic ex. valves if faulty.
  - No ↓ Check the indoor ex. valve during cooling and the outdoor ex. valve during heating.
- Check if connections of electric wiring and refrigerant piping between the indoor and outdoor unit are wrong.

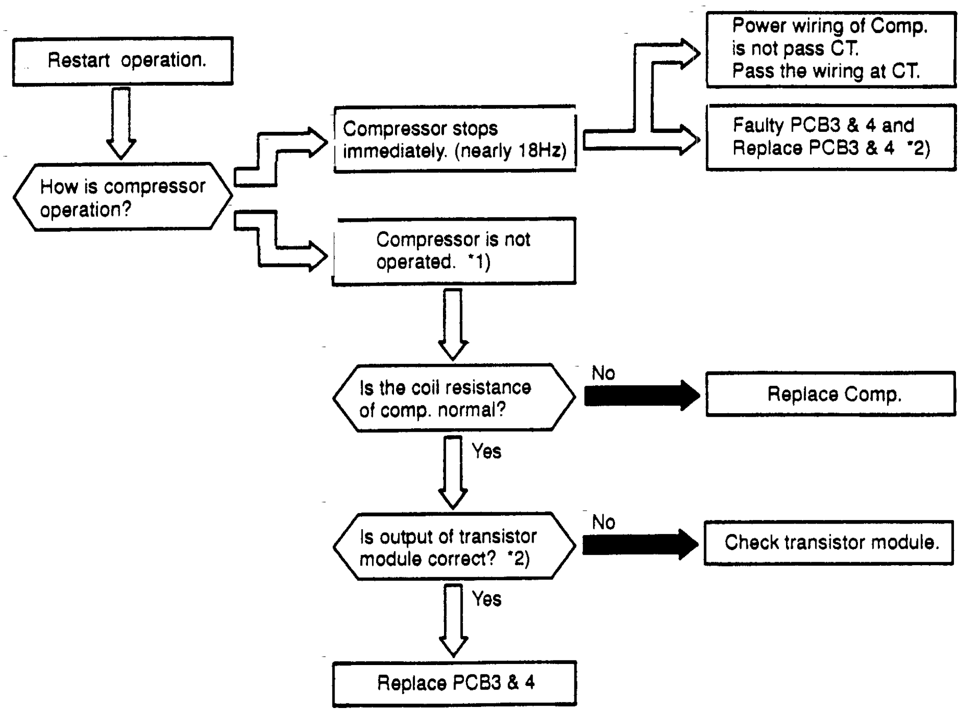
| Phenomenon | Cause | Check Item | Action (Turn OFF Main Switch) |
|---|---|---|---|
| Excessively Low Suction Pressure (in Vacuum) | Shortage of Ref. | Check ref. charged volume or check for leakage. | Repair leakage and correctly charge. |
| | Closed Valve | Check stop valve. | Open stop valve. |
| | Abnormal Low Pressure Sensor | Check connector for PCB. | Replace pressure sensor if faulty. |
| | Incorrect Connection between Indoor Unit and Outdoor Unit | Check electrical system and ref. cycle. | Correctly connect between indoor unit and outdoor unit. |
| | Locked Ex. Valve | Check connector for PCB. | Repair or replace connector of PCB or ex. valve. |

| Alarm Code 51 | Abnormality of Current Transformer (0A Detection) |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★ This alarm code is indicated when the current transformer is abnormal (0A detection) and its state occurs more than 3 times in 30 minutes.

Condition of Activation ··· When the frequency of compressor is maintained at 15~18Hz after compressor is started, one of the absolute value of running current at phase U+, U–, V+ and V– is less than 0.5A.

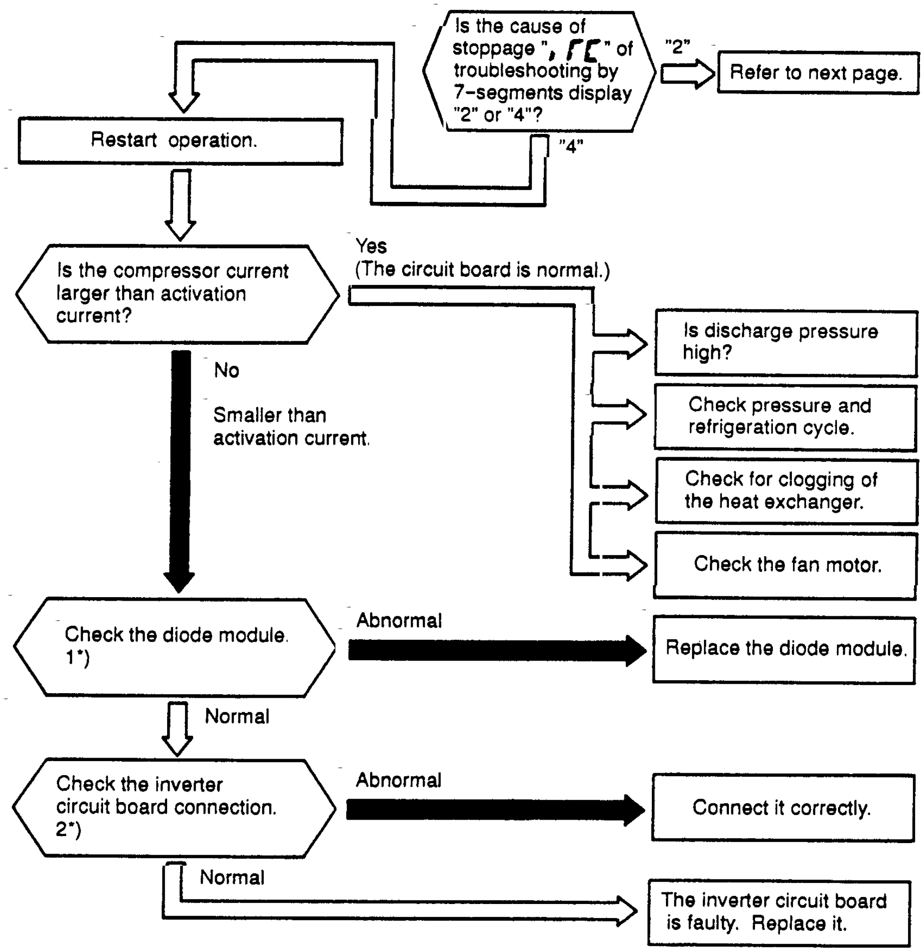



*1): Number of operating comp. is shown 7-segment.
*2) Perform the high voltage discharge work refer to 1.3.7 before checking and replacing the inverter parts.

| Alarm Code 52 | Protection Activation Against Instantaneous Overcurrent of Inverter (1) |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★This alarm code is indicated when the electronic thermal relay for the inverter is activated 3 times including 3 in 30 minutes. Retry operation is performed up to the occurrence of 2 times.
conditions: Inverter current is 105% of the rated current (14A), ① continuously for 30 seconds or ② the accumulated time reaches 3.5 minutes in 10 minutes .

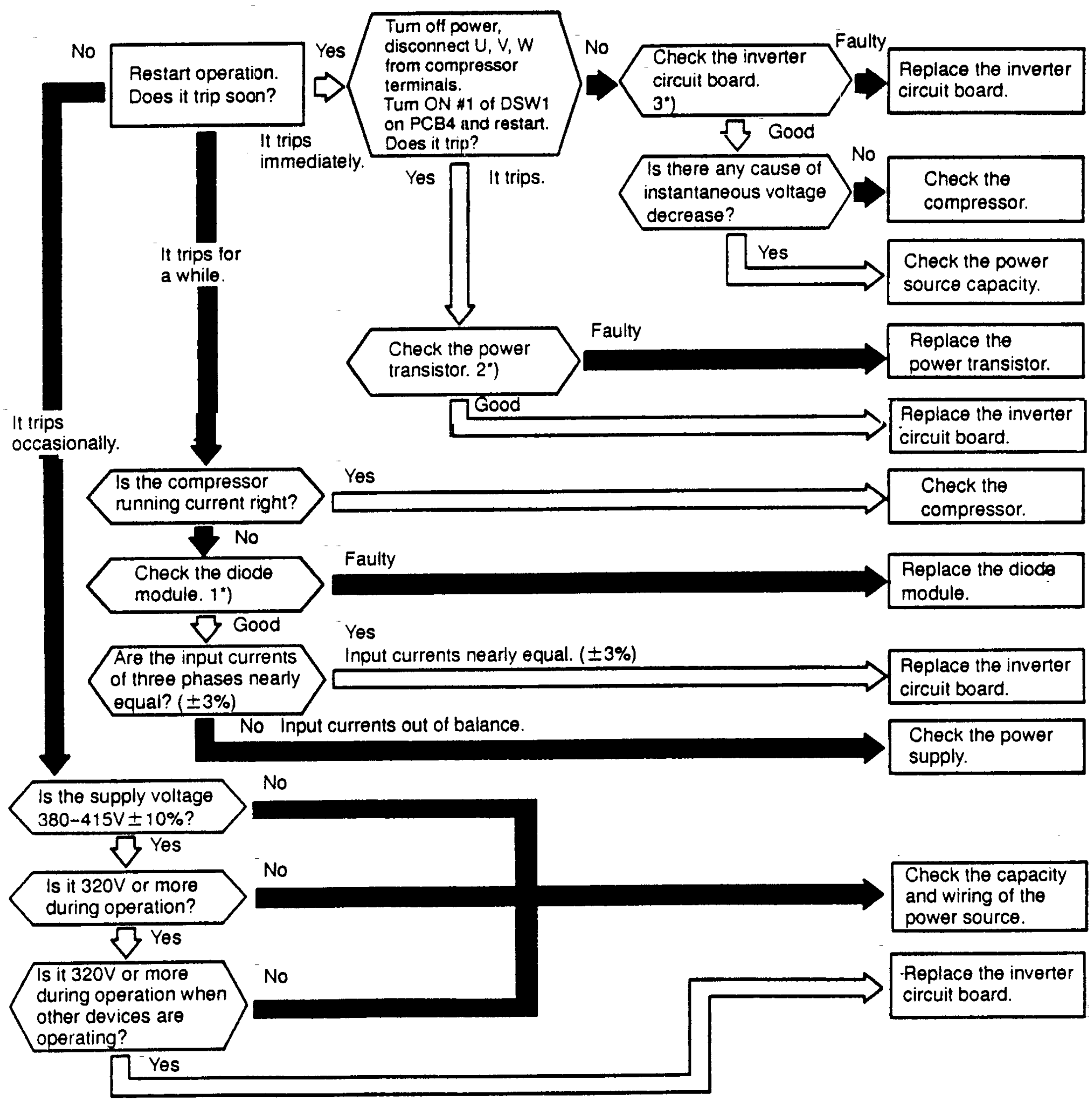



NOTES:
*1): Regarding the checking method for diode module, refer to page 1-92.
*2): Regarding the checking of inverter components, refer to page 1-93 regarding electrical discharge.

| Alarm Code 52 | Protection Activation Against Instantaneous Overcurrent of Inverter (2) |
|---|---|

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★ This alarm code is indicated when the instantaneous overcurrent tripping occurs 3 times including 3 in 30 minutes. Retry operation is performed up to the occurrence of 2 times.
conditions: Inverter current is 150% of the rated current.




NOTES:

1*): Regarding the checking method for the diode module, refer to page 1-92.
2*): Regarding the checking method for power transistor, refer to page 1-91.
3*): Regarding the checking of inverter components, refer to page 1-93 regarding electrical discharge.

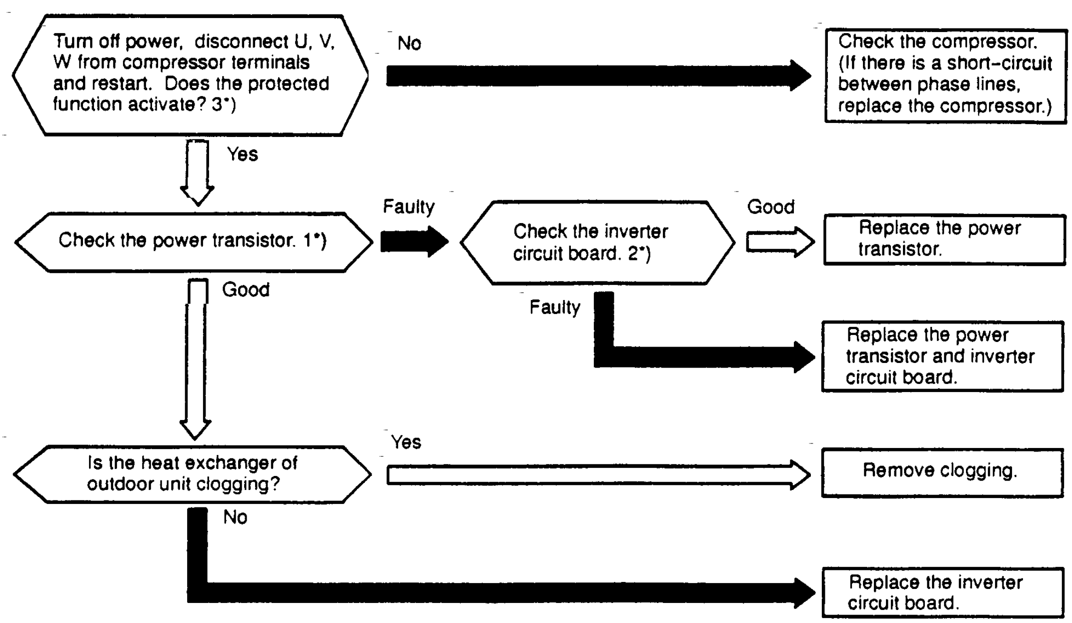
| | |
|---|---|
| Alarm Code 53 | Protection Activation of Transistor Module |

- "RUN" light flickers and "ALARM" is indicated on the remote control switch.
- The unit No. and alarm code is alternately indicated on the set temperature section, or the unit No. and alarm code are indicated on the display of the outdoor unit PCB.

★ Transistor module have detected function of abnormality.

★ This alarm is indicated when the transistor module detect the abnormality 3 times in 30 minutes including 3. Retry operation is performed up to the occurrence of 2 times.

Conditions: Abnormal Current to the Power Transistor such as Short Circuited or Grounded
Abnormal Temperature of the Power Transistor
Control Voltage Decrease




NOTES:

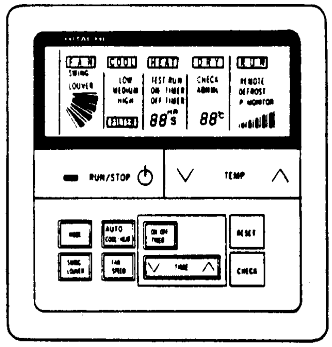
1*): Regarding the checking method for the power transistor, refer to page 1-91.

2*): Regarding the checking of inverter components, refer to page 1-93 regarding electrical discharge.

3*): Turn ON the No.1 switch of the dip switch DSW1 on PCB3 when restarting with disconnecting the terminals of the compressor. After troubleshooting, turn OFF the No.1 switch of the dip switch DSW1 on PCB3.

## 1.2.3 Troubleshooting in Check Mode by Remote Control Switch

- Use the remote control CHECK switch in the following cases.
  - (1) When the RUN lamp is flashing.
  - (2) To trace back the cause of trouble after restarting from stoppage with the RUN lamp flashing.
  - (3) To check during normal operation or stoppage.
  - (4) To monitor the temperatures of intake and discharge air.



Check Mode → Check Mode 1: Current data will be displayed.
Check Mode → Check Mode 2: Data held immediately before failure occurs will be displayed.

Check Mode

CHECK — Press for more than 3 seconds.

- Indication will delay as transmission between the remote control switch and indoor unit takes about 10 seconds.
- All data may be displayed as "FF" or "-1". These transient data produced temporarily by software do not affect device functions at all. (The alarm code may also be indicated as "FF".)

Unit Number and Alarm Code Displayed

A0 02

- Alarm code identifying the last fault that has occurred in the indicated unit.
- Unit number of connected unit or unit number for which checking mode was selected previously.

(A) Within 7 seconds to check another unit

- Forward: Press the ∧ switch to rise from A1 to A2 to A3 ···
- Backward: Press the ∨ switch to descend from AF to AE to Ad ···

After 7 seconds

Check Mode 1 (See 1-62 for details.)

b 1 22

P 1 25

To view the previous indication: ∨ TEMP ∧ — Press

To view the next indication: ∨ TEMP ∧ — Press

CHECK — Press for more than 3 seconds.

Press ∧ to view the next data.
Press ∨ to view the previous data.

Unit Number and Alarm Code Displayed

A0 02

See (A)

After 7 seconds

Check Mode 2 (See 1-65 for details.)

9 1 23

∨ TEMP ∧ — Press

9F 20

CHECK — Press

Check Mode Released

- In Check Mode 2, Data of the first three units connected serially to a remote control switch are available.
- You can press the CHECK switch to release Check Mode 2. Check Mode 1 cannot be released even if you press the CHECK switch.

## (1) Contents of Check Mode 1

The next indication is shown by pressing ∧ the part of "TEMP" switch. If the ∨ part of "TEMP" switch is pressed the previous indication is shown.

Temperature Indication

Indication of Category Code
Indication of Temperature, etc.

| No. | Item | Category Code | Temperature, etc. |
|---|---|---|---|
| 1 | Indoor Unit Temp. Setting (℃) | b 1 | 22 |
| 2 | Indoor Unit Intake Air Temp. at Thermistor (℃) | b 2 | 20 |
| 3 | Indoor Unit Discharge Air Temp. at Thermistor (℃) | b 3 | 55 |
| 4 | Indoor Unit Heat Exchanger Liquid Pipe Temp. (℃) | b 4 | 20 |
| 5 | Temperature at Remote Sensor (℃) | b 5 | 25 |
| 6 | Outdoor Unit Ambient Air Temp. (℃) | b 6 | 10 |
| 7 | Indoor Unit Heat Exchanger Gas Pipe Temp. (℃) | b 7 | 25 |
| 8 | Outdoor Unit Evaporating Temp. during Heating (℃) | b 8 | 02 |
| 9 | Control Information | b 9 | - - |
| 10 | Discharge Gas Temp. at the Top of Comp. Chamber (℃) | b A | 41 |

to next page

Normal ·····Temperature displayed

Abnormal
- - - (Thermistor Open-circuited or PCB Defected
- F F (Thermistor Short-circuited or PCB Defected

See "1.3.1 Self-Checking of PCB's using Remote Control Switch" in pages 1-81, 82.

(NOTE 1) During transient period such as starting time, etc. "- -" may be indicated for a limited time.

(No. 5) This is indicated only when a remote sensor is connected. The indication of "- -" is normally indicated.

(No. 9) This indicates the internal information for the remote control switch. This does not have any specific meaning.

(No. 10) In case that temperature is higher than 100℃, two digits flicker.

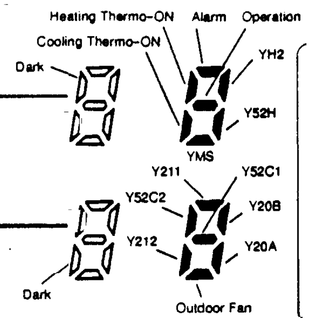
(ex.) 123℃ → 23 Flickering

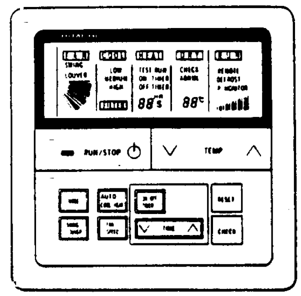
(During two compressors operation, average temperature of two compressors is indicated.)

In case that temperature is higher than 126℃, "26" flicker.

### Indication on Micro-Computer Input/Output

| | | |
|---|---|---|
| 11 | Micro Computer Input/Output in Indoor Unit | C 1 \| 4 |
| 12 | Micro-Computer Input/Output in Outdoor Unit | C 2 \| : |



Symbols with a letter Y are relays on PCB

### Indication of Unit Stoppage Cause

| | | |
|---|---|---|
| 13 | Cause of Stoppage | d 1 \| 0 1 |

| | |
|---|---|
| 00 | Operation OFF, Power OFF |
| 01 | Thermo-OFF |
| 02 | Alarm (NOTE 1) |
| 03 | Freeze Protection, Overheating Protection |
| 05 | Instantaneous Power Failure at Outdoor Unit, Reset |
| 06 | Instantaneous Power Failure at Indoor Unit, Reset |
| 07 | Stoppage of Cooling Operation due to Low Outdoor Air Temperature, Stoppage of Heating Operation due to High Outdoor Air Temperature (NOTE 4) |
| 08 | Compressor Quantity Changeover, Stoppage |
| 09 | Demand of 4-Way Valve Changeover Stoppage (FX Only) |
| 10 | Demand, Enforced Stoppage |
| 11 | Retry due to Compressor Ratio Decrease |
| 12 | Retry due to Low Pressure Increase |
| 13 | Retry due to High Pressure Increase |
| 14 | Retry due to Abnormal Current of Constant Compressor |
| 15 | Retry due to Abnormal High Temperature of Discharge Gas, Excessively Low Suction Pressure |
| 16 | Retry due to Decrease of Discharge Gas Superheat |
| 17 | Retry due to Inverter Tripping |
| 18 | Retry due to Voltage Decrease |
| 19 | Expansion Valve Opening Change Protection |
| 20 | Operation Mode Changeover of Indoor Unit (NOTE 3) |

(NOTE 1) Even if stoppage is caused by "Alarm", "02" is not always indicated.

(NOTE 2) Explanation of Term,
Thermo-ON: A condition that an indoor unit is requesting compressor to operate.
Thermo-OFF: A condition that an indoor unit is not requesting compressor to operate.

(NOTE 3) In the FX2 system, "20" will be indicated at the indoor unit operation changeover, and the unit will automatically restart by the Thermo-ON.
In the FS2 system "20" will be indicated at the difference mode between indoor units.

(NOTE 4) If transmission between the inverter printed circuit board and the control printed circuit board is not performed during 30 seconds, the outdoor unit is stopped.
In this case, stoppage is d1-05 cause and the alarm code "04" may be indicated.

(NOTE 5) If transmission between the indoor unit and the outdoor unit is not performed during 3 minutes, indoor units are stopped.
In this case, stoppage is d1-06 cause and the alarm code "03" may be indicated.

### Abnormality Occurrence Counter

| | | |
|---|---|---|
| 14 | Abnormality Occurrence Times | E 1 \| 0 1 |
| 15 | Instantaneous Power Failure Occurrence Times in Indoor Unit | E 2 \| 0 0 |
| 16 | Transmission Error Occurrence Times between Remote Control Switch and Indoor Unit | E 3 \| 0 0 |
| 17 | Abnormality Occurrence Times on Inverter | E 4 \| 0 0 |

Countable up to 99.
Over 99 times, "99" is always indicated.

(NOTE 1) If a transmitting error continues for 3 minutes, one is added to the occurrence times.

(NOTE 2) The memorized data can be cancelled by the method indicated in 1.3.1 "Self-checking of PCBs using Remote Control Switch" in pages 1-81, 82.

### Indication of Automatic Louver Condition

| | | |
|---|---|---|
| 18 | Louver Sensor | F 1 \| 0 0 |

to next page

## TROUBLESHOOTING

**Compressor Pressure/Frequency Indication**

| | | | |
|---|---|---|---|
| 19 | Discharge Pressure (High) ($kg/cm^2G$) | H1 18 | |
| 20 | Suction Pressure (Low) ($kg/cm^2G$) | H2 04 | |
| 21 | Control Information | H3 44 | This is an indication for internal information for the remote control switch. This does not have any specific meaning. |
| 22 | Operation Frequency (Hz) | H4 44 | The total frequency is indicated when two compressors are running. |

**Indoor Unit Capacity Indication**

| | | | |
|---|---|---|---|
| 23 | Indoor Unit Capacity | J1 08 | Indication 08 × Factor $\frac{1}{8}$ = Indoor Unit Capacity 1HP. Except, the capacity code for 0.8HP unit and 1.5HP unit are respectively 6 and 13. |
| 24 | Outdoor Unit Code | J2 Fn | "n" indicates total number of indoor units; n= 1~9, A (10), b (11), C (12), d (13), E (14), F (15), U (16) |
| 25 | Refrigerant Cycle Number | J3 01 | |
| 26 | Refrigerant Cycle Number | J4 00 | |

**Expansion Opening Indication**

| | | | |
|---|---|---|---|
| 27 | Indoor Unit Expansion Valve Opening (%) | L1 20 | |
| 28 | Outdoor Unit Expansion Valve MV1 Opening (%) | L2 99 | In case of models without Expansion Valve (MV2), the same figure is indicated. |
| 29 | Outdoor Unit Expansion Valve MV2 Opening (%) | L3 99 | In case of models without Expansion Valve (MV2), the same figure is indicated. |
| 30 | Outdoor Unit Expansion Valve MVB Opening (%) | L4 00 | indicates control information except FX2 Series |

**Estimated Electric Current Indication**

| | | | |
|---|---|---|---|
| 31 | Compressor Running Current (A) | P1 25 | The total current is indicated when two compressors are running. |

Returns to Temperature Indication

**Temperature Indication**

(2) Contents of Check Mode 2

The latest data of the first three indoor units only connected serially are indicated when more than three indoor units are connected to one remote control switch.
By pressing the ∧ part of "TEMP" switch, the next display is indicated. If the ∨ part of "TEMP" switch is pressed, the previous display is indicated.

Temperature Indicator

Indication of Category Code
Indication of Temperature, etc.

| | | | |
|---|---|---|---|
| 1 | Indoor Unit Intake Air Temp. at Thermistor (˚C) | 91 | 23 |
| 2 | Indoor Unit Discharge Air Temp. at Thermistor (˚C) | 92 | 50 |
| 3 | Indoor Unit Heat Exchanger Liquid Pipe Temp. (Freeze Protection) (˚C) | 93 | 25 |
| 4 | Outdoor Air Temp. (˚C) | 94 | 12 |
| 5 | Indoor Heat Exchanger Gas Pipe Temp. (˚C) | 95 | 25 |
| 6 | Evaporating Temp. at Heating (˚C) | 96 | 03 |
| 7 | Control Information | 97 | - - |
| 8 | Discharge Gas Temp. at the Top of Comp. Chamber (˚C) | 98 | 45 |

to next page

Normal ····Temperature displayed

Abnormal
- - (Thermistor Open-circuited or PCB Defected
- F F (Thermistor Short-circuited or PCB Defected

See "1.3.1 Self-Checking of PCB's using Remote Control Switch" in pages 1-81, 82.

(NOTE 1) During a transient period such as starting time, etc., "--" may be indicated for a limited time.

(NOTE 2) All the data during the check mode 2 will be changed to "00" if a power failure occurs during alarm indication.

(Item 7) This is an indication for internal information for the remote control switch.
This does not have any specific meaning.

(Item 8) The average temperature is indicated when two compressors are running.

## Compressor Pressure/Frequency Indication

| No. | Item | Display | |
|---|---|---|---|
| 9 | Discharge Pressure (High) (kg/cm²G) | 99 | 18 |
| 10 | Suction Pressure (Low) (kg/cm²G) | 9A | 04 |
| 11 | Control Information | 9b | 44 |
| 12 | Operating Frequency (Hz) | 9C | 44 |

No. 11: This is an indication for internal information for the remote control switch. This does not have any specific meaning.

No. 12: The total frequency is indicated when two compressor are running.

## Expansion Opening Indication

| No. | Item | Display | |
|---|---|---|---|
| 13 | Indoor Unit Expansion Valve Opening (%) | 9d | 20 |
| 14 | Outdoor Unit Expansion Valve MV1 Opening (%) | 9E | 99 |

## Estimated Electric Current Indication

| No. | Item | Display | |
|---|---|---|---|
| 15 | Compressor Running Current (A) | 9F | 20 |

No. 15: The total value is indicated when two compressors are running.

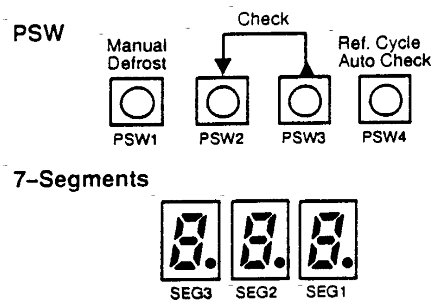
Returns to Temperature Indication

## Temperature Indication

### 1.2.4 Troubleshooting by 7-Segment Display

(1) Checking Method by 7-Segment Display

By using the 7-segments and check switch (PSW) on the PCB1 in the outdoor unit, total quantity of combined indoor units, 7-segments operation conditions and each part of refrigeration cycle, can be checked.




PSW3 ◀ Backward    PSW2 ▷ Forward

- To start checking, press the "PSW2" switch for more than 3 seconds.
- To proceed checking, press the "PSW2" switch for less than 2 seconds.
- To proceed reversely, press the "PSW3" switch for less than 2 seconds.
- To cancel this checking, press the "PSW2" switch for more than 3 seconds. The display will be changed to the indication one step before. Then, press the "PSW2" switch once again for more than 3 seconds.

| No. | Checking Item | Display | Checking Contents | | (Unit) |
|---|---|---|---|---|---|
| 1 | Total of Thermo-ON Indoor Units Capacity | oP | 26 | 000 ~ 240 | (× $\frac{1}{8}$ HP) |
| 2 | Running Frequency of Inverter Compressor MC1 | H1 | 100 | 000 ~ 115 | (Hz) |
| 3 | Number of Running Compressor | CC | 2 | 0 ~ 2 | |
| 4 | Control Information (For FS2 Series) | rE | 20 | -127 ~ 127 Internal Information of Outdoor Unit's PCB | |
| 5 | Control Information (For FS2 Series) | rEo | 20 | -127 ~ 127 Internal Information of Outdoor Unit's PCB | |
| 6 | Air Flow Ratio | Fo | 16 | 0 ~ 16 (FS2 Series)<br>0 ~ 4 (FX2 Series) | |
| 7 | Outdoor Expansion Valve MV1 Opening | oE1 | 38 | 0 ~ 100 | (%) |

from Previous Page

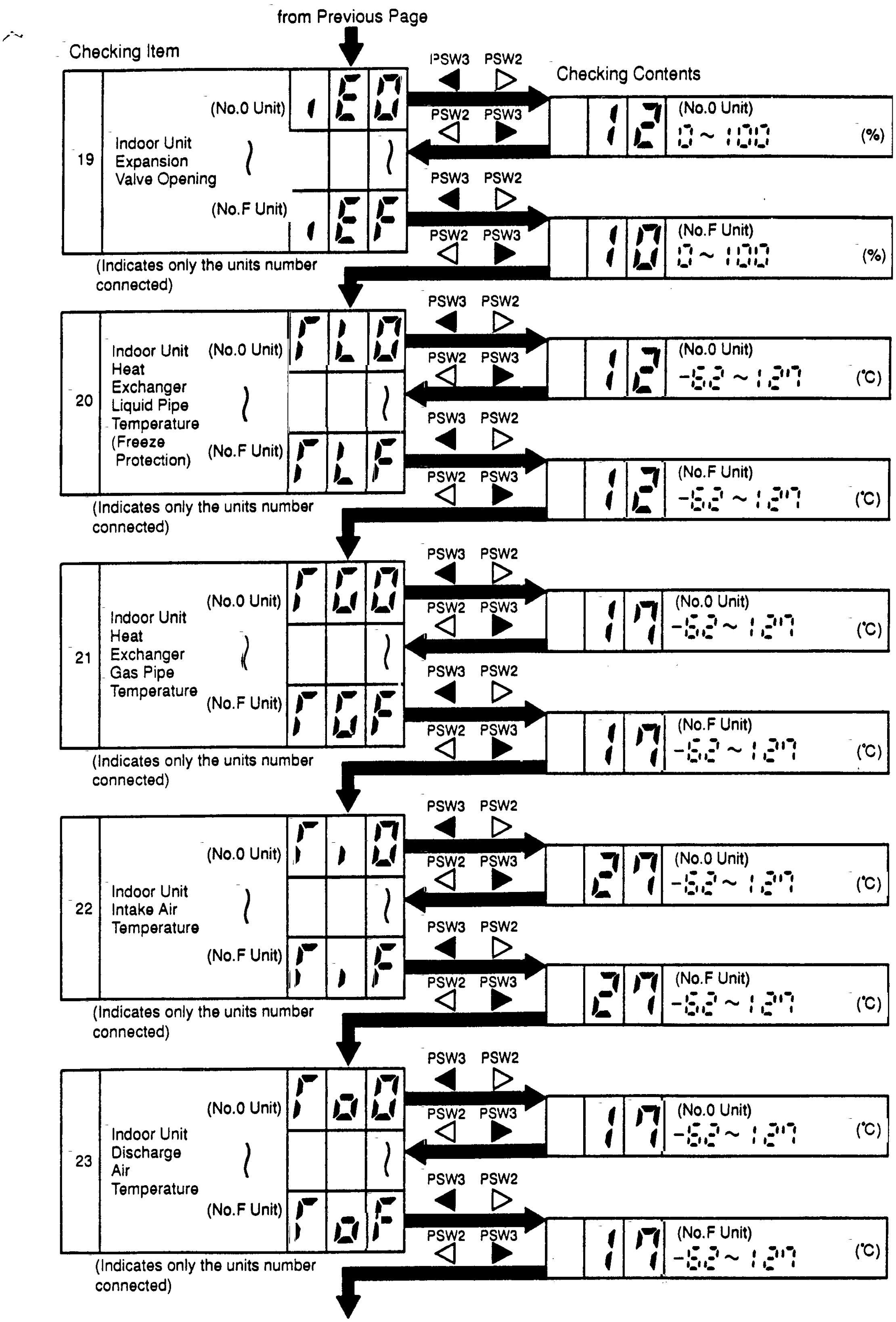
PSW3 ◀ ▷ PSW2 / PSW2 ◁ ▶ PSW3

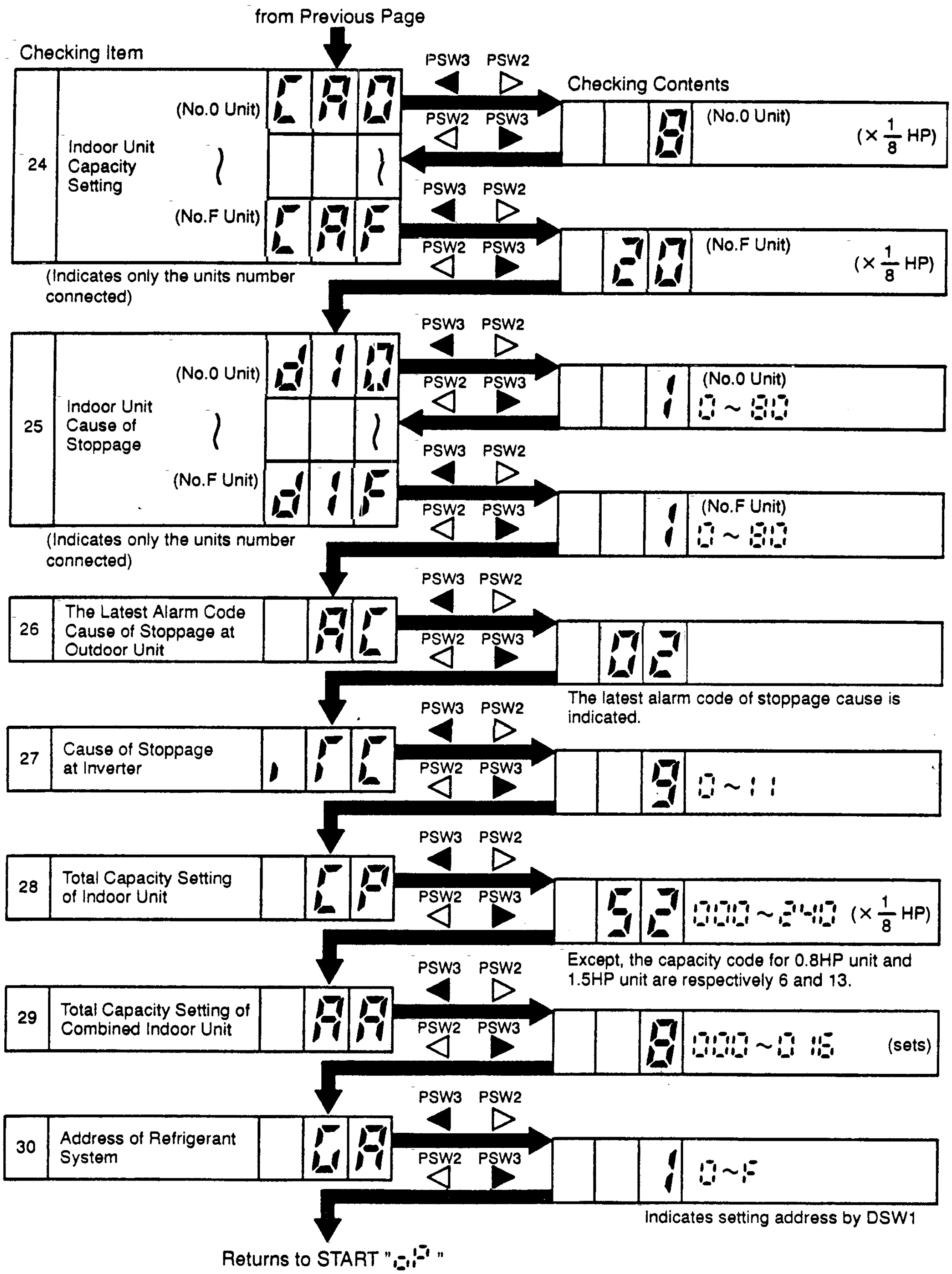
| No. | Checking Item | Display | Checking Contents (example) | Range | Unit |
|---|---|---|---|---|---|
| 8 | Outdoor Expansion Valve MV2 Opening (For FX2 Series) | oE2 | 100 | 0 ~ 100 | (%) |
| 9 | Outdoor Expansion Valve MVB Opening (For FX2 Series) | oEb | 20 | 0 ~ 100 | (%) |
| 10 | Discharge Pressure (High) | Pd | 18 | 000 ~ 030 | (kg/cm$^2$G) |
| 11 | Suction Pressure | PS | 4.0 | -.9 ~ 9.9 | (kg/cm$^2$G) |
| 12 | Discharge Gas Temperature on the Top of Compressor MC1 (TD1) | rd1 | 100 | 1 ~ 142 | (℃) |
| 13 | Discharge Gas Temperature on the Top of Compressor MC2 (TD2) (For 8, 10HP) | rd2 | 90 | 1 ~ 142 | (℃) |
| 14 | Evaporating Temperature 1 at Heating | rE1 | 6 | -42 ~ 88 | (℃) |
| 15 | Evaporating Temperature 2 at Heating | rE2 | 2 | -42 ~ 80 | (℃) |
| 16 | Ambient Air Temperature (TO) | ro | 35 | -42 ~ 80 | (℃) |
| 17 | Estimated Running Current of Compressor MC1 | A1 | 10 | 000 ~ 255 | (A) |
| 18 | Estimated Running Current of Compressor MC2 (For 8, 10HP) | A2 | 15 | 000 ~ 255 | (A) |

For items 12 and 13:
0 : Open-circuited
255 : Short-circuited

For items 14, 15 and 16:
-127 : Open-circuited
127 : Short-circuited

For items 17 and 18: Indicates running current of primary side of inverter

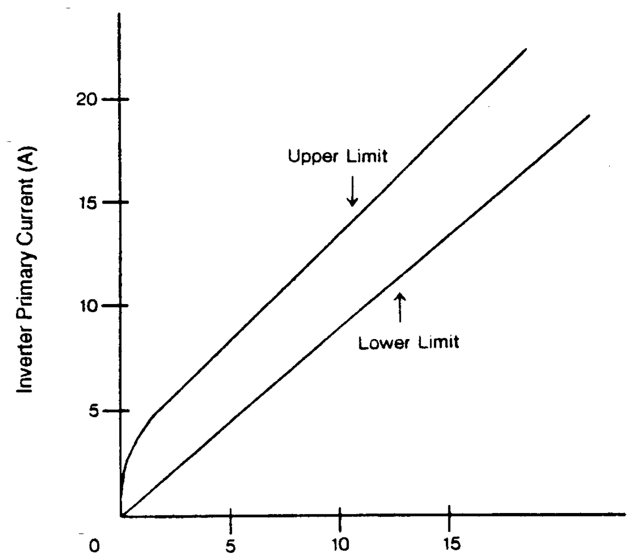

from Previous Page
Checking Item
Checking Contents
PSW3 PSW2
PSW2 PSW3
19
Indoor Unit Expansion Valve Opening
(No.0 Unit)
(No.F Unit)
0 ~ 100 (%)
(Indicates only the units number connected)
20
Indoor Unit Heat Exchanger Liquid Pipe Temperature (Freeze Protection)
(No.0 Unit)
(No.F Unit)
-63 ~ 127 (°C)
(Indicates only the units number connected)
21
Indoor Unit Heat Exchanger Gas Pipe Temperature
(No.0 Unit)
(No.F Unit)
-63 ~ 127 (°C)
(Indicates only the units number connected)
22
Indoor Unit Intake Air Temperature
(No.0 Unit)
(No.F Unit)
-63 ~ 127 (°C)
(Indicates only the units number connected)
23
Indoor Unit Discharge Air Temperature
(No.0 Unit)
(No.F Unit)
-63 ~ 127 (°C)
(Indicates only the units number connected)


from Previous Page
Checking Item
Checking Contents
PSW3 PSW2
PSW2 PSW3
24
Indoor Unit Capacity Setting
(No.0 Unit)
(No.F Unit)
(× 1/8 HP)
(Indicates only the units number connected)
25
Indoor Unit Cause of Stoppage
(No.0 Unit)
(No.F Unit)
(Indicates only the units number connected)
26
The Latest Alarm Code Cause of Stoppage at Outdoor Unit
The latest alarm code of stoppage cause is indicated.
27
Cause of Stoppage at Inverter
28
Total Capacity Setting of Indoor Unit
(× 1/8 HP)
Except, the capacity code for 0.8HP unit and 1.5HP unit are respectively 6 and 13.
29
Total Capacity Setting of Combined Indoor Unit
(sets)
30
Address of Refrigerant System
Indicates setting address by DSW1
Returns to START

(2) Running Current of Compressor

- Inverter Primary Current

The inverter primary current is estimated from the running current of the compressor MC1 which is indicated on 7-segments, as shown in the chart below.




Running Current of Compressor MC1
indicated on 7-segment

- Indicated Running Current of Compressor MC2

The running current of the compressor MC2 is detected by current sensor.

- Cause of Stoppage for Inverter

| Code | Cause | Cause of Stoppage for Corresponding Unit | Remark | |
|---|---|---|---|---|
| | | | Indication during Retry | Alarm Code |
| 1 | Automatical Stoppage of Transistor Module (IPM Error) (Over Current, Decrease Voltage, Increase Temperature) | 17 | P 17 | 53 |
| 2 | Instantaneous Over Current | 17 | P 17 | 52 |
| 4 | Electronic Thermal Activation | 17 | P 17 | 52 |
| 5 | Inverter Voltage Decrease | 18 | P 18 | 06 |
| 6 | Increase Voltage | 18 | P 18 | 06 |
| 8 | Abnormal Current Sensor | 17 | P 17 | 51 |
| 9 | Instantaneous Power Failure Detection | 18 | - | - |
| 11 | Reset of Micro-Computer for Inverter | 18 | - | - |

## TROUBLESHOOTING

● Protection Control Code on 7-Segment Display

a) The protection control indication can be seen on 7-segments when a protection control is activated.

b) The 7-segment continues ON while function is working, and goes out when released.

c) When several protection controls are activated, code number with higher priority will be indicated (see below for the priority order).

① Higher priority is given to protection control related to frequency control than the other.

<Priority Order>

* Pressure Ratio Control
* High-Pressure Rise Protection
* Current Protection
* Discharge Gas Temperature Rise Protection
* Low-Pressure Fall Protection
* Oil Return Control

② In relation to retry control, the latest retrial will be indicated unless a protection control related to frequency control is indicated.

| Code | Protection Control | Code | Protection Control |
|---|---|---|---|
| P01 | Pressure Ratio Control | P12 | Low-Pressure Rising Retry |
| P02 | High-Pressure Rise Protection | P13 | High-Pressure Rising Retry |
| P03 | Current Protection | P14 | Over Current Retry of Constant Compressor |
| P05 | Discharge Gas Temperature Rise Protection | P15 | Vacuum/Discharge Gas Temperature Rising Retry |
| P06 | Low-Pressure Fall Protection | P16 | Discharge Gas SUPERHEAT Falling Retry |
| P07 | Reversing Valve Changeover Control (FX2 Only) | P17 | Inverter Trip Retry |
| P08 | Oil Return Control | P18 | Voltage Falling Retry |
| P11 | Pressure Ratio Falling Retry | | |

○ Retry indication continues for 30 minutes unless a protection control is indicated.

○ Retry indication disappears if the stop signal comes from all rooms.

NOTE:

The protection control code being indicated on 7-segment display is changed to an alarm code when the abnormal operation occurs. Also, the same alarm code is indicated on the remote control switch.

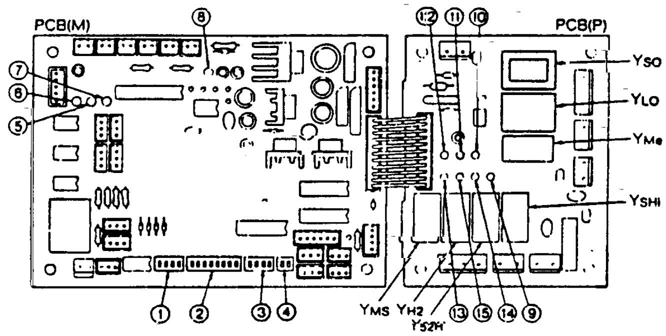
| Code | Protection Control | Activation Condition | Remark |
|---|---|---|---|
| P01 | Pressure Ratio Control | Compression Ratio≧9<br>⇒ Frequency Decrease<br>(Pd/(Ps+1.3))≦2.2<br>⇒ Frequency Increase | Ps: Suction Pressure |
| P02 | High-Pressure Rise Protection | Pd≧23.5kg/cm$^2$G<br>⇒ Frequency Decrease | Pd: Discharge Pressure |
| P03 | Current Protection | Inverter Output Current≧14.5<br>⇒ Frequency Decrease | - |
| P05 | Discharge Gas Temperature Rise Protection | Temperature at the top of compressor is high.<br>⇒ Frequency Decrease<br>(Temp. is different depending on frequency.) | - |
| P06 | Low-Pressure Fall Protection | Low pressure is low.<br>⇒ Frequency Decrease<br>(Pressure is different depending on Ambient Temp.) | - |
| P07 | Reversing Valve Changeover Control (FX2 Only) | When Changing;<br>ΔP < 5kgf/cm$^2$G<br>⇒ Frequency Increase<br>ΔP > 13kgf/cm$^2$G<br>⇒ Frequency Decrease | ΔP = Pd - Ps |
| P08 | Oil Return Control | Frequency less than 40Hz is maintained for one hour.<br>⇒ Frequency≧40Hz | - |
| P11 | Pressure Ratio Falling Retry | Compression Ratio<br>(Pd/(Ps+1.3)<1.8) | 3 times operation in one hour, 43 alarm is indicated. |
| P12 | Low-Pressure Rising Retry | Ps>9kg/cm$^2$G | 3 times operation in one hour, 44 alarm is indicated. |
| P13 | High-Pressure Rising Retry | Pd>27kg/cm$^2$G<br>(In case of 20~30Hz,<br>Pd>25kg/cm$^2$G) | 3 times operation in one hour, 45 alarm is indicated. |
| P14 | Over Current Retry of Constant Compressor | Running Current≧13.1 (8FSA2)<br>Running Current≧16.2 (10FSA2) | 3 times operation in 30 minutes, 39 alarm is indicated. |
| P15 | Vacuum/Discharge Gas Temperature Rising Retry | a.Ps<0.2kg/cm$^2$G is more than 12 minutes.<br>b.Discharge Gas Temp.≧132℃ is more than 10 minutes.<br>c.Discharge Gas Temp.≧140℃ is more than 5 seconds. | 3 times operation in one hour, 47 or 08 alarm is indicated. |
| P16 | Discharge Gas SUPERHEAT Falling Retry | Discharge Gas SUPERHEAT less than 10℃ is maintained for one hour. | 2 times operation in two hours, 07 alarm is indicated. |
| P17 | Inverter Trip Retry | Automatical Stoppage of Transistor Module, Electronic Thermal Activate or Abnormal Current Sensor | 3 times operation in 30 minutes, 51, 52 or 53 alarm is indicated. |
| P18 | Voltage Falling Retry | Voltage Falling or Rising at Circuit of Inverter, Capacitor (CB) | 3 times operation in 30 minutes, 06 alarm is indicated. |

1.2.5 Function of DSWs and LEDs

(1) Printed Circuit Board in Indoor Unit

( * Following figure shows a separated-board type PCB.)

PCB(M): Controlling Board PCB(P): Power Supply Board



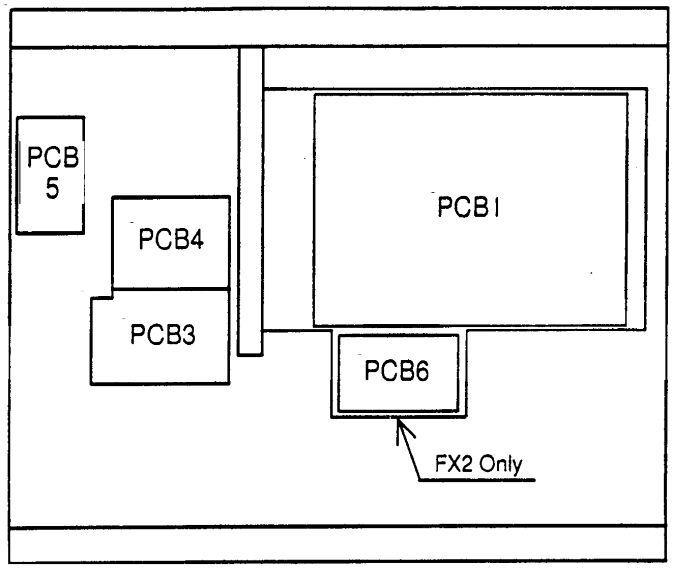

* Some unit has an one-board type PCB which has the same functions and devices with the separated-board type.

■ Dip Switch and LED Functions on Indoor Unit Printed Circuit Board

| Name of Printed Circuit Board | Part Name | Contents of Functions |
|---|---|---|
| Controlling Board: PCB(M) | ① DSW1 | Setting of Indoor Unit Number |
| | ② DSW2 | Setting of Optional Functions<br>A. Self-Diagenosis<br>B. Remote ON/OFF Control<br>C. Automatic Restart after Short Power Failure<br>D. Remote Sensor Control<br>E. Setting of Filter Indication Interval |
| | ③ DSW3 | Setting of Indoor Unit Capacity Code |
| | ④ DSW4 | Setting of Capacity Adjustment |
| | ⑤ LED1 (Red) | This LED1 indicates the transmission state between the indoor unit and remote control switch.<br>Normal Condition: Flickering<br>Abnormal Condition: Activated or Deactivated |
| | ⑥ LED2 (Yellow) | This LED2 indicates the transmission state between the indoor unit and outdoor unit.<br>Normal Condition: Flickering<br>Abnormal Condition: Activated or Deactivated |
| | ⑦ LED3 (Green) | This LED3 indicates the transmission state between the indoor unit and central station.<br>When Disconnected:<br>Normal Condition: Deactivated<br>Abnormal Condition: Activated<br>When Connected:<br>Normal Condition: Flickering<br>Abnormal Condition: Activated or Deactivated |
| | ⑧ LED4 (Green) | This LED4 indicates the power supply (5V) for micro-computer.<br>Normal Condition: Activated<br>Abnormal Condition: Deactivated |
| Power Supply Board: PCB(P) | ⑨ LD1 | Relay YSHi ON: Activated |
| | ⑩ LD2 | Relay YMe ON: Activated |
| | ⑪ LD3 | Relay YLo ON: Activated |
| | ⑫ LD4 | Relay YSo ON: Activated |
| | ⑬ LD5 | Relay YMs ON: Activated |
| | ⑭ LD6 | Relay Y52H ON: Activated |
| | ⑮ LD7 | Relay YH2 ON: Activated |

(2) Printed Circuit Board in Outdoor Unit

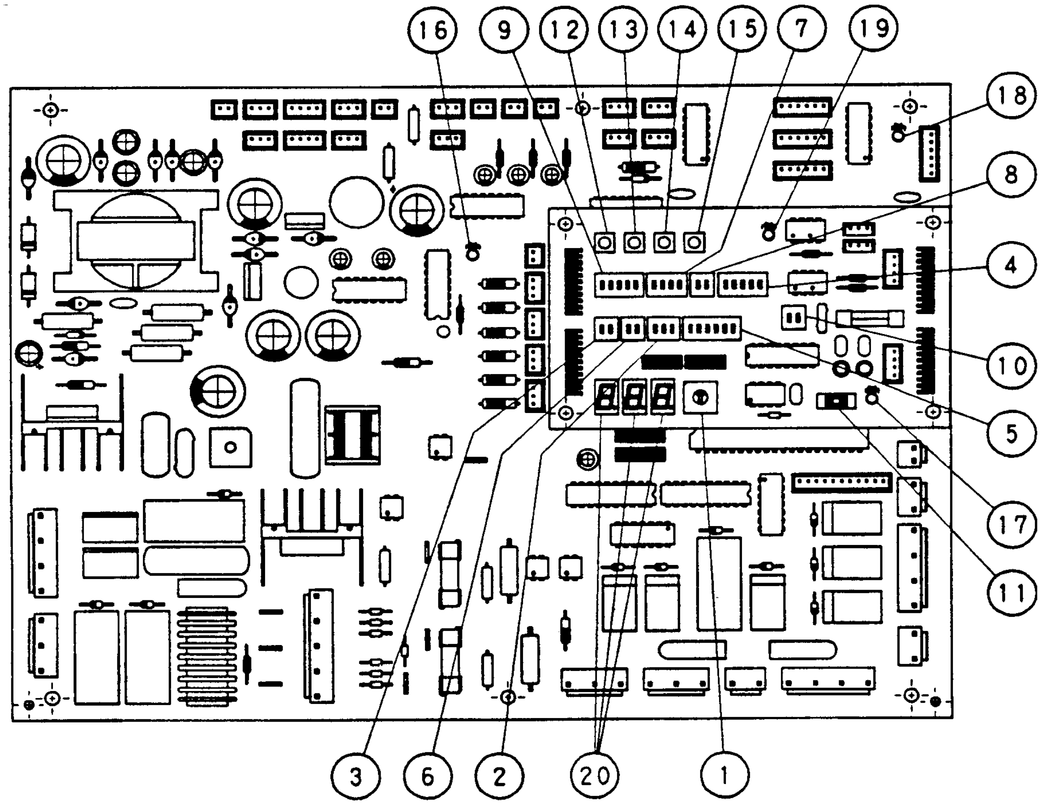
●Arrangement




●Purpose

| Symbol | PCB | Purpose |
|---|---|---|
| PCB1 | for Control | 1. Transmitting between Indoor Unit and Outdoor Unit<br>2. Processing for Sensor Input<br>3. Processing for Dip Switch Input<br>4. Operation Control for Above items 1 to 3.<br>Compressor Operation Control, Bypass Valve Control, Fan Control and Overcurrent Control<br>5. 7-Segment Indication<br>6. Processing of Safety Device Input<br>7. Processing of Relay Output<br>8. Reverse Phase Detection for Power Source |
| PCB3<br>PCB4 | for Inverter | 1. Inverter power part is driven by instruction of PCB1 and compressor is driven.<br>2. Overcurrent Control<br>3. Protection Control for Inverter Part |
| PCB5 | for "SNUBBER" | 1. Restraining of Surge Voltage added to Transistor Module (IPM)<br>2. Absorbing for Switching Noise of Transistor Module (IPM) |
| PCB6 | Sub-PCB<br>(FX2 Only) | For Control of 4-Way Valve and Solenoid Valve |

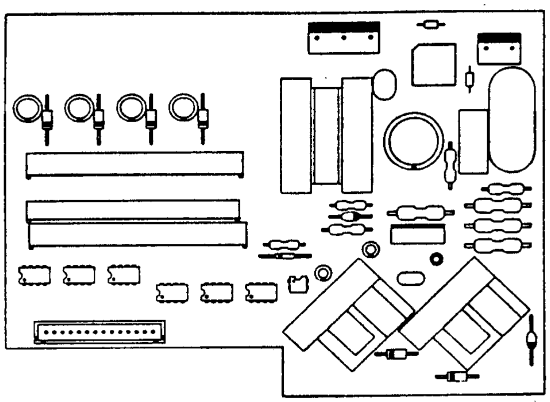
a. Control Printed Circuit Board: PCB1



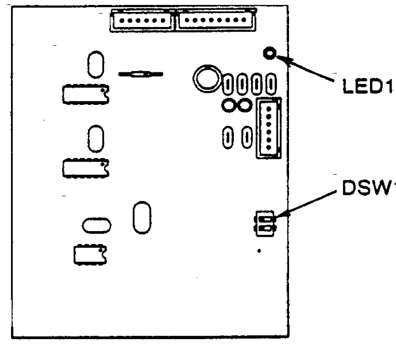
■ Dip Switches and LED Functions on Outdoor Unit Printed Circuit Boards

| Name of Internal Circuit Board | Part Name | Contents of Functions |
|---|---|---|
| Controlling Board: PCB1 | ① RSW | Setting of Outdoor Unit Number |
| | ② DSW2 | Setting of Capacity Code<br>Outdoor unit capacity is set according to nominal capacity (HP). |
| | ③ DSW3 | Setting of Height Difference<br>The height difference between outdoor and indoor units is set. |
| | ④ DSW4 | A. Test Running for Cooling or Heating<br>An outdoor unit can be run for testing. When testing has been finished, reset the function.<br>B. Forced Stoppage of Compressor<br>When performing test running or inspection, compressors can be forcedly stopped to ensure safety. |
| | ⑤ DSW5 (Optional Function) | A. Changeover of Defrosting Condition<br>The defrosting operation for normal areas or cold areas can be changed over. |
| | ⑥ DSW6 | Setting of Piping Length<br>The total piping length between the outdoor unit and indoor unit is set. |
| | ⑦ DSW7 | Setting of Emergency Operation |
| | ⑧ DSW8 | Test Operation and Service Setting II<br>No setting is required. |
| | ⑨ DSW9<br>⑪ SW1 | Transmission Setting<br>The transmission form of indoor units is set. |
| | ⑩ DSW10 | No specific Meaning |
| | ⑫ PSW1 | Manual Defrosting Switch<br>The defrosting operation is manually available under the forced defrosting area. |
| | ⑬ PSW2<br>⑭ PSW3 | Check Switches<br>When checking units, checking items can be selected by these switches. |
| | ⑮ PSW4 | No Specific Meaning |
| | ⑯ LED1 (Red) | Power Source for PCB1<br>Normal Condition: Deactivated<br>Abnormal Condition: Activated |
| | ⑰ LED2 (Green) | This LED2 indicates the transmission state between the PCB1 and PCB3 & 4.<br>Normal Condition: Flickering<br>Abnormal Condition: Activated or Deactivated |
| | ⑱ LED3 (Yellow) | This LED3 indicates the transmission state between the indoor unit and outdoor unit.<br>Normal Condition: Flickering<br>Abnormal Condition: Activated or Deactivated |
| | ⑲ LED4 | In case of Central Station II is connected, LED4 is flickering. |
| | ⑳ SGE1<br>SGE2<br>SGE3 | These indicate the following: "alarm", "protective safety device has tripped" or "checking items". |

b. Inverter Printed Circuit Board: PCB3 and PCB4

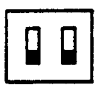


PCB3



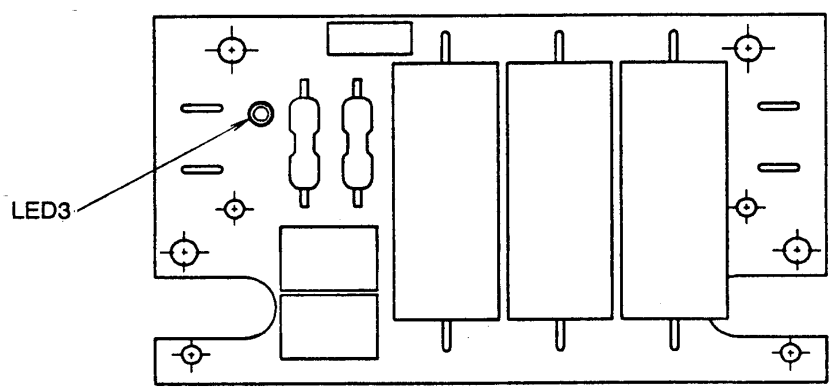

PCB4

* Dip Switch and LED Functions on Outdoor Unit Inverter Printed Circuit Board

| Switch Name<br>DSW1 | Function |
|---|---|
| ON  OFF | Normal: No.1 and No.2 of DSW1 are OFF.<br>Emergency Operation or Troubleshooting:<br>No.1 of DSW1 is ON<br>No.2 of DSW1 is OFF<br>The above setting aims not to trip even if the CT detects 0A. |

| Name of Printed Circuit Board | Function |
|---|---|
| LED1 (Red) | This indicates the state of transmission.<br>Flickering: Normal Transmission<br>Activated or Deactivated: Abnormality in Transmission Circuit |

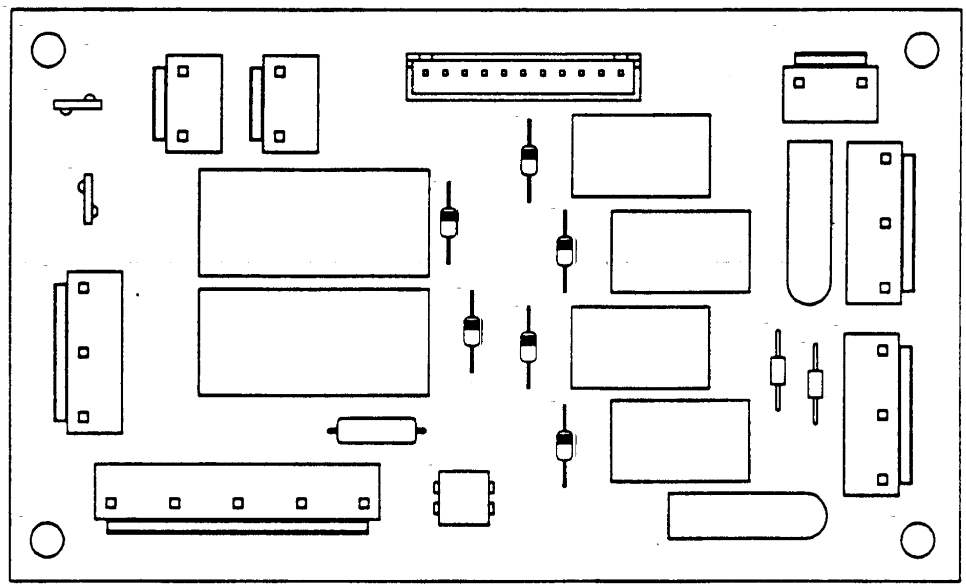
c. "SNUBBER" Board: PCB5




* LED Function on Outdoor Unit Printed Circuit Board

| Name of Printed Circuit Board | Function |
|---|---|
| LED3 (Red) | This indicates the voltage between both terminal of capacitor CB1 and CB2 for inverter part.<br>Activated: The voltage between both terminals of capacitor, CB is 50V$\pm$20V or greater.<br>Deactivated: The voltage between both terminals of capacitor, CB is 50V$\pm$20V or smaller. |

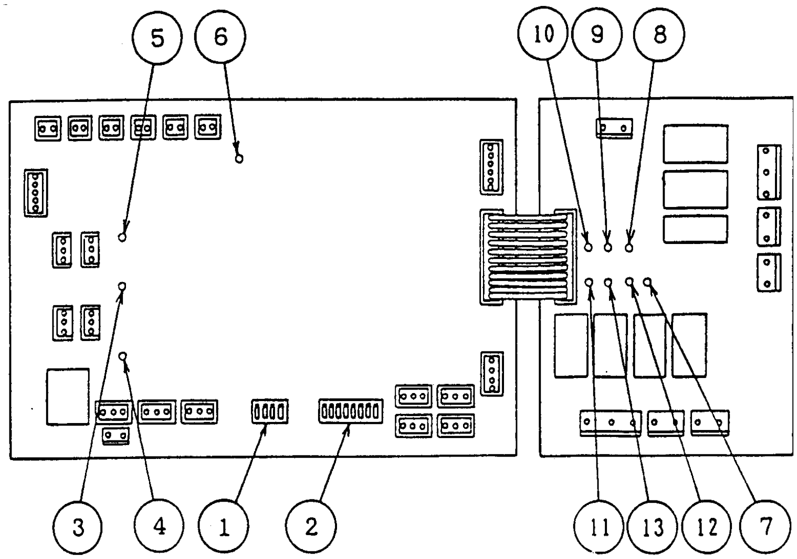
d Sub-PCB: PCB6 (FX2 Only)



(3) Printed Circuit Board in CH Unit

PCB2: Controlling Board

PCB1: Power Supply Board



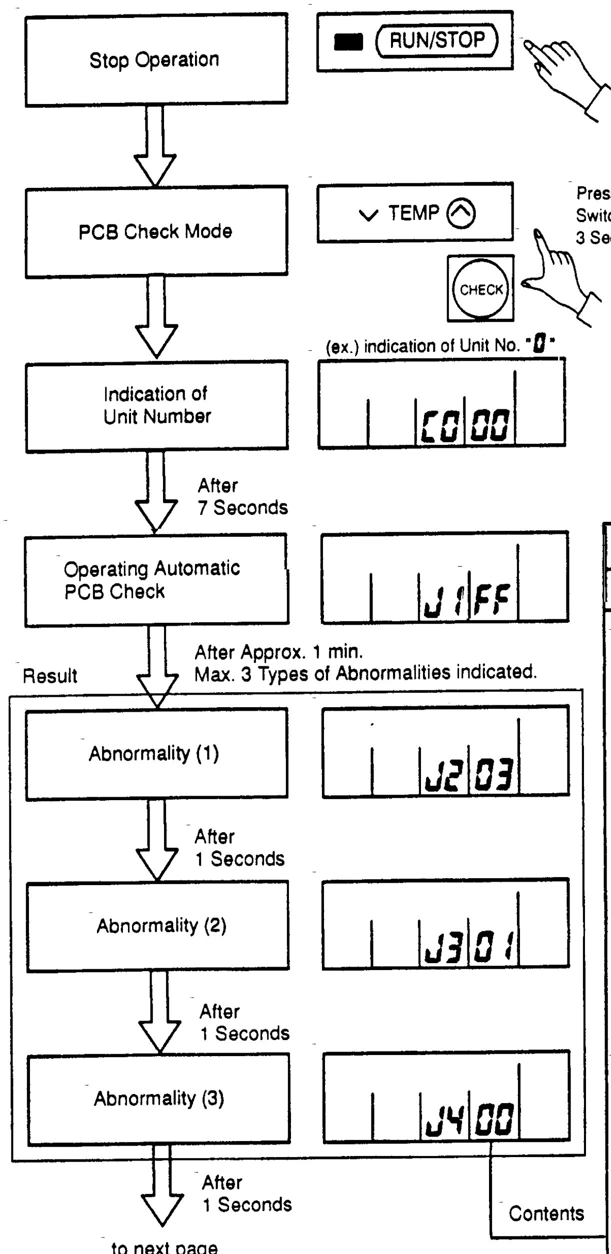

■ Dip Switches and LED Functions on CH Unit Printed Circuit Boards

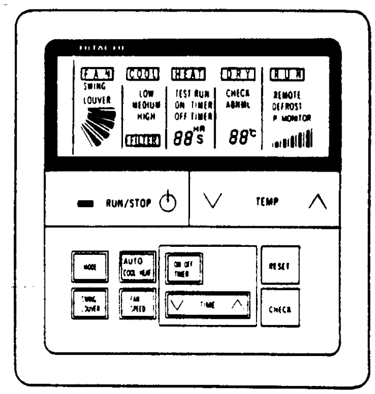
| Name of Printed Circuit Board | Part Name | Contents of Functions |
|---|---|---|
| Controlling Board: PCB2 (Low Voltage) | ① DSW1 | Performing of Self-checking Function |
| | ② DSW2 | Setting of Self-checking Mode<br>When only switch #1 is OFF, Self-checking of PCB is available.<br>ON / OFF 1 2 3 4 5 6 7 8 |
| | ③ LED1 (Red) | This LED1 indicates the transmission state between the indoor unit and CH unit.<br>Normal Condition: Flickering<br>Abnormal Condition: Activated or Deactivated |
| | ④ LED2 (Yellow) | This LED2 indicates the transmission state between the outdoor unit and CH unit.<br>Normal Condition: Flickering<br>Abnormal Condition: Activated or Deactivated |
| | ⑤ LED3 (Green) | This LED3 is not applied.<br>Normal Condition: Deactivated<br>Abnormal Condition: Activated |
| | ⑥ LED4 (Green) | This LED4 indicates the power supply (5V) for micro-computer.<br>Normal Condition: Activated<br>Abnormal Condition: Deactivated |
| Power Supply Board: PCB1 (High Voltage) | ⑦ LD1 | Solenoid Valve SVS2 ON (Relay YSHi): Activated |
| | ⑧ LD2 | Solenoid Valve Changeover Operation ON: Deactivated<br>Abnormality of Power Supply to PCB: Deactivated (Alarm Code "03") |
| | ⑨ LD3 | Transmission State Between Indoor Unit and CH Unit<br>Abnormal: Deactivated (Alarm Code "03") |
| | ⑩ LD4 | Transmission State Between Outdoor Unit and CH Unit<br>Abnormal: Deactivated (Alarm Code "03") |
| | ⑪ LD5 | Solenoid Valve SVS1 ON (Relay YMS): Activated |
| | ⑫ LD6 | Solenoid Valve SVD2 ON (Relay Y52H): Activated |
| | ⑬ LD7 | Solenoid Valve SVD1 ON (Relay YH2): Activated |

## 1.3 Procedure of Checking Each Main Parts

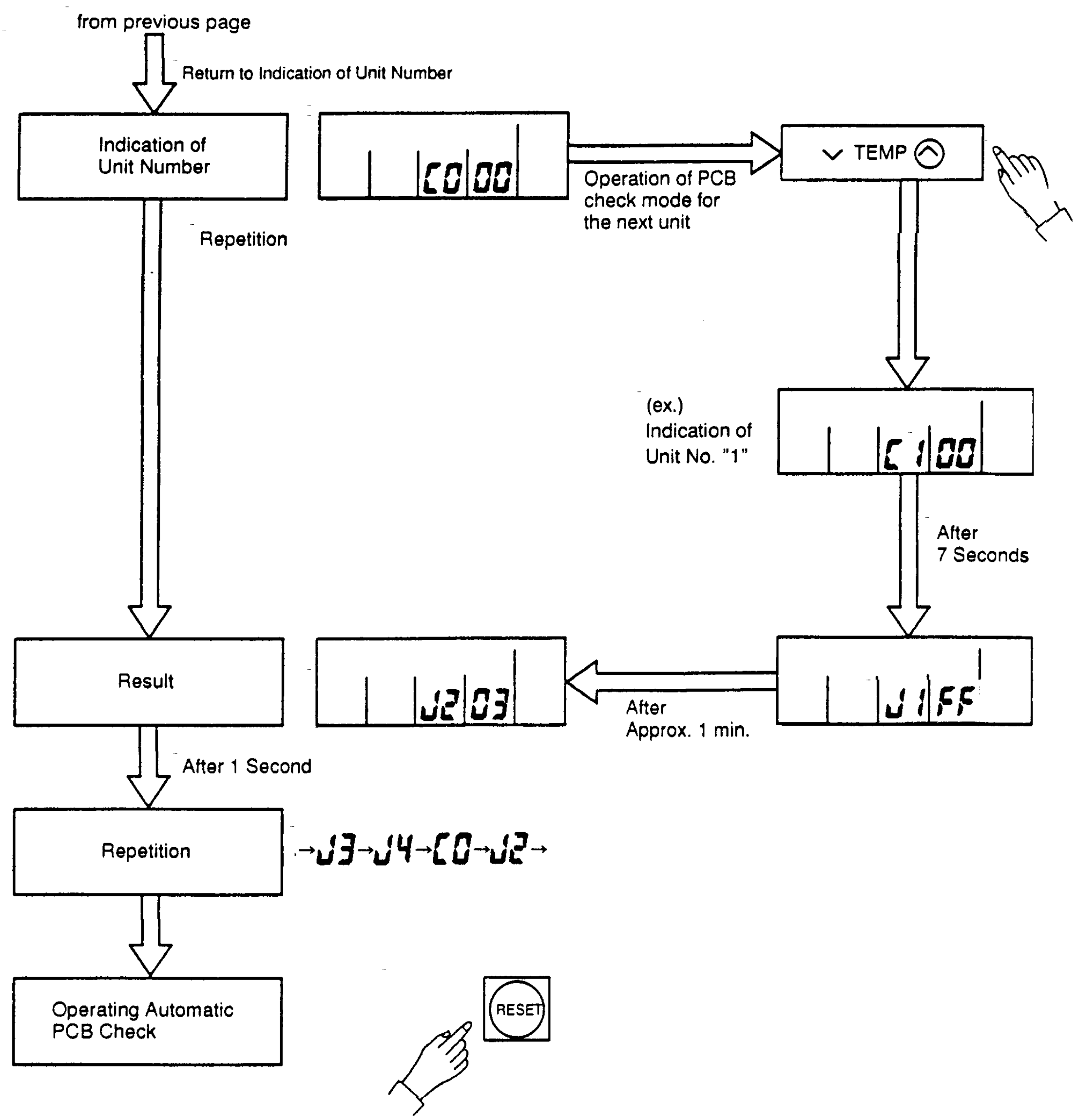
### 1.3.1 Self-Checking of PCBs using Remote Control Switch

The following troubleshooting procedure is utilized for function test of PCBs in the indoor unit and outdoor unit.




| Indication | Contents | |
|---|---|---|
| 00 | Normal | |
| | Abnormality (Open-circuit, Short-circuit, etc.) in circuit for | |
| 01 | · Intake Air Temp. Thermistor | Indoor Unit PCB |
| 02 | · Discharge Air Temp. Thermistor | Indoor Unit PCB |
| 03 | · Liquid Pipe Temp. Thermistor | Indoor Unit PCB |
| 04 | · Remote Thermistor Abnormality | Indoor Unit PCB |
| 05 | · Gas Pipe Temp. Thermistor | Indoor Unit PCB |
| 06 | · Remote Sensor | Indoor Unit PCB |
| 08 | · Transmission of Central Station | Indoor Unit PCB |
| 09 | · Louver Sensor | Indoor Unit PCB |
| 0A | · EEPROM | Indoor Unit PCB |
| 0b | · Zero Cross Input | |
| EE | · Transmission between Indoor Unit and Remote Control Switch During This Checking Operation | |
| 07 | · Transmission of Outdoor Unit | Outdoor Unit PCB |
| F4 | · Outdoor Fan Internal Thermistor | Outdoor Unit PCB |
| F5 | · High Pressure Switch | Outdoor Unit PCB |
| F6 | · Protection Signal Detection Circuit Phase Detection | Outdoor Unit PCB |
| F7 | · Transmission of Inverter | Outdoor Unit PCB |
| F8 | · High Pressure Sensor | Outdoor Unit PCB |
| F9 | · Comp. Discharge Gas Temp. Thermistor | Outdoor Unit PCB |
| Fb | · Low Pressure Sensor | Outdoor Unit PCB |
| FC | · Heat Exchanger Evaporation Temp. Thermistor | Outdoor Unit PCB |
| Fd | · Ambient Air Temp. Thermistor | Outdoor Unit PCB |
| FF | | |




NOTE:

(1) C0 00 — If this indication is continued and "J1 FF" is not shown, this indicates that each one of indoor units is not connected to the remote control switch. Check the wiring between the remote control switch and indoor unit.

(2) In this troubleshooting procedure, checking of the following part of the PCB's is not available.
PCB in Indoor Unit: Relay Circuit, Dip Switch, Option Circuit
PCB in Outdoor Unit: Relay Circuit, Dip Switch, Option Circuit

(3) If no result is given using this troubleshooting procedure (J1 FF), there is a possibility of abnormal transmission of the central station (short-circuit: TL-TG).

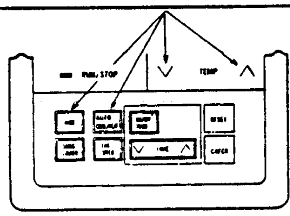
(4) In the case that this troubleshooting is performed in the system using the central station, indication of the central station may change during this procedure. However, this is not abnormal.

3.2 Self-Checking of Remote Control Switch

Cases where CHECK switch is utilized.
(1) If the remote control switch reads out malfunction
(2) For regular maintenance check

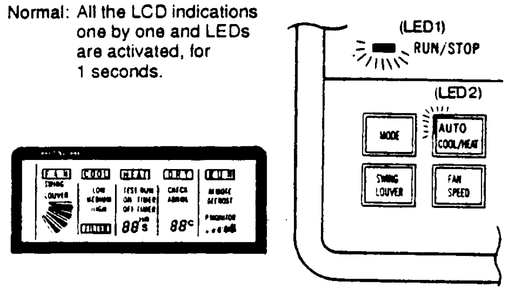
NOTES:
1. AUTO COOL/HEAT Switch is available for FX2 system only.
2. SWING LOUVER Switch is available for indoor unit with swing louver only.

| Step | Action |
|---|---|
| 1 | Turn ON the power source. |
| 2 | Simultaneously depress the following 4 switches. (During Operation, they can be depressed. |



Checking of LCD, LED and Indicating Circuit

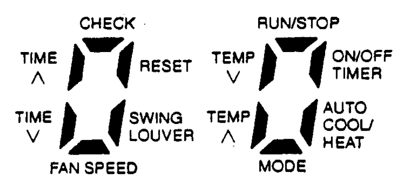
Normal: All the LCD indications one by one and LEDs are activated, for 1 seconds.




| Step | Action |
|---|---|
| 3 | Depress all the touch type switches one by one. Any one of the switches can be depressed. |

Checking of touch type switch function
By depressing a particular touch type switch, the corresponding segment is activated.

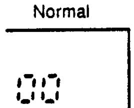
Temperature Indicated.




Depress the switch corresponding at "AUTO HEAT/COOL".

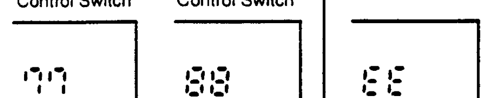
After 15 seconds

Normal

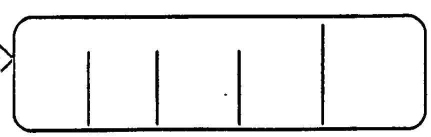



| Step | Action |
|---|---|
| 4 | The transmission circuit is checked automatically. |

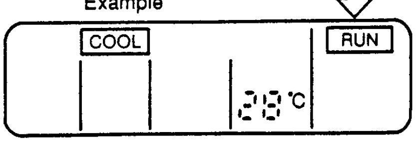
| Normal | | Abnormal |
|---|---|---|
| Main Remote Control Switch | Sub Remote Control Switch | |




| Step | Action |
|---|---|
| 5 | Depress the Reset switch for initialization. |
| | The indications return after 15 seconds, when the unit is operating. |



Example



### 1.3.3 Self-Checking of Indoor Unit PCB

(1) Self-Checking by Relays and LED on Indoor Unit PCB

- To Check Abnormality on Indoor Unit PCB due to Malfunction

- To check Abnormality on Indoor Unit PCB Based on Results of Checking by CHECK switch on the remote control switch and Self-Checking Function

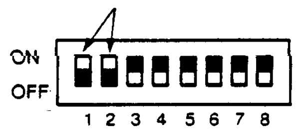
■Procedure

Refer to page 1-74 for location and function of DSW and LED.

① Turn OFF the main power switch.

② Disconnect connectors CN12 and CN2, and set dip switch DSW2 as shown below.

Set #1 and #2 at OFF side.




Dip Switch DSW2

NOTE: Before turning ON the power, see checking procedure mentioned in next page.

③ Turn ON the main power switch.
Check Mode starts. (See next page.)

Ⓐ Analog Test
Ⓑ Relay Test
Ⓒ LED Test

④ After completion of self-checking, turn OFF the power and reset the dip switch as before

## 1.3.4 Self-Checking of CH Unit PCB

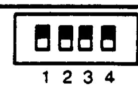
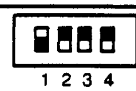
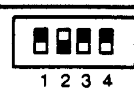
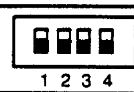
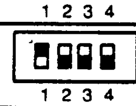
Troubleshoot CH unit PCB if abnormalities are found

(1) Self-Checking Procedures

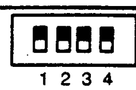
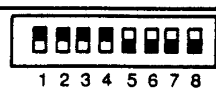
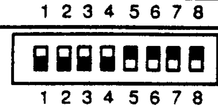
Refer to page 1-80 for location and function of DSW and LED.

① Turn OFF the main power switch.
② Set switch only #1 of DSW2 at OFF position.
③ Turn ON the main power switch. Check to ensure that LED4 is ON.
④ Start the test according to next item (2).
⑤ End the test.
⑥ Turn OFF the main power switch.
⑦ Reset the dip switches as before.

(2) Contents of Test

| Test Name | PSW1 Setting | Contents of Test |
|---|---|---|
| ① DSW 「H」 Test | ON OFF  | Checking on ON State of DSW |
| ② Relay Test | ON OFF  | Checking on ON/OFF Function of Relays on PCB1 |
| ③ Transmission Test | ON OFF  | Checking on Transmission Circuit to Outdoor Unit and to Indoor Unit |
| ④ DSW 「L」 Test | ON OFF  | Checking on OFF State of DSW |
| ⑤ Reset Test | ON OFF  | Resetting of Micro-computer |

① DSW 「H」 Test

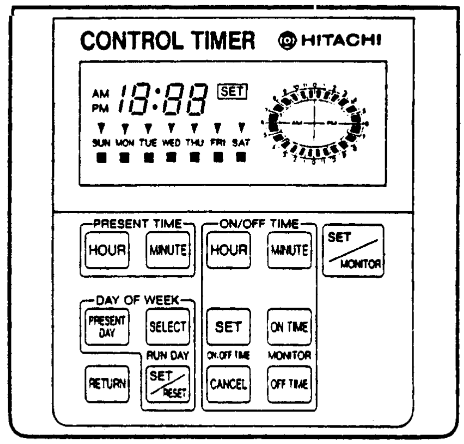
| DSW | Setting | Checking Item |
|---|---|---|
| DSW1 | ON OFF  | Switch #1 to #4 ON State<br>Normal: LD4 on PCB1 Activated |
| DSW2 | ON OFF  | Switch #1 to #4 ON State<br>Normal: LD3 on PCB1 Activated |
| DSW2 | ON OFF  | Switch #5 to #8 ON State<br>Normal: LD2 on PCB1 Activated |

1.3.5 Self-Checking of 7-Day Timer (PSC-3T)

This procedure is utilized for regular maintenance check, and if the 7-day timer is malfunction.

① Turn ON the main power switch.

② Checking of LCD and Indicating Circuit:
Simultaneously depress the following 4 switches, [HOUR] of "PRESENT TIME" [PRESENT DAY] of "DAY OF WEEK" [ON TIME] [OFF TIME] of " MONITOR".
During operation, they can be depressed.




All the LCD indications activated, for 5 seconds.

All the LCD indications deactivated, for 1 second.

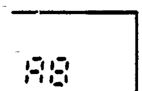
NORMAL:

All the LCD indications activated, for 5 seconds

All the LCD indications deactivated, for 1 second

③ Checking of Touch Type Switch Function:
Depress all the touch type switches one by one
Any one of the switches can be depressed.

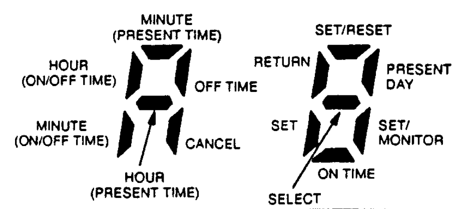
NORMAL:



After 15 Seconds

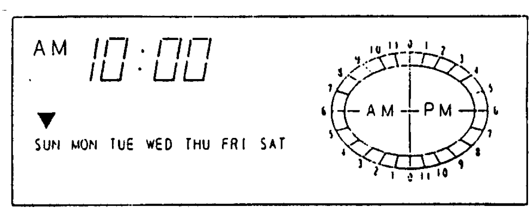
By depressing a particular touch type switch, the corresponding segment is activated.

Indication




④ Completion of self-checking, "77" is indicated, for 3 seconds.

⑤

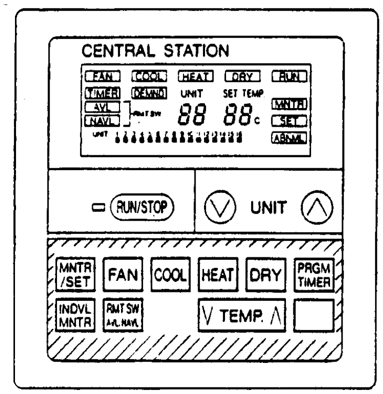


### 1.3.6 Self-Checking of Central Station (PSC-3S1)

This procedure is utilized for regular maintenance check, and if the central station is malfunction.

① Turn ON the main power switch.

② Checking of LCD and Indicating Circuit:
Simultaneously depress the following 4 switches, 〔FAN〕 〔COOL〕 and 〔▽〕 〔△〕 of "UNIT"
During operation, they can be depressed.




NORMAL:

All the LCD indications activated, for 5 seconds.

All the LCD indications deactivated, for 1 second.

All the LCD indications activated, for 5 seconds

All the LCD indications deactivated, for 1 second

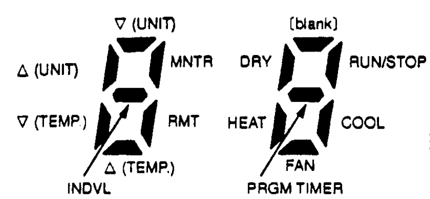
③ Checking of Touch Type Switch Function:
Depress all the touch type switches one by one.
Any one of the switches can be depressed.

NORMAL:

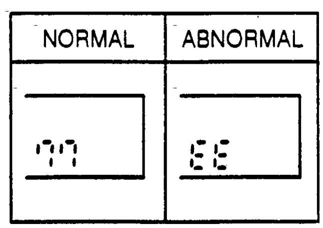
After 15 Seconds

By depressing a particular touch type switch, the corresponding segment is activated.

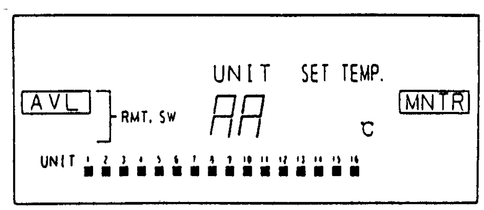
Temp. Indication Part




④ Checking of Transmission Circuit:
Automatically checked, and the result will be indicated as figure right, for 3 seconds.




⑤ Return to the indication before the self-checking.



.3.7 Procedure of Checking Other Main Parts

(1) High Voltage Discharge Work for Replacing Parts

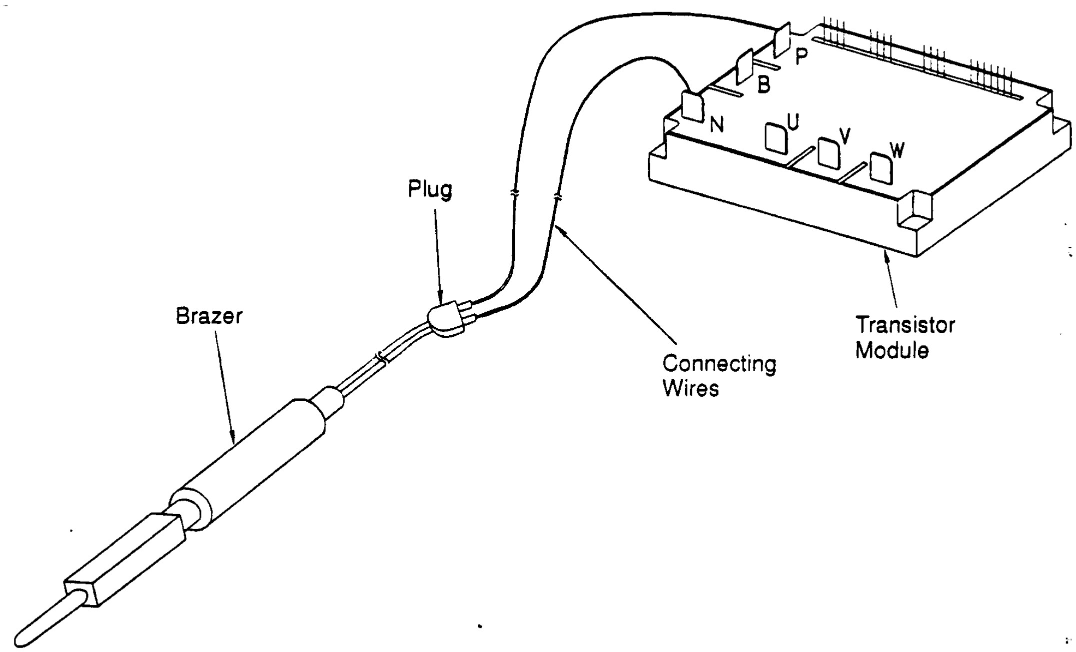
[Perform this high voltage discharge work to avoid an electric shock

Procedure

(a) Turn OFF the main switches and wait for three minutes. Check to ensure that no high voltage exists. If LED3 is ON after start-up and LED3 is OFF after turning OFF power source, the voltage will decrease lower than DC50V.

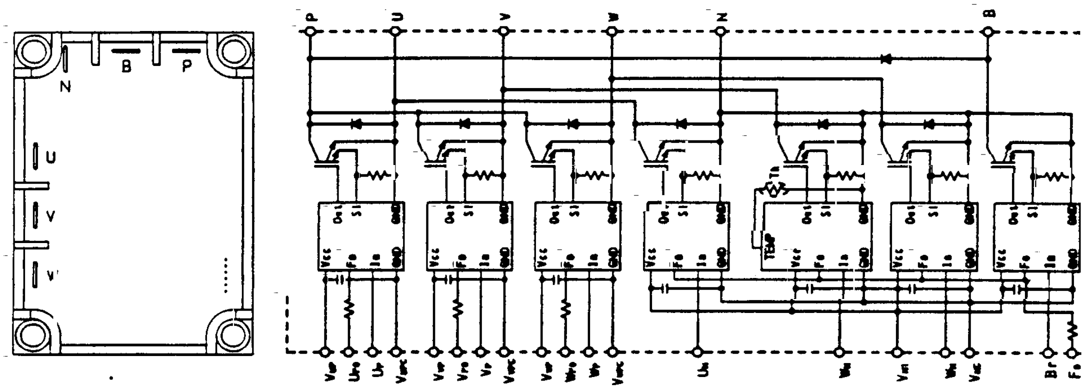
(b) Connect connecting wires to an electrical brazer.

(c) Connect the wires to terminals, P and N on transistor module. →Discharging is started, resulting in hot brazer. Pay attention not to short-circuit between terminal P and N .



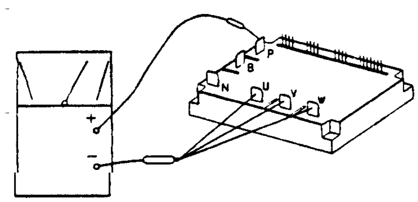

(d) Wait for 2 or 3 minutes and measure the voltage once again. Check to ensure that no voltage is charged.

(2) Checking Procedures Transistor Module
Outer Appearance and Internal Circuit of Transistor Module

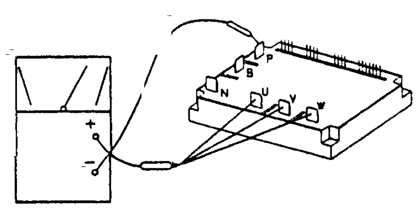


Remove all the terminals of the transistor module before check.
If items (a) to (e) are performed and the results are satisfactory, the transistor module is normal
Measure it under 1 kΩ range of a tester. Do not use a digital tester.

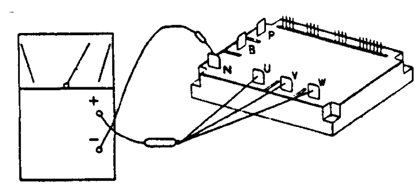
(a) By touching the + side of the tester to the P terminal of transistor module and the − side of tester to U, V and W of transistor module, measure the resistance. If all the resistances are from 1 to 5 kΩ, it is normal.



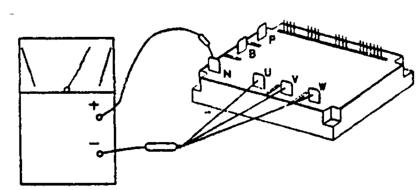
(b) By touching the − side of the tester to the P terminal of transistor module and the + side of tester to U, V and W of transistor module, measure the resistance. If all the resistances are greater than 100 kΩ, it is normal.



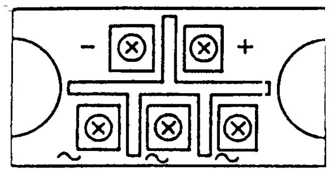
(c) By touching the − side of the tester to the N terminal of transistor module and the + side of tester to U, V and W of transistor module, measure the resistance. If all the resistances are from 1 to 5 kΩ, it is normal.

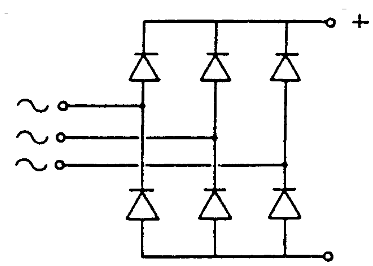


(d) By touching the + side of the tester to the N terminal of transistor module and the − side of tester to U, V and W of transistor module, measure the resistance. If all the resistances are greater than 100 kΩ, it is normal.



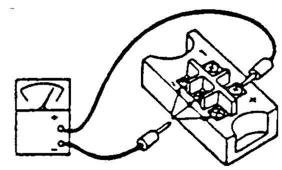
(3) Checking Procedures on Diode Module
Outer Appearance and Internal Circuit of Diode Module



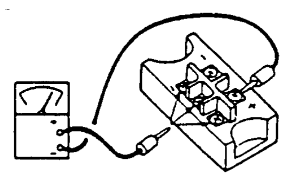




If items (a) to (d) are performed and the results are satisfactory, the diode module is normal
Measure it under 1 kΩ range of a tester. Do not use a digital tester.

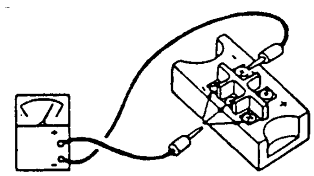
(a) By touching the + side of the tester to the + terminal of the diode module and the – side of tester to the ~ terminals (3 NOs.) of the diode module, measure the resistance. If all the resistances are from 5 to 50 kΩ, it is normal.



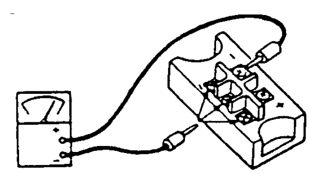
(b) By touching the – side of the tester to the + terminal of the diode module and the + side of tester to the ~ terminals(3 Nos.) of the diode module, measure the resistance. If all the resistances are greater than 500 kΩ, it is normal.



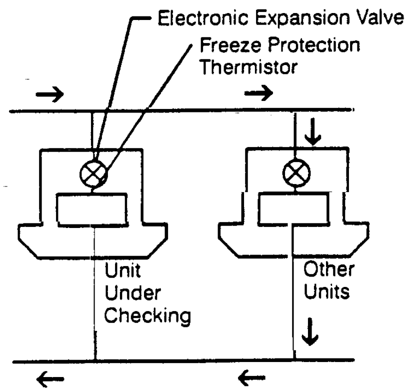
(c) By touching the – side of the tester to the – terminal of the diode module and the + side of tester to the ~ terminals(3 Nos.) of the diode module, measure the resistance. If all the resistances are from 5 to 50 kΩ, it is normal.



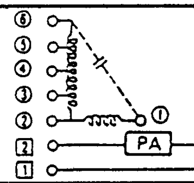
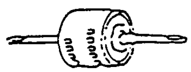
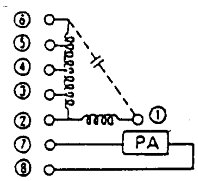
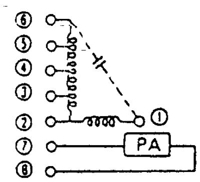
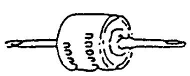
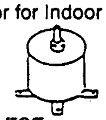
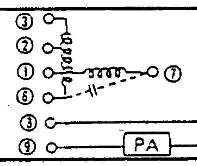
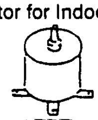
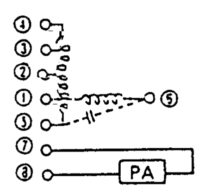
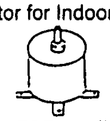
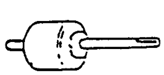
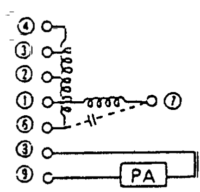
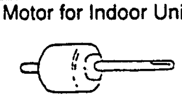
(d) By touching the + side of the tester to the – terminal of the diode module and the – side of tester to the ~ terminals(3 Nos.) of the diode module, measure the resistance. If all the resistances are greater than 500 kΩ, it is normal.



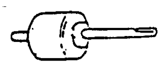
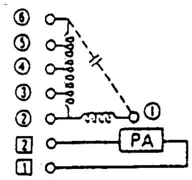
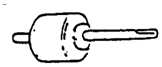
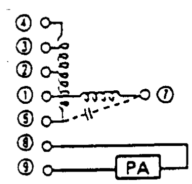
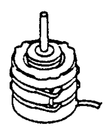
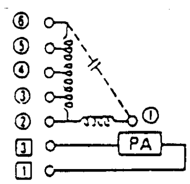
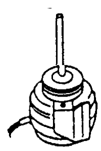
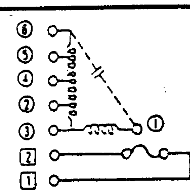
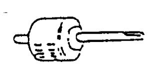
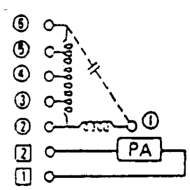
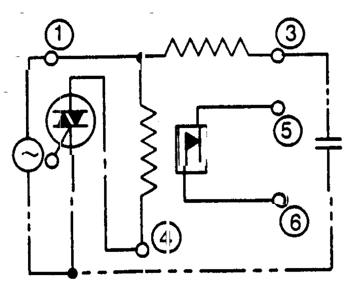
(4) Checking Method of Electronic Expansion Valve

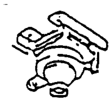
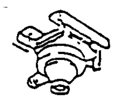
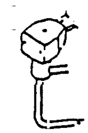
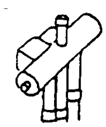
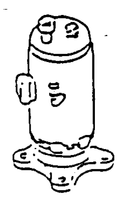
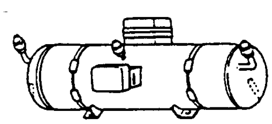
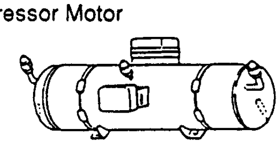
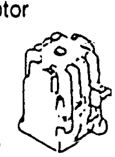
| | Indoor Unit Electronic Expansion Valve | Outdoor Unit Electronic Expansion Valve |
|---|---|---|
| Locked with Fully Open | Check for the liquid pipe temperature during heating operation.<br>It is abnormal if the temperature does not increase. | It is abnormal if the liquid pipe pressure does not increase during cooling operation. |
| Locked with Slightly Open | It is abnormal under the following conditions; The temperature of freeze protection thermistor becomes lower than the suction air temperature when the unit under checking is stopped and other units are under cooling operation. | It is abnormal if the liquid pipe pressure does not increase and the outlet temperature of the expansion valve decreases after the cooling operation is started. |
| Locked with Fully Closed | <br> | It is abnormal under the following conditions; After heating operation for more than 30 min., the discharge gas temperature of compressor is not 10°C higher than the condensing temperature and there is no other faults such as excessive charge of refrigerant, etc. |

(5) Checking of Electrical Coil Parts

| Name of Parts | Model | Electrical Wiring Diagram | Wiring No. | Resistance (Ω) |
|---|---|---|---|---|
| Fan Motor for Indoor Unit  for RPI-0.8FS, 1FS, 1.5FS | ENO-KPPA 55W |  | ①-② ②-③ ③-④ ④-⑤ ⑤-⑥ | 44.8 41.1 21.0 4.3 2.9 at 25℃ |
| Fan Motor for Indoor Unit  for RPI-2FS | ENO-KPPA 160W |  | ①-② ②-③ ③-④ ④-⑤ ⑤-⑥ | 12.7 12.8 4.4 6.6 3.7 at 25℃ |
| Fan Motor for Indoor Unit  for RPI-2.5FS | ENO-KPPA 180W | | ①-② ②-③ ③-④ ④-⑤ ⑤-⑥ | 23.7 10.4 4.1 4.7 4.2 at 25℃ |
| Fan Motor for Indoor Unit  for RPI-4FS | ENO-KPPA 200W |  | ①-② ②-③ ③-④ ④-⑤ ⑤-⑥ | 23.6 7.8 6.3 8.6 5.2 |
| Fan Motor for Indoor Unit  for RPI-5FS | ENO-KPPA 200W | | ①-② ②-③ ③-④ ④-⑤ ⑤-⑥ | 23.6 7.8 6.3 8.6 5.2 |
| Fan Motor for Indoor Unit  for RCI-1FSE | CL2361 12W |  | ⑥-① ①-② ②-③ ①-⑦ | 397.5 66.91 52.93 389.3 at 25℃ |
| Fan Motor for Indoor Unit  for RCI-1.5FSE | CL3060 23W |  | ⑤ ① ① ② ② ③ ③ ④ ① - ⑥ | 139.8 31.94 31.94 42.62 115.1 at 25℃ |
| Fan Motor for Indoor Unit  for RCI-2FSE, 2.5FSE | CP3030 30W | | ⑤·① ①·② ②·③ ③·④ ①·⑦ | 102.2 16.27 47.80 51.78 154.9 at 25℃ |
| Fan Motor for Indoor Unit  for RCD-1FS | ENO-KPPA 20W |  | ⑥·① ①·② ②·③ ③·④ ①·⑦ | 156.18 46.9 25.5 67.4 111.8 at 25℃ |
| Fan Motor for Indoor Unit  for RCD-1.5FS | CP2559 20W | | ⑥·① ①·② ②·③ ③·④ ①·⑦ | 113.4 31.29 27.68 74.58 133.9 at 25℃ |

HITACHI

| Name of Parts | Model | Electrical Wiring Diagram | Wiring No. | Resist-ance (Ω) |
|---|---|---|---|---|
| Fan Motor for Indoor Unit  for RCD-2FS | ENO-KPPA 65W |   | ①-② ②-③ ③-④ ④-⑤ ⑤-⑥ | 64.5 22.2 16.1 6.6 13.2 at 25°C |
| Fan Motor for Indoor Unit  for RCD-2.5FS | CP3045 85W |   | ⑥-① ①-② ②-③ ③-④ ①-⑦ | 88.85 34.25 37.02 109.2 107.8 at 25°C |
| Fan Motor for Indoor Unit  for RPK-1.5FS | ENO-KPPA 15W |   | ①-② ②-③ ③-④ ④-⑤ ⑤-⑥ | 316.5 54.3 37.8 41.1 117.5 at 25°C |
| Fan Motor for Indoor Unit  for RPF-1.5FS, RPFI-1.5FS | TNO-KP 20W |  | ⑥-① ①-② ②-③ ①-⑦ | 161.4 61.1 26.3 51.7 73.8 at 25°C |
| Fan Motor for Indoor Unit  for RPC-2.5FS(E) | CH3531 65W |   | ⑤-⑥ ④-⑤ ③-④ ②-③ ①-② | 5.4 8.8 4.5 10.8 32.5 at 25°C |
| Fan Motor for Indoor Unit  for RAS-5FS2, RAS-5FX2 | CX5004 150W |  | ④-① ①-③ | 30.37 32.51 |
| Fan Motor for Indoor Unit  for RAS-8FS2, RAS-8FX2 | CX5004 150W | | ④-① ①-③ | 30.37 32.51 |
| Fan Motor for Indoor Unit  for RAS-10FS2, RAS-10FX2 | CX5004 150W | | ④-① ①-③ | 30.37 32.51 |

<table>
<tr><th>Name of Parts</th><th colspan="3">Model</th><th colspan="2">Resistance (Ω)</th></tr>
<tr><td>Drain-up Motor<br><br>for RCI-1FSE, 1.5FSE, 2FSE, 2.5FSE</td><td colspan="3">PCD-4N230HS-3</td><td colspan="2">135<br>at 20℃</td></tr>
<tr><td>Drain-up Motor<br><br>for RCD-1FS, 1.5FS, 2FS, 2.5FS</td><td colspan="3">PCD-4N230HS-2</td><td colspan="2">135<br>at 20℃</td></tr>
<tr><td>Solenoid Valve for Gas Bypass<br><br>for RAS-5FS2, 8FS2, 10FS2,<br>RAS-5FX2, 8FX2, 10FX2</td><td colspan="3">ST10PA</td><td colspan="2">1,900<br>at 20℃</td></tr>
<tr><td rowspan="3">4-Way Valve<br><br>for RAS-5FS2, 8FS2, 10FS2,<br>RAS-5FX2, 8FX2, 10FX2</td><td>RAS-5FS2<br>RAS-5FX2</td><td colspan="2">CHV-0403<br>+CHV-01AJS03C1</td><td colspan="2" rowspan="3">1,740<br>at 20℃</td></tr>
<tr><td>RAS-8FS2<br>RAS-10FS2</td><td colspan="2">VH60100<br>+LB62012</td></tr>
<tr><td>RAS-8FX2<br>RAS-10FX2</td><td colspan="2">VH60100<br>+LB60012</td></tr>
<tr><td rowspan="3">Compressor Motor<br><br>for RAS-5FS2, RAS-5FX2,<br>RAS-8FS2, RAS-10FS2</td><td>RAS-5FS2<br>RAS-5FX2</td><td colspan="2">401DHV</td><td colspan="2">at 20℃</td></tr>
<tr><td>RAS-8FS2</td><td rowspan="2">401DHVM</td><td>400DHM</td><td rowspan="2">1,746<br>at 75℃</td><td>4,006<br>at 75℃</td></tr>
<tr><td>RAS-10FS2</td><td>500DHM</td><td>3,319<br>at 75℃</td></tr>
<tr><td>Compressor Motor<br><br>for RAS-8FX2</td><td colspan="3">803TLV</td><td colspan="2">for Inverter: 1.68<br>for Constant Speed:<br>3.93<br>at 20℃</td></tr>
<tr><td>Compressor Motor<br><br>for RAS-10FX2</td><td colspan="3">1003TLV</td><td colspan="2">for Inverter: 1.68<br>for Constant Speed<br>2.90<br>at 20℃</td></tr>
<tr><td>Contactor for Compressor Motor<br><br>for RAS-5FS2, 8FS2, 10FS2,<br>RAS-5FX2, 8FX2, 10FX2</td><td colspan="3">A25</td><td colspan="2">520<br>at 20℃</td></tr>
</table>

(6) Checking of Compressor

# CHECK LIST ON COMPRESSOR

CLIENT: ____________ MODEL: ____________ DATE: ____________

Serial No. ____________ Production Date ____________ ____________

| No. | Check Item | Check Method | Result | Remarks |
|---|---|---|---|---|
| 1 | Are THM2(CN103) and THM3(CN104) correctly connected?<br>THM2, THM3: Discharge Gas Thermistor<br>(THM3 is only for RAS-8FS2, 8FX2, 10FS2 and 10FX2) | ① Are wires between THM2 and CN103, and THM3 and CN104 correctly connected by viewing?<br>② Check to ensure that 7-segment indication of Td1 is higher than Td2 when No.1 comp. is operating.<br>Td1: Temperature of THM2<br>Td2: Temperature of THM3 | | |
| 2 | Are thermistors, THM2 and THM3 disconnected? | ① Check to ensure that thermistor on the top of comp. is correctly mounted by viewing?<br>② Check to ensure that actually measured temp. are greatly different from the indication(Td1, Td2) during check mode. | | |
| 3 | Are connectors for current sensor (CN110 and current sensor connector)? | ①Check to ensure that indication A1 and A2 are 0 during compressor stopping.<br>②Check to ensure that indication A1 and A2 are not 0 during compressor running.(However, A2 is 0 during stopping of No. 2 comp.) | | |
| 4 | Is current sensor faulty? | | | |
| 5 | Is current sensing part on PCB3 faulty? | | | |
| 6 | Is the direction of current sensor(CTU, CTV) reverse? | Check the direction ➧ by viewing. | | |
| 7 | Are power source wires, U and V inserted correctly into current sensor? | Check to ensure that wires are correctly inserted. | | |
| 8 | Are ex. valves(MV1, MV2 and MVB) correctly connected? | Check to ensure that MV1~CN115, MV2~CN116 and MVB~CN117 are correctly connected. | | |
| 9 | Are ex. valve coils(MV1, MV2 and MVB) correctly mounted? | Check to ensure that each coil is correctly mounted on the valve. | | |
| 10 | Are the refrigeration cycle and electrical wiring system incorrectly connected? | Check to ensure that refrigerant is flowing into indoor units by operating one refrigerating cycle only from the outdoor unit. | | |
| 11 | Is opening of ex. valve completely closed(locked)? | Check the following by the check mode of outdoor unit.<br>①Liquid Pipe Temp.(TL)<Air Intake Temp.(Ti) during Cooling Operation<br>②Liquid Pipe Temp.(TL)>Air Intake Temp.(Ti) during Heating Operation | | |
| 12 | Is opening of ex. valve fully opened (locked)? | Check to ensure that liquid pipe temp. is lower than air intake temp. of stopping indoor unit when other indoor units are operating under cooling operation. | | |
| 13 | Are the contacts for comp. magnetic switch CMC faulty? | Check the surface of each contact(R, S and T) by viewing. | | |
| 14 | Is there any voltage abnormality among L1-L2, L2-L3 and L3-L1? | Check to ensure that voltage imbalance is smaller than 3%.<br>Please note that power source voltage must be within 380V or 415V ±10%. | | |
| 15 | Is the comp. oil acidified during compressor motor burning? | Check to ensure that the oil color is not black. | | |

## Additional Information for "CHECK LIST ON COMPRESSOR"

| Check Item | Additional Information(Mechanism of Compressor Failure) |
|---|---|
| 1 & 2 | The liquid refrigerant return volume to the compressor is controlled by the discharge gas temperature Td1 when only No.1 compressor is operating. If Td1 and Td2 are reversely connected, the liquid refrigerant return volume will become small by detecting the temperatures even if the actual discharge gas temperature is high. Therefore, this abnormal overheating operation will result in insulation failure of the motor winding. |
| 3, 4 & 5 | Overcurrent control(operating frequency control) is performed by detecting current by the current sensor.<br>In this case, winding insulation failure will occur, since control is not available in spite of actually high current. |
| 6, 7 | The current sensor checks phase and adjusts output electrical wave in addition to the above mentioned items. If fault occurs, the output electrical wave becomes unstable giving stress to the motor winding, resulting in winding insulation failure. |
| 8, 9 | During a cooling operation, Pd is controlled by MV1 and MV2, and Td and SH are controlled by MVB.<br>During a heating operation, Td and SH are controlled by MV1 and MV2, and MVB is fully closed.<br>If expansion valves are incorrectly connected, correct control is not available, resulting in compressor seizure depending on liquid refrigerant returning conditions or motor winding insulation failure depending on overheating conditions. |
| 10 | If the refrigeration cycle and electrical system are incorrectly connected, abnormally low suction pressure operation is maintained or abnormally high discharge pressure operation is maintained, resulting in giving stress to the compressor, since their correct control is not available. |
| 11 | ditto |
| 12 | The compressor may be locked due to the liquid return operation during the cooling operation. |
| 13 | In the case that the contacting resistance becomes big, voltage imbalance among each phase will cause abnormal overcurrent. |
| 14 | In this case, overcurrent will occur, efficiency will decrease or the motor winding will be excessively heated. |
| 15 | In the case, it will result in motor burning or compressor seizure. |

## 1.4 Test Run

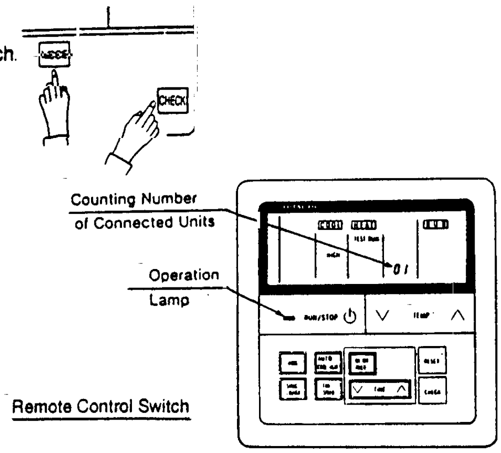
### 1.4.1 Test Run Mode by Remote Control Switch

I Turn ON the power source of the units.

II Procedure for "TEST RUN" mode of remote control switch.
Depress the MODE and the CHECK switches together for more than 3 second.

→ If "TEST RUN" and the counting number of the connected units to the remote control switch (for example "01") are indicated on the remote control switch, the connection of remote control cable is correct.
This indication will last for approximately 1 minute.

→ If no indication appears or the number of the units indicated is smaller than the actual number of the units, some abnormalities exist.




III

| Remote Control Switch Indication | Fault | Inspection Points after the Power Source OFF |
|---|---|---|
| No Indication | · The power source is not turned ON.<br>· The connection of the remote control cable is incorrect.<br>· The connecting wires of power supply line are incorrect or loosened. | 1. Connection between the connector and the wires: Red wire–No.1, Black wire–No.2, White wire–No.3<br>2. Connecting points of Remote Control Cable<br>3. Contact of Connectors of Remote Control Cable<br>4. Connection Order of each Terminal Boards<br>5. Screw Fastening of each Terminal Boards |
| Counting number of connected units is incorrect. | · The setting of unit number is incorrect.<br>· The connection of control cables between each indoor units are incorrect. (When one remote control switch controls multiple units.) | 6. Dip Switch Setting on Printed Circuit Board<br>7. Wire Connecting Order of Bridge Cable<br>8. Connecting Points of Bridge Cable<br>9. Contact of Connectors of Bridge Cable |

IV Select TEST RUN MODE by depressing MODE Switch. (COOL or HEAT)

V Depress RUN/STOP switch.

→ The "TEST RUN" operation will be started. (The "TEST RUN" operation will be finished after 2 hours unit operation or by depressing the RUN/STOP switch again.)

→ If the units do not start or the operation lamp on the remote control switch is flickered, some abnormalities exist.

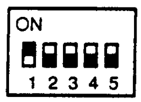
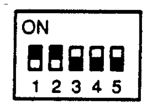
VI

| Remote Control Switch Indication | Unit Condition | Fault | Inspection Points after the Power Source OFF |
|---|---|---|---|
| The operation lamp flickers. (1 Time/1 Sec.) And the Unit No. and Alarm Code "03" flicker. | The unit does not start. | The connecting wires of operating line are incorrect or loosened. | Connecting Order of each Terminal Boards.<br>Screw Fastening of each Terminal Boards. |
| The operation lamp flickers. (1 Time/2 Sec.) | The unit does not start. | The connection of remote control cable is incorrect. | This is the same as item III 1,2 and 3. |
| Indication or Flicker different to above. | The unit does not start, or starts once and then stops. | The connection of the thermistors or other connectors are incorrect. Tripping of protector exists, etc. | Check by the alarm code table in the service manual. (Do it by service people.) |
| Normal | The outdoor fans rotate reversely. | The connecting order of power supply line is incorrect. | Connecting Order of the Terminal Board: TB1 in the Outdoor Unit. |
| Normal | The outdoor fans do not start. | Some wires of power supply line is disconnected. | Connecting Point of Power Supply Line. Contact Outdoor Fan Motor Connector. |

1 . 4 . 2 Test Run Mode by Outdoor Unit

(1) Purpose: This function is used to check operation conditions of components by test operation from the outdoor unit by a service engineer.

(2) Operation Method: After setting the dip switch, DSW4 on the outdoor unit PCB, turn ON the main switch. After power supply, wait for more than 5 minutes due to automatic address setting by the micro-computer, and then, start the test operation.

| Cooling Operation | Heating Operation |
|---|---|
|   |   |

a. All the indoor units connected to the outdoor unit are automatically started under the same operation mode such as cooling or heating.

b. During this test running, the remote control switch indicates "TEST" on the display. If temperature setting is not performed, the following setting is given.

- Cooling Operation
  Fan Speed: Setting Speed before Test Run
  Temperature: 28
- Heating Operation
  Fan Speed: Setting Speed before Test Run
  Temperature: 22

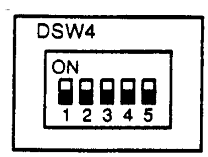
c In the case that the operation mode such as operation controlled by remote control switches or central station, the normal test operation is not available.
◎Remote Control Switch: Switch OFF the "RUN/STOP switch.
◎Central Station: Stop the central control and give instruction for individual operation.
After the test operation by the outdoor unit is started, If the above mentioned controller are used, operation mode may be changed or this test operation may be automatically stopped. Therefore, do not touch any controllers.

d. If the system is stopped due to an alarm during this test operation, turn OFF the main power switch and remove the cause of the alarm and then, supply power to the system so that the test operation can be re-started.

(3) Cause of Stoppage

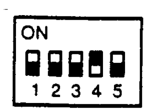
a. Turning OFF Main switch to Outdoor Unit (Power OFF to Outdoor Unit PCB)
b. Alarm
c. Switching RUN/STOP Switch OFF
d. Resetting the dip switch, DSW4

Check to ensure that the dip switch, DSW4 is reset as it was after this test operation.

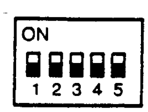



(4) Compressor Forced Stop

a. If #4 of DSW4 is turned ON during operating comp., immediately comp are stopped and indoor units are Thermo-OFF.




b. If #4 of DSW4 is turned OFF, comp. are operated after cancelling enforced 3 minutes stoppage.




NOTE: Do not turn ON/OFF frequently the compressor.

1.4.3 Checking List (FS2 Series)

# CHECK LIST ON TEST OPERATION

CLIENT: ____________ INSTALLER: ____________ DATE: ____________

____________ O.U. Serial No.: ____________ Checker ____________

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| I.U. Model | | | | | | | | |
| I.U. Serial No. | | | | | | | | |

I.U. Indoor Unit, O.U.: Outdoor Unit

Piping Length: m Additional Refrigerant Charge: kg

(1)General

| No. | Check Item | Result |
|---|---|---|
| 1 | Was the dip switch, DSW6 for piping length in O.U. set? | |
| 2 | Was the dip switch, DSW3 for piping lift in O.U. set? | |
| 3 | Is the transmitting wire contacting to power lines? | |
| 4 | Was an earth wire connected? | |
| 5 | Is there any short circuit? | |
| 6 | Is there any voltage abnormality among each phase(L1-L2, L2-L3, L3-L1)? | |

(2)Refrigeration Cycle

a Operation (Cooling/Heating)

| No. | Check Item | Result |
|---|---|---|
| 1 | Operate all the indoor units ("TEST RUN" mode). | |
| 2 | Operate all the indoor units at "HIGH" speed. | |
| 3 | In case that the constant compressor is turned ON and OFF repeatedly, stop one indoor unit (small capacity one). | |

b. Sampling Data

| No. | Check Item | Result |
|---|---|---|
| 1 | After the operation for more than 20 min. | |
| 2 | Check Pd and Td. Is Td · SH 20 to 40 deg.? | |
| 3 | Is Ps 2.0 to 5.0? | |
| 4 | Is Pd 12 to 22? (If the outdoor temperature is high, Pd becomes high.) | |

(3)Check Item after Sampling Data

a. Cooling Operation (It is applicable when outdoor temperature is higher than 15°C

| No. | Check Item | Standard | Causes | Result |
|---|---|---|---|---|
| 1 | Is H1(Compressor Frequency) + (CC(Numbers of Running Compressor)−1) × (※) abnormally low or high? (It is applicable when Intake Air Temp. is 3 deg. higher than Setting Temp.) | Running Horsepower of Indoor Units × 15Hz | ·Low: Insufficient Refrigerant<br>·High: Excessive Refrigerant<br>·DSW for I.U. Capacity; Incorrect Setting | |
| 2 | Is fan actually running at "HIGH" speed when Fo(Air Flow Rate of Fan) is「15」or「16」? | − | ·Fan Motor; Failure<br>·PCB; Failure<br>·Condenser; Failure | |
| 3 | Is Td1 higher than Td2 when only No. 1 compressor is running(when CC(Numbers of Running Compressor) is「1」)? | − | ·Td Thermistor; Incorrect Connection or Incorrect Mounting | |
| 4 | Is the total of iE(Indoor Ex. Valves Opening) abnormally low or high? | Total of iE: Horsepower of Outdoor Unit × (5~30) | ·Low: Excessive Refrigerant<br>·High: Insufficient Refrigerant, | |
| 5 | Is TL(Liquid Pipe Temp. of I.U. Heat Exchanger) lower than Ti(Intake Air Temp. of I.U.)? | It is normal when TL−Ti <−5. | ·TL Thermistor; Failure<br>·Ex. Valve; Fully Closed<br>·Short-circuit | |
| 6 | Is TG(Gas Pipe Temp. of I.U. Heat Exchanger) lower than Ti(Intake Air Temp. of I.U.)? | It is normal when TG−Ti <−5. | ·TG Thermistor; Failure<br>·Ex. Valve; Fully Closed or Slightly Open<br>·Short-circuit | |
| 7 | Is there any excessive difference among I.U. at SHTG−TL of I.U. heat exchanger? (It is applicable when Intake Air Temp. is 3 deg. higher than Setting Temp.) | It is normal if the difference among units is within 7 deg. | ·TL/TG Thermistor; Failure<br>·Ex. Valve; Fully Open, Slightly Open or Fully Closed | |
| 8 | Is there any I.U. with the I.U. heat exchanger SHTG−TL excessively different from other units' value? (It is applicable when Intake Air Temp. is 3 deg. higher than Setting Temp.) | It is normal if SH is within 3 deg. lower than other units. | ·Ex. Valve; Locked with Fully Open<br>· | |
| 9 | Is there any I.U. with SH excessively lower than other units' value, under the condition of iE(I.U. Ex. Valve)「100」? | It is normal if SH is within 3 deg. higher than other unit | ·Ex. Valve; Locked with Slightly Open or Closed<br>· | |
| 10 | Is the difference between Discharge Air Temp. and Intake Air Temp. more than 7 deg.? | − | − | |

b. Heating Operation (It is applicable when outdoor temperature is higher than 0°C.)

| No. | Check Item | Standard | Causes | Result |
|---|---|---|---|---|
| 1 | Is oE1(O.U. Ex. Valves Opening) abnormally low or high when Td - SH is 15 to 30 deg.? | oE1 = 30~75% | ·Low: Insufficient Refrigerant<br>·High: Excessive Refrigerant<br>·DSW for I.U. Capacity; Incorrect Setting | |
| 2 | Is Pd「16」to「22」?<br>(Pd is high when the indoor temperature is high.) | – | ·Low: Solenoid Valve 20A Leakage<br>·High: Excessive Gas Pipe Resistance | |
| 3 | Is H1(Compressor Frequency) + (CC(Numbers of Running Compressor)-1) × (※) abnormally low or high? (The lower is the room temp. and outdoor temp., the higher is the above value.) | – | ·Low: Excessive Refrigerant<br>·High: Insufficient Refrigerant, Excessive Pipe Resistance | |
| 4 | Is Ps「2」to「5」?<br>(Only under the condition that electrical expansion valve 20A is OFF.) | – | ·Low: O.U. Short-circuit<br>·Low/High: O.U. Fan Thyristor Failure or Outdoor Air Sensor; Failure | |
| 5 | Is the temperature difference between I.U.* more than 15 deg. when iE(I.U. Ex. Valve) is 100?<br>*The temperature difference between I.U. means the following;<br>b3(Discharge Gas Temp.) - b2(Intake Air Temp.) indicated on the remote control switch by check mode. However, this is applicable only when b2(Intake Air Temp.) - b1(Setting Temp.) is higher than 3 deg. | – | ·Failure such as P.C.B., Wiring, Coil, Valve<br>·Excessive Pipe Resistance<br>·Thermistor Failure for Discharge Air | |

NOTES: 1. The symbol with an underline ___ indicates checking item and the mark 「 」 indicates checking data.

2. Regarding (※), the following value should be applied; 8FS2···50/60, 10FS2···62/75 (This is not applicable to 5FS2.)

1.4.4 Checking List (FX2 Series)

## CHECK LIST ON TEST OPERATION

CLIENT: ____________ INSTALLER: ____________ DATE: ____________

____________ O.U. Serial No. ____________ Checker ____________

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| I.U. Model | | | | | | | | |
| I.U. Serial No. | | | | | | | | |

I.U. Indoor Unit, O.U.: Outdoor Unit

Piping Length: m Additional Refrigerant Charge: kg

(1)General

| No. | Check Item | Result |
|---|---|---|
| 1 | Was the dip switch, DSW6 for piping length in O.U. set? | |
| 2 | Was the dip switch, DSW3 for piping lift in O.U. set? | |
| 3 | Is the transmitting wire contacting to power lines and operation lines? | |
| 4 | Was an earth wire connected? | |
| 5 | Check for any short circuit? | |
| 6 | Is there any voltage abnormality among each phase(L1-L2, L2-L3, L3-L1)? | |

(2)Refrigeration Cycle

NOTE: Perform checking after all the indoor units are operated at the "HIGH" speed and the system becomes stable.

a. Common

| No. | Check Item | Result |
|---|---|---|
| 1 | Is A1 (compressor running current) is greater than 「1」 when compressor is running? | |
| 2 | Is Td1 (discharge gas temp.) higher than Td2 when only No. 1 comp. is operating? | |
| 3 | Is there any abnormal sound or vibration? | |

b. Cooling Operation (It is applicable when outdoor temperature is higher than 15℃.)

| No. | Check Item | Result |
|---|---|---|
| 1 | Is oEb (O.U. Ex. Valve Opening) within "7" to "40" when Td – SH is 15 to 30 degC.? | |
| 2 | Is Ps 「3」 to 「5」 when solenoid valve, SVA is OFF? | |
| 3 | Is the H1 + H2 (compressor frequency) abnormally low or high? If outdoor temp. or indoor temp. is high, H1 + H2 becomes high.<br>H2 is "50" when constant comp. is running. | |
| 4 | Is Pd 「11」 to 「22」 ? | |
| 5 | Is solenoid valve SVA OFF? | |

| No. | Check Item | Result |
|---|---|---|
| 6 | Is indoor unit heat exchanger SH 2 to 10 deg. when iE (I.U. ex valve) is less than 「7」 or 「100」 ?<br>Indoor unit heat exchanger SH indicates the following;<br>b7 (I.U. Heat Exchanger Gas Temp.) - b4 (I.U. Heat Exchanger Liquid Temp.) which are indicated on the remote control switch<br>However, The above is applicable only when the data can meet the following.<br>b2 (Intake Air Temp.) - b1 (Setting Temp.) >3 deg. | |
| 7 | Are TL (I.U. heat exchanger liquid temp.) and Ti (I.U. Intake Air Temp.) almost same when iE (I.U. ex. valve) is 「2」 (at stoppage or thermostat OFF)? | |

c. Heating Operation (It is applicable when outdoor temperature is higher than 0°C.)

| No. | Check Item | Result |
|---|---|---|
| 1 | Is oE1 + oE2 (O.U. Ex. Valves (No. 1 & 2) Opening) abnormally low of high when Td - SH is 15 to 30 degC.?(If operating frequency is high, oE1 + oE2 will become high.) | |
| 2 | Is Pd 「16」 to 「22」 ? | |
| 3 | Is the H1 + H2 (compressor frequency) abnormally low or high? If outdoor temp. or indoor temp. is low, H1 + H2 becomes high. | |
| 4 | Is Ps 「1」 to 「5」 when solenoid valve SVA is OFF? | |
| 5 | Check for leakage on liquid bypass ex. valve(oEb) by touching | |
| 6 | Is the temperature difference between I.U. more than15 deg. when iE (I.U. ex valve) is 「100」 ?<br>The temperature difference between I.U. means the following;<br>b3 (Discharge Gas Temp.) - b2 (Intake Air Temp.) indicated on the remote control switch by check mode. However, this is applicable only when<br>b2 (Intake Air Temp.) - b1 (Setting Temp.) > 3 deg. | |

NOTE: The symbol with an underline indicates checking item and the mark 「 」 indicates checked data.

(3) Others